

How a PGM Map Gets Made

The division of labour between an authoring agent
and pgm-studio

Two maps are built from first principles in these pages: a Capture the Wool board, composed by the studio and then adapted into a place, and a destroy-the-monument board authored from nothing at all. Every stage of both is shown as it came out of the machine. At each stage the paper asks the same question. *Who decided this?*

Purpose, Audience and Method

pgm-studio is a program for making Minecraft PvP maps of the kind a PGM server loads. It is driven over HTTP, which means the thing operating it need not be a person sitting at a screen. For most of the maps described here, the operator was a language model with a terminal and a list of endpoints. This paper is an account of what such a machine actually does, and of what the studio does for it.

That question turns out to be the interesting one. A map is not produced by a generator that takes a seed and returns a finished world, and it is not produced by an author typing coordinates either. It is produced by two parties handing work back and forth, and the line between them moves from stage to stage. The studio arranges a board and the agent decides what the ground is. The agent states a theme and the studio decides which block lands on which cell. The agent asks for something, the studio refuses, and the refusal is where a good deal of the map comes from. Naming that line precisely, stage by stage, is the whole of what follows.

The paper is in three parts, and they can be read in any order.

Part I. The System

What a map is made of, the four documents that describe one, and the division of labour stated plainly. Three short chapters; read these first whatever else you read.

Part II. The Instruments

One chapter per capability of the studio: the composer, the rules, shapes, relief, layers, terrain, houses, dressing, the export, and reading a world back. This is the reference half, meant to be dipped into rather than read through.

Part III. Two Maps, End to End

The two boards, every stage pictured, and a chapter on what an authoring session actually looks like from the outside, reconstructed from a recording of every call the agent made.

A reader who plays these maps and wants to know where they come from can read Part I and then Part III, and skip the middle entirely. A reader who wants to drive the studio wants Part II. Terms are defined where they are first used, and collected again in the glossary at the back.

One discipline is worth stating up front, because it constrains what this paper is permitted to say. Nothing here claims that a map *plays* a particular way unless that claim traces to one of two sources: a ruling by the repository's author, cited as such, or a measurement over the corpus of roughly 350 published PGM maps, quoted with its number. The source code can say what the software does and the corpus can say what mapmakers have done. Neither can say what is good. Where the honest answer was that nobody had measured it, the claim is absent rather than invented.

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PART I

The System

What a map is made of, the four documents that describe one, and the division of labour stated plainly. Three chapters; read these first whatever else is read.

Maps, Objectives and the Two Authors

1.1 The Anatomy of a PGM Map

PGM is a server plugin that runs competitive Minecraft matches. It loads a map, which is two things bundled together: a Minecraft world, meaning the blocks, the terrain and the buildings; and a file called `map.xml` that says what the world is *for*. Without the XML the world is scenery. The XML is what states that this team spawns here, that this region cannot be built in, that the match ends when that block is broken.

Objectives come in a handful of kinds, and two of them carry this paper. A **Capture the Wool** map, universally abbreviated CTW, hides coloured wool in rooms, and a team wins by carrying the right wool back to its own monument. A **destroy-the-monument** map, DTM, puts a block or a small structure in each team's ground, and a team wins by breaking the enemy's. Others exist: a core that leaks lava when broken, a control point held over time. The studio builds them all, but CTW and DTM are the two worked examples here.

1.2 Team Counts by Objective

The number of teams a map can hold is not one question but two, and they have different answers. One is about the objective and belongs to PGM. The other is about symmetry and belongs to the studio.

Capture the Wool works at two teams or four, and four-team CTW maps have been built. It works at four because each team has a *distinct* objective: capturing the other three teams' wools. Control points, whose support is recent, take four teams or more, usually an even number, simply because maps are usually symmetric.

Destroyables and cores are different, and the reason is worth stating properly because it is easy to mistake for a software limit. It is not one. The difficulty is **ownership**. A destroyable cannot be assigned to a team the way a wool can. If a team's monument is destroyed then that team has, in a sense, lost. But that does not determine a *winner*, and with more than two teams there is no well-defined one. PGM maps work around it in two ways. Some make the match blitz, so the last surviving team wins. Others move the players of a team that has lost its monument onto another team, so that one team can truly win. Neither of those is raw DTM or DTC. A destroy map without one of those modes layered on it is therefore a two-team map in practice.

The Studio's Team Ceiling

The studio's automatic symmetry fan tops out at four teams, because `rot_90` is the only four-image mode it has. That is what caps the studio, rather than anything about the objectives. An agent driving the API directly can author anything, a six-team map included. What it gives up is the automatic fan, so each team's ground has to be authored rather than mirrored into place.

The blocks and the XML have to agree with each other, and that is harder than it sounds. A wool room's position in the XML is a coordinate; whether that coordinate is inside a room a player can

reach is a fact about the blocks. A build region drawn in the XML means nothing if the ground it covers was never placed. Most of what a map-authoring program does is keeping those two descriptions true of one another.

1.3 The Studio: Scope and Interface

pgm-studio is an ASP.NET Core application with a browser client, a database, and an HTTP API of 174 paths carrying 231 operations. It authors maps: it plans a layout, draws its ground, states what the map is played for, and writes out the world and the `map.xml` a server can load.

The browser client exists and is used. But the API is the whole surface, and every map described in this paper was made without opening the client once. A caller with a terminal can compose a board, reshape its ground, state its objectives, build the world and read the result back, entirely over HTTP. That property is what makes the rest of this paper possible, and it was designed in rather than discovered. The documentation for each tool carries that tool's endpoints, on the principle that a description which cannot be acted on is a description of nothing.

1.4 The Authoring Agent

For most of the maps in the repository this paper draws on, the operator is a language model running in a terminal. It reads documentation, writes a Python script that emits two JSON documents, posts them, reads the studio's answer and corrects. It has no eyes on the world. It cannot walk around the map. Every belief it holds about what it has built comes from a number the studio handed back, and a good deal of the discipline described in Part III exists because that is a genuinely difficult position to author from.

This paper calls that operator **the agent** throughout, and the program it drives **the studio**. The two words are used precisely and always mean those two things.

1.5 The Central Argument

It is tempting to describe a system like this as a generator with a human in the loop, or as an editor with some automation in it. Neither is right, and the reason neither is right is the substance of this paper.

The studio is not a generator, because it does not produce maps. Its composer produces an *arrangement*: where the hub sits, how far the wool rooms stand off it, which side the front line faces. An arrangement is not a map. It has no elevation, no curve, no building, no second storey, no material. Every composed board is flat, rectangular and grey. What it buys is the one thing that is hard to invent and easy to get wrong, and the repository's own brief on the subject puts it as sharply as it can be put: a composed board is a suggestion, not a plan.

The studio is not an editor either, because a very large amount of what ends up in the world was never stated by the agent. The agent says a theme should follow the ground's slope, and the studio decides which of five materials lands on each of several hundred thousand cells. The agent says a house stands here, and the studio decides its storey heights, its roof pitch, where its door goes and which way the door faces. The agent says the two halves of the board are mirror images, and the studio decides what that means for every shape, every prop and every region. These are not defaults in the sense of values that could have been typed out. They are derivations, and the agent could not have produced them by hand in any reasonable time.

The honest description is therefore the third one: two parties, a boundary between them, and the boundary in a different place at every stage. At the board level the studio arranges and the agent

adapts. At the ground level the agent states intent and the studio computes geometry. At the paint level the agent picks a rule and the studio applies it. At the objective level the agent decides almost everything, and the studio's task is to refuse the decisions that cannot be built.

That last one deserves its own sentence, because it is the part least visible from outside. A large fraction of the studio consists of gates: 183 rules across 28 families, each of which can stop a request and say why. They are not validation in the defensive sense. They are the accumulated knowledge of what makes a map fail, expressed as things the software will not let happen, and an agent authoring against them is being taught as much as it is being checked.

1.6 Structure of This Paper

The paper works through the stack once, in the order a map passes through it, and at each stage says which party decided what. Then it builds two maps.

The first is a CTW board in the *micro* size band, pulled straight off the composer and then adapted: reshaped, given elevation, given buildings, given a second storey, until it is a place rather than a diagram. The second is a DTM board authored from nothing, with no composed arrangement to start from, by an agent given no brief beyond the objective. Both are shown at every stage of the pipeline, from the bare rectangles the composer emitted to the finished world.

The last chapter is about the agent rather than the map. While the DTM board was being authored, every HTTP call its author made was recorded: the route, the status, the rules that fired, the time it took. That recording is what the chapter is written from, so the account of how an authoring session goes is taken off the wire rather than from the agent's own description of itself.

The Four Levels of Description

2.1 A Map as Four Documents

The first thing to understand about the studio is that it does not hold a map. It holds four descriptions of one, at four different grains, each owned by a different tool and stored separately. The documentation states it flatly:

A map is not one document. It is four, each describing the whole map at a different grain, each owned by a different tool and stored separately.

Table 2.1 is that list. Read it top to bottom as increasing resolution: the board knows what things are and not where they are exactly; the world knows where every block is and nothing about what any of it is for.

Table 2.1. The four levels. Each is a description of the whole map.

Level	What it states	Document	Tool
The board	Rectangles on a coarse cell grid, what each one is for, and where the objectives sit	PlanModel (*plan.json)	Plan
The ground	The real geometry at block resolution: outlines, heights, relief, paint, props	SketchLayout	Sketch
The play	Teams, spawns, protections, build regions, objectives and how they are captured	MapIntent	Configure
The map	The voxel world and the map.xml a PGM server loads	VoxelWorld + MapXml	— (built)

These are grains, not drafts

The temptation is to read the four as stages of completeness, a rough sketch refined into a finished thing. That reading is wrong and leads directly to authoring mistakes, so the documentation heads it off:

A plan is not a rough draft of a layout: it states things a layout cannot, such as that this rectangle is a wool room, or that these two pieces share a defence wall, and cannot state things a layout can, like a curve or a one-block step. The same is true upward: a layout knows where every block of ground is and has no idea what any of it is for. That is why a finished map needs all four and why no one of them is ‘the’ map.

This is worth dwelling on because it explains the shape of everything downstream. A wool room is a concept that only exists at the board level. Once the board compiles, there is no wool room in the geometry. There is a polygon, at a height, with a theme on it. Conversely a coastline is a concept that only exists at the ground level; no arrangement of rectangles can state one. An agent that tries to express a curve in the plan is fighting the model, and an agent that tries to state an objective in the layout is doing the same thing from the other side.

The flow is one-way

A plan compiles into a layout and an intent; a layout rasterizes into a world; an intent projects into the map document; the document writes out as `map.xml`. Nothing reads back up. There is no path from geometry to a plan, and none from a finished `map.xml` to an intent. Both omissions are deliberate.

Nothing infers upward. A finished map cannot be decompiled into the plan that would have produced it, and the studio does not try; the documentation's position is that hypothesising what a finished map's plan would have been is a person's job rather than a tool's. The practical consequence for an agent is that corrections are made *upstream and re-run*, not patched downstream. If the ground is wrong, the thing to change is the document that produced it.

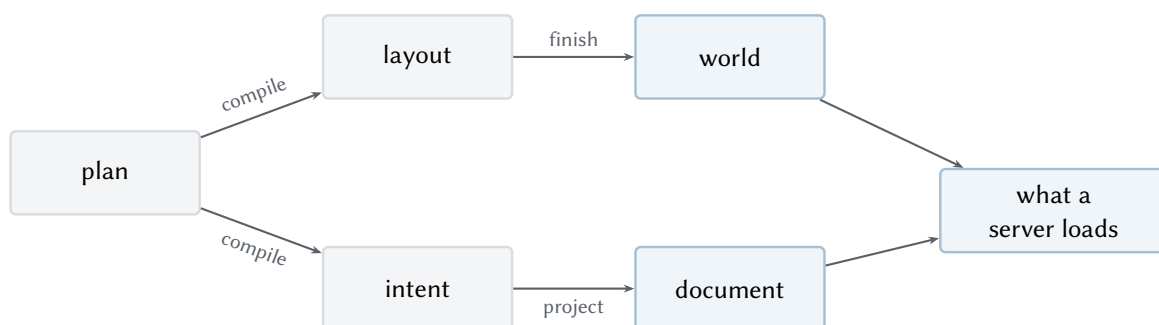


Figure 2.1. The five hand-offs. The two branches rejoin at the end rather than staying independent: building the world *resolves* parts of the intent, and the `map.xml` is rendered from that resolved copy.

2.2 The Five Hand-Offs

Figure 2.1 is the whole pipeline. Each arrow is a call.

Plan to layout and intent. `POST /api/plan/compile` turns one plan into both downstream documents, and it is pure: the same plan compiles to the same pair on the server and in the browser. One thing happens here that surprises people reading a compiled board for the first time. Abutting pieces of equal height fuse into single polygons, so what arrives at the ground level is a board rather than a grid of rectangles. A plan of eleven rectangles can compile to four shapes.

Layout onto a map. `PUT /api/map/{slug}/sketch/from-plan` merges rather than replaces. Themes, room shells, dressing and author-corrected heights are carried onto the fresh geometry. Relief is the exception, and the reason is worth stating because it is the clearest example in the system of the studio refusing to guess: relief is keyed by group id, group identity is derived from the geometry, and a recompile that re-fuses the board produces different groups. Hand-authored terrain would have nowhere correct to land. So that case answers **409**, one SK1 finding per group it would orphan, and `?force=true` accepts the loss. It is the author's call, not the server's.

Intent onto a map. `PUT /api/map/{slug}/intent/from-plan` carries much less: the authors and contributors the stored intent already held, and nothing else. Two slices deliberately do not ride across. `islandTeams` is a positional derivation, and carrying it would relabel territory rather than preserve an answer. `symmetry` is left behind because its presence is what switches the orbit expander on. A rebuild therefore clears both, and the World and Teams phases have to be walked again.

Layout to world. POST /api/map/{slug}/sketch/finish rasterizes the layout into world geometry and moves the map to the configure stage. This is the only stage transition the studio performs at runtime.

Intent to document to map.xml. PUT /api/map/{slug}/intent stores the intent and projects it into the PGM document (teams, kits, regions, filters, apply-rules and spawns) in one idempotent pass. GET /api/map/{slug}/xml renders that document; GET /api/map/{slug}/export gives the world.

2.3 The Three-Document Load

An agent reading the five hand-offs might reasonably conclude that authoring a map means making five calls in order. It does not, and this is the single most consequential piece of practical knowledge in the whole pipeline.

POST /api/map/from-documents takes a plan, a layout and an intent together, stores a whole map from them, and answers the slug it landed under. The compile, the rasterize and the projection all run inside it. A map already stored under that slug is replaced, because the documents name one map, so loading them twice is a reload rather than a second map.

The documentation is explicit that this is the authoring call and not merely the import one, and equally explicit about what the staged path costs an agent that walks it instead: six calls for one store, a fresh slug originated on every correction, and the need to know that the intent's projection lands after the metadata write. The staged path is what the browser does. A driver never needs it.

The layout and the intent are required; the plan is not. A board drawn on a grid has no plan, so a layout emitter states none. Omitting either of the required two answers **400** naming which.

This call is also the way *back* in, and that matters more than it sounds. No route in the studio reads a map.xml at all: POST /map/import-folder refuses a folder carrying one outright, and import-url extracts only the region files. So a map authored against one studio could reach another only as a world, arriving without its plan, its drawing or its intent, and never able to be re-planned. The three documents carry more than the world does.

Request Body Conventions

Twenty-five of the studio's write endpoints take a bare document and fourteen take a wrapper, so an author guessing gets it right slightly more than half the time. The rule that resolves it: if the endpoint is asking *about a document*, post the document unwrapped; if it is asking to *file a document under a name*, post the record that names it. A wrong guess is silent, because an unknown property is dropped rather than reported. Posting {"style": {...}} where a bare style was wanted answers **200** with a preview of the defaults.

2.4 The Library

Beside the four sits one more thing that is not a level at all: the library of materials, themes, house parts and room styles. It knows nothing about maps, having neither slug nor stage, and the tools that build worlds reach into it to pick a recipe. What they take is a *copy*, so a library edit can never rebuild a map that already shipped. A theme changed today does not alter a world exported last week.

2.5 Stage and Layer

A map row carries a **stage** and, separately, the **layers** it holds. A stage is where a map has got to. A layer is a document it carries, and it keeps carrying it: a map built from a plan still has its plan after it has been sketched, configured and exported.

A stage is a progress marker and not a lock. A configured map may be re-planned, and no endpoint refuses on the basis of the stage a map stands at. What the stage is for is saying which of the moves already open is the one the author was about to make.

The studio answers both questions at runtime. `GET /api/map/{slug}/state` gives the stage, the artifacts, and the moves they allow, each with its route. A move is offered because the documents it reads are stored, not because the stage is right. Its pair, `GET /api/map/{slug}/findings`, answers what is wrong with the map right now from every gate the stored documents can reach, and names the gates it did *not* ask along with the route that pays for them. An agent that has lost track of where it is asks those two.

2.6 Responsibilities Held at a Single Level

A few things belong to exactly one level, and an agent looking for them at the wrong one will not find them.

Terrain paint and dressing are the ground level's alone. The rasterizer is what makes the world, so a finish authored a level above it would be authored before the thing it finishes exists. The observer spawn is the play level's alone: a plan puts it at the origin at a computed height, which is why an unattended spectator stands at 0, 0. Water lanes are the board level's alone; they survive a save at the play level and generate their region, but nothing at that level draws them. Room shells come from the ground level's theme phase rather than from the play level, which is the one most often looked for in the wrong place.

The Division of Labour

3.1 The Boundary and Its Source

The repository states its own answer in one sentence, in the brief that governs authoring from a composed board:

A composed board is a **suggestion**, not a plan.

And then, more usefully, the mechanism behind it:

It emits a plan, not a layout. The plan is where the board's *arrangement* lives. Everything that makes a place, whether the shape of the ground, the second storey, the edge that is not a straight line, the wall or the island, is downstream of it and is yours.

That is the cleanest statement of the boundary anywhere in the system, and it maps exactly onto chapter 2's four levels. The studio's generative half works at the board level and stops there. Everything at the ground level, the play level and the world is authored. That does not mean typed out by hand, because the studio computes enormous amounts of it, but it does mean that no part of it happens unless the agent asks for it.

3.2 Contributions of the Studio

3.2.1 The Arrangement

The composer knows how a CTW board is put together: a hub with a spawn hung off it, one or two wool approaches hanging off the hub's free surface, a frontline fronting it, a mid band between the two fanned images with a row of stepping stones in it, and a land budget that keeps a wool board about a third land. The brief's verdict on why that is worth having is unsentimental. It is the thing that is hard to invent and easy to get wrong, and a board drawn from scratch that ignores it comes out illogical in ways no gate catches.

It is equally explicit about where that stops. Table 3.1 is the list, and every row of it is a thing the agent must supply or the board will not have.

3.2.2 Derived Furniture

Downstream of the plan, a great deal appears that the agent never wrote. When a CTW board is stored, the studio works out the monuments, places them beside the right spawns, and generates the regions, filters and apply-rules that make them function. One run records the scale of this plainly: a plan stating four wool rooms and their markers produced four wools, four monuments, 22 regions, 34 filters and 11 apply-rules, with no authoring and no correction.

The same is true of values that are derived rather than placed. The build ceiling is the tallest terrain column plus twenty, so raising any terrain silently raises the ceiling over the whole board. A walls entry names an interface and nothing else; the studio decides that a wall is bedrock, two thick, three courses, across the full interface width, stamped on the attack side. Pieces are fanned through

Table 3.1. What the composer does not do, and what therefore arrives as the agent’s work. Every row is stated in the repository’s own brief.

The composer does not	So the board arrives
vary the crossing’s ground	with one lateral row of stones at one depth: no tiered island, no staggered pair, nothing off the band’s centre line
place defence walls	"walls": [], always. A defence wall is authored, never composed
state any elevation	with no piece carrying a surface; the whole board flat at <code>globals.surface 9</code>
draw anything but rectangles	as axis-aligned cell rects, with no curve, no spur and no taper anywhere
model an intra-team build zone	with exactly one zone, the mid band
carry layers	with no storeys at all: no bottom lane, no top lane, no tunnel, no undercroft
make four-team boards	two teams only, whatever is asked for
balance its own wools	routinely lopsided (section 4.6)

the board’s symmetry automatically, though as chapter 5 will show, not everything is, and knowing which is which has cost several runs a rebuild.

3.2.3 Substitution and Complaint

One case is worth isolating because it shows the studio’s temperament. A run specified obsidian for a fifteen-block destroyable, which is out of band. The studio did not refuse. It built the map in ender stone, raised the complaint DC3 saying so, and wrote the `map.xml` declaring what was actually laid. A refusal stops the request; a complaint lets it through and states what changed. That distinction runs through the whole API, and an agent that reads only the status code throws half the answer away.

3.3 Decisions of the Agent

3.3.1 The Identity Sentence

The most consistently enforced agent decision in the corpus is also the least technical: before a single shape is drawn, the agent writes one sentence saying what the board is. The warmup discipline states it as a rule: decide what the board is about before deciding what it is made of, keep it to one thing, write the sentence down, and if it cannot be written the board is not ready.

A worked example, from a run that built four boards at once:

A sand spit, flat and pale and open nearly everywhere, with one ring of dune on it that you have to walk round, two tide-worn stones in the middle at a ten-block grain, and a water lane down the west flank that opens forty-five minutes in.

This turns out to be load-bearing in a way nobody designed. One orchestrated run had three agents killed by an interrupt at twenty-eight minutes. Two of them had written their identity sentences into their reports before authoring, and those survived. The third had not, and its three remaining identities were lost with it and had to be invented again. Writing the identity down first is a durability property as well as a design one.

3.3.2 Palette Across a Set

Where several boards are built together, the agent chooses each board's ground, built and accent tones against the others, because, as one run puts it, five boards can each be internally coherent and still be five greys. The reasoning is recorded per board as a position in a range: one board the brightest and highest contrast, another the coldest and bluest and lowest contrast.

3.3.3 Size

Every one of the twelve boards in the most recent composed-board runs resized the card it was pulled from. The composer sizes to its band; the agent sizes to the place it has decided on.

3.3.4 Gameplay Intent Without an Oracle

This is the category the repository treats most carefully, and it is the reason this paper is constrained the way it is. Questions about how a map plays are not derivable from the code or the corpus. Every run report from the first onward therefore carries a section titled *open gameplay questions, decided without an oracle*, in which the agent records the judgement it had to make, the answer it picked, and its confidence. Three real examples:

How high may a mid island stand over the fronts? Gallowsholt's causey is four blocks proud, so a player bridging out of his own front has to place a block to get onto it. That makes it a prize and makes the first team there hard to shift. Decided: four.

May two wools be the same distance and completely different to raid? Longstrand's are 200 and 192 blocks from the enemy's door; one is inside a ring of dune you must go round and one is on open strand. Decided yes, and it is the thing I would most like judged.

Is a two-block bench across most of a board playable? ... This is the decision I am least sure of.

The practice exists because of a specific failure. An earlier run measured that destroyables and cores float a few blocks above the terrain, reasoned from first principles that this made every generated destroy map unwinnable, and filed and committed the claim. The measurement was right and the conclusion was invented. Floating is deliberate: a core on the ground cannot leak, and a destroyable on the ground is trivially covered, so the studio's generator floats one three to five blocks and an author's cap bounds it at twelve. The corpus, measured afterwards, does not float them at all (chapter 12), which makes the invented claim doubly wrong. Neither the corpus nor the code would have corrected it. One question would have.

3.4 The Intermediate Category

Between what the studio supplies and what the agent decides sits a third category, and it is where the system is most itself.

3.4.1 Placement as Query

An agent's natural instinct is to place things by eye: this tree here, that boulder there, a bench against the wall. The most recent structure-focused run records what happens when it does.

Every prop I placed by eye was declined. ... Everything that shipped came from POST ... /sketch/seats or the candidate loop. Final tally: 78 placed, 0 declined.

The studio knows where a prop can stand, because it knows the ground, the clearance, what else is there and what the passage rules require around a building. The agent does not, because the agent cannot see. So the productive move is not to place and be corrected; it is to ask what the legal seats are and choose among them. This inverts the usual relationship between an author and a tool, and it is the single most transferable piece of craft in the whole corpus.

3.4.2 Instruction by Evaluator

The studio scores a board against 31 terms, and the scores are not merely a grade. One run records the evaluator changing a design rather than a document: a monument authored on an open pan read a goal-distance ratio of 1.875 against the G01 band of 3.0–4.0.

... which is not a placement error. It is the board saying the goal is in the contested space rather than a short walk forward of the spawn. Moving it to the brow gave 3.22 and produced the topology the gameplay documentation describes.

The agent did not learn a number. It learned what the number was about.

3.4.3 Convergence as a Failure Mode

This is the finding that most deserves to be in a paper about generated maps, because it is the failure mode of the whole approach and it was caught by accident.

An orchestrated run put three agents on three different ground families, with no contact between them. They converged on nearly identical goal geometry: spawn separation 184 blocks, goal 40.5–44.0 from its own spawn, G01 between 3.24 and 3.59, lateral offset 11–21. The explanation is not imitation, because there was none available.

... because G01's band of [3.0, 4.0] on a lane-shaped board puts the goal between $L/5$ and $L/4$ from its own spawn, and the centre line is the cheapest way to satisfy it.

A constraint with a cheapest satisfying solution will be satisfied that way by every author who does not deliberately pay more. The rules that keep bad maps from being built will, left to themselves, also make good maps resemble each other. Nothing in the system detects this; it took three independent runs and somebody noticing.

3.4.4 Rules as Band Lookups

One further finding reframes the rule system entirely, and an agent that does not know it will generalise wrongly from every board it has built:

A rule id is not a fixed constraint: FR6, G8 and CT12 all read **authored bands** whose values depend on what the board is played for, and three boards of experience on one objective family taught me nothing about the next.

This is why the studio publishes `GET /api/rules/terms`, so that a driver can ask what band is enforced on *this* board rather than writing down a number that was true of the last one.

3.5 The Boundary by Stage

Table 3.2 states the division of labour across the whole pipeline. It is the table the rest of the paper elaborates, and each of its rows is a chapter in Part II.

Table 3.2. Who decides what, by stage. **Blue** is the studio; **orange** is the agent.

Stage	The studio decides	The agent decides
Arrangement	hub, spawn, wool approaches, frontline, mid band, land budget, size band	which candidate, and everything the composer does not state
Ground shape	how outlines fuse, how a bend is drawn, where a relief field goes	the identity, the sizes, the reshaping, where two grounds meet
Elevation	the solved height field from the marks it is given	the marks, the grain, the pushes, the scopes
Layers	the painter's bottom-up order, what a layer occludes	how many storeys, what is under what, what is a made thing
Paint	which material lands on which cell, given a band axis	the theme, the axis, where the bands cut
Structures	storey heights, roof pitch, door side, porch, beams	which style, which footprint, which seat from the offered set
Dressing	the legal seats	which seats to take, and the styles
Objectives	monuments, regions, filters, apply-rules, resolved volumes	what the map is played for, and where the goals sit
Refusal	what may not be built, and why	what to do about it

PART II

The Instruments

One chapter per capability of the studio, in the order a map passes through them. This is the reference half, meant to be consulted rather than read through.

The Composer and the Size Ladder

4.1 Composition: Definition and Verbs

The studio’s documentation refuses the word *generate* for this stage, and the reason is precision rather than pedantry: the word has been used both for the whole pipeline and for the narrow step that turns an intent into `map.xml`. There are five verbs, each meaning one thing. **Emit** fills one box with one base shape. **Derive** reads structure back out of geometry. **Compose** builds the plan. **Evaluate** scores a plan. **Realize** compiles a plan into a sketch and an intent and exports it.

Emit and derive are a forward/inverse pair, and the fact that they are required to agree is the shape model’s correctness test: composition emits, verification derives, and a disagreement is a defect in one of them.

The whole of composition is one chain, and it runs in one direction.

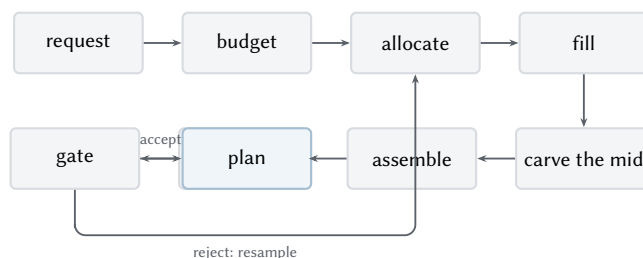


Figure 4.1. Composition. Nothing later reopens what something earlier settled, which is what makes any property of the output traceable to the one step that fixed it.

Two commitments shape everything downstream. Generation runs **from the hub outward** in a relative frame and embeds into world coordinates late, so every box but the hub is positioned against its neighbour rather than against the map. And composition is **allocate-then-fill**: every footprint is placed before any terrain exists, and filling never grows a box outward to accommodate what is being put in it.

4.2 Player Count and the Size Bands

The composer takes five inputs (players per team, team count, symmetry, seed and the cell scale), and only one of them drives the board’s size. The player count names a **size band**, and two counts inside one band compose to the same budget, because in the model’s view they are the same map.

The striking thing about table 4.1 is the fourth column. Across a four-times range in player count, the land each player gets barely moves: the fit over the corpus is $\text{land} = 176 \times \text{players}^{1.12}$ with $r = 0.83$ on log-log, which is about 250 square blocks per player at every size. Mapmakers converged on that figure without anybody writing it down, and the studio reads it back off their work rather than asserting it.

A count above centi’s range is clamped into it, because centi is the top of the ladder. That is a deliberate decision: there is no size band above it worth composing for.

Table 4.1. The size-band ladder. Land per team is measured over 331 CTW corpus maps carrying both a team count and a player cap; the budget in cells is that land divided by the cell area.

Band	Players per team	Land/team blocks ²	Land/player blocks ²	Maps measured	Budget cells	Unit share cells
nano	6–13	2250	225	77	140.6	126.6
micro	14–21	4025	242	90	251.6	226.4
milli	22–31	7075	270	65	442.2	398.0
centi	32+	8730	256	84	545.6	491.1

4.2.1 The Four-Block Grid

The plan works on a coarse cell grid, and the default cell is four blocks. The reason is that the corridor widths the corpus actually contains have to be expressible on it. Measured modal local thickness runs 8, 10, 14, 16 blocks up the bands. At cell 4 the reachable widths are 4, 8, 12, 16, 20 and both ends of that ladder land on one. At cell 5 they are 5, 10, 15, 20 and neither does.

Table 4.2. Corridor widths by band, in blocks and in cells at the default scale.

Band	Map corridor	in cells	Wool corridor	in cells
nano	12	3	10	2 (floor)
micro	14	4	12	3
milli	16	4	14	4
centi	16	4	14	4

The corpus also supplies the floor: ground narrower than eight blocks accounts for between 1.5 and 4 percent of any map, which the model reads as the width authors build down to and no further. The design vocabulary resting on it is simple: one lane is a chokepoint, two the unstable middle, three multi-access.

4.3 Constituents of a Composed Board

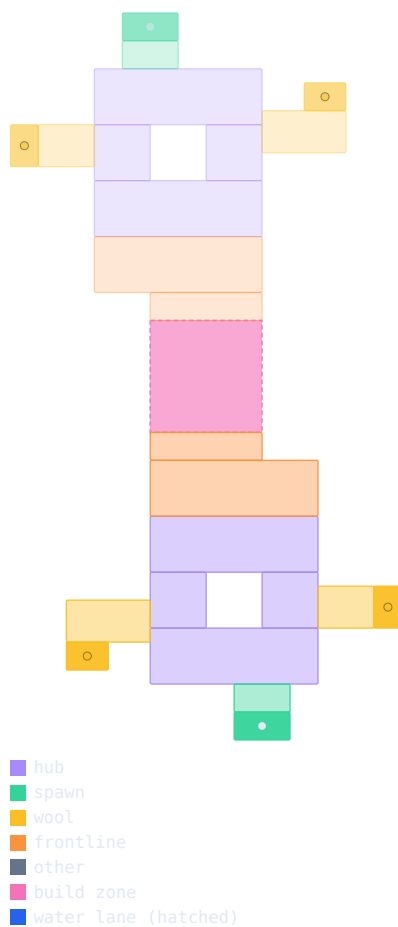
4.3.1 Boxes: The Units of Allocation

Allocation deals in **boxes**: typed bounding envelopes, each carrying a footprint and a land target. There are five kinds: spawn, hub, wools, frontline and mid. A box is an envelope and not a fill target: its contents must touch its edges and stay connected, but need not fill it solid. Boxes exist only during composition and are a model of how an author works rather than a property of maps.

Two currencies are being spent at once, and the distinction between them is the mechanism the whole budget rests on. **Land** is walkable terrain area, set by the player count. **Footprint** is total box area, meaning terrain plus build zone plus gap, and is set by the partition. A build zone costs footprint but not land. Converting a terrain piece into a build zone therefore keeps the size and drops the land, so fragmenting a board conserves footprint and spends land.

4.3.2 The Hub as Constraint Source

The hub is placed first and is the unit's only absolute rectangle; everything else is positioned against a neighbour. It takes what the budget leaves after the others have had their share, and never less



The board the worked example of ?? begins from, drawn by the studio itself. Eighteen players a side, two teams, `rot_180`, cell 4, seed 5. It scores 0, meaning no term falls outside its band.

Every element of table 4.3 is visible. The **hub** is the large violet body, here in its *ring* form, and the white square inside it is the hole that CT8 calls the rotation device. The **spawn** hangs off the hub's far edge. The two **wool** approaches, both of the *i* family, stand off its flanks. The **frontline** fronts it in orange, in its *single* form. The pink band across the middle is the **mid**: the only build zone a composed board carries.

The whole of it is one authored unit repeated through the board's symmetry. Nothing in the picture is curved, nothing is at a different elevation, and nothing is made of anything. That is the state a board arrives in, and ?? is the account of what an author does to it next.

Figure 4.2. A composed board, as `GET /api/compose` draws it. The studio renders this card itself; the figure is its SVG, unedited.

than a third of it. It is always wider than deep, the sampled aspect running from 1.3 to 2.4, because a hub grows wider rather than squarer.

Table 4.3. The budget shares. A spawn is deliberately absent: it is not a share at all but a box, costing what a room and a run-up cost at the map’s corridor width.

Constant	Value	What it means
MidShare	0.10	a tenth of <i>each</i> unit’s land, so the crossing’s own allowance is a fifth of one team’s budget
FrontlineShare	0.22	a frontline is the ground the mid is met on, and takes about a fifth
WoolShare	0.12	each wool about an eighth
HubMinShare	0.30	the hub takes what is left, never under a third
SpendFloor / SpendCeiling	0.70 / 1.30	the band the built land must land in, or the attempt resamples

Having chosen a size, the hub chooses a **body**: a ring, a G, a P, a twin, a double-hole and so on. The body is what publishes, per free run of its edge, the width that run can carry. Those publications are **offers**, and an offer’s width is a capacity: it bounds the search rather than filtering its output.

4.3.3 Seating and Single-Integer Search

Each neighbour, whether a wool approach, the spawn or the frontline, arrives as a request naming a side, a kind, a depth, an along-extent and a rolled family. The allocator then chooses a **seat**: a single integer, the offset in the hub’s edge-local coordinates at which the neighbour’s along-extent begins. Because the search runs in one integer, nothing is ever built and then moved.

What a joint records is a **grant**: the corridor width a consumer was given across an abutment. An offer is a capacity and a grant is a selection, and one run can carry two docks at two widths. Keeping those two words apart is load bearing; an earlier version called both of them the same thing and hid the fact that they are different quantities.

4.3.4 The Crossing

Between the two fanned halves lies the **mid band**: the neutral crossing. Laterally it spans exactly the hull of the opposing front faces and docks flush against them, so the crossing is one shape with the fronts it connects; in depth it is the gap. An empty crossing is 30 blocks front to front. A hop, the distance a player can cross without building, is 12.

Across the gap the composer lays a row of one to three **stones**. A stone is shared ground, and that is a property of where it sits rather than of anything it declares. Their depth follows a grain: a broad-grained row runs 16, 24, 24, 24 blocks up the bands, while a fine-grained one is exactly one corridor wide. Row counts run one at nano and micro, one or two at milli, and up to three at centi.

4.4 Determinism and the Descriptor

The seed drives every random draw the composer makes, so the same seed under the same composer version reproduces the same board byte for byte. That is why a composed board is addressed by a **descriptor** (players, teams, symmetry, cell, seed, composer version and schema) rather than by storing its plan. A card in the browse feed carries its descriptor and its rendered SVG, and pinning one re-composes the exact plan server-side.

The composer version is part of the identity for a reason worth noting: a card composed by an older composer reproduces a *different* board from the same seed. The descriptor records which composer built it, and a stored plan whose composer has moved on is flagged as stale rather than silently re-composed into something else.

4.5 Refusals of the Browse Feed

GET /api/compose scans seeds forward from a cursor, composes each, and sieves. The structural sieve runs first because classifying a board on its tiny box masks is cheaper than evaluating it, so a structural reject skips the evaluator and the render entirely.

Two parameters are refused outright rather than accommodated. A symmetry the composer does not build for, meaning anything but `rot_180` and `mirror_z`, answers 400 with an RQ1 finding naming the field. So does a team count other than two.

Four-Team Boards and the Composer

The composer builds two-team boards only. That is a limit of the composer as it stands. It is narrower than the studio's own ceiling of four (section 1.2), and narrower again than Capture the Wool, which takes two or four. A four-team CTW board is authored at the plan tier instead. Until recently the endpoint read no team parameter at all and answered a two-team board at 200 for any count asked of it, a request quietly misanswered rather than refused. It now refuses, on the principle that a board which is not the one asked for is worse than no board.

4.6 Wool Asymmetry in Composed Boards

One property of composed boards is consistent enough to have been measured, and consequential enough that an adapting agent has to deal with it every time.

On a two-wool board, the two wools are routinely *not* equidistant from the spawn that has to attack them. Straight-line cell distance from the attacking spawn to each wool room, measured off boards pinned from the live composer:

Table 4.4. Wool distances from the enemy spawn, off boards pinned from the live composer. One board in six is balanced.

Request	Seed	wool-a	wool-b	Worst ratio
p20 rot_180	0	138	94	1.47×
p20 rot_180	1	132	98	1.35×
p20 rot_180	2	142	93	1.53×
p30 rot_180	0	138	99	1.39×
p30 rot_180	3	129	100	1.29×
p20 rot_180	3	127	118	1.08×

The shape of it is always the same: wool-a is seated deep behind the hub at the far end of the board, and wool-b is a short spur off the hub's flank near the front line. One wool is a raid and the other is a walk.

What makes this interesting rather than merely a defect is where the cause turns out to lie. An agent working through four boards traced it past the wools themselves:

The lopsided wool is caused by where the composer hangs the spawn, and fixing it moves the spawn.

The derivation is geometric rather than rule-driven. Equal distance from the *enemy* spawn forces the two wools to comparable depth. Equal distance from the team's *own* spawn, given equal depth, then forces symmetry in x about that spawn's own x . A spawn on a flank, which is where the composer puts it on every board pulled under both symmetries, therefore cannot carry two balanced wools at all.

That is a good illustration of what adaptation means here. The composer's output is not wrong; it is under-determined in a way that shows up as a gameplay asymmetry, and correcting it requires moving a piece the composer considered settled. An agent that treats the composed plan as fixed inherits the imbalance.

The Rule System and Its Refusals

5.1 Two Distinct Rule Sets

A reader who goes looking for ‘the rules’ will find two collections with overlapping names, and confusing them makes the system incomprehensible. They answer different questions.

The first is the **composer’s law**: 97 identifiers under 12 headings, spanning 13 id prefixes, held in a single document that is embedded in the application and parsed rather than transcribed. It says what a well-formed CTW board *is*: how wide a lane runs, how far a wool room stands from a spawn, what a build zone may touch. It is a design document with identifiers attached, and most of its rules are enforced during composition, where the thing being judged still has a name.

The second is the **finding catalogue**: 183 rules across 28 families, served live at `GET /api/rules`. It is the set of every rule that something, somewhere, can raise as a complaint about a document. Its scope is the whole studio rather than the composer, covering sketch geometry, dressing, house styles, export and the request itself.

The two overlap without either containing the other. The composer’s law carries rules that nothing raises as a finding, and those are deliberately not served: the question the catalogue exists to answer is *what is this finding*, and a rule nothing raises has no finding to explain. Measured against the running studio, 41 of the law’s 97 ids are answered by the catalogue. Everything else in the catalogue belongs to a gate the composer never touches.

The full catalogue, rule by rule, is appendix A.

5.2 Anatomy of a Finding

Every gate in the studio answers in one shape. A client rendering a refusal never has to know which gate produced it before it can read one, and this uniformity is the single most useful property the API has for a machine operator.

A finding is three things: a **rule id**, a **sentence**, and **what it is about**.

The id is machine-legible and stable forever, so it outlives the task that added it and never doubles as a task-tracking identifier. The sentence carries the measured numbers, on an explicit principle. The word *invalid* tells an author nothing they can fix, so a refusal says *this one is 2×8* rather than *this one is too small*. What it is about is a **field** where a document field can be named, and **subjects** where the fault indicts things an editor could highlight. A finding may carry either, both or neither.

One further field matters more to a machine than to a person. A finding states its **edit** where the fix is mechanical: a document, a path, an operation and a value. A reader applies it rather than re-deriving it from the rule’s prose. It is, as the documentation puts it, the half of a finding a model acts on where the sentence alone is inspected and left.

What a finding deliberately does *not* carry is the rule’s classification. The category and the concerns list are fixed by the id rather than by the raising site, so they live on the rule and a caller joins them from the catalogue. Putting them on the finding would put one fact at 96 sites.

Table 5.1. The finding catalogue by family, generated from the running studio. A family is the field the API answers with, not the id's prefix: the four build-zone rules are numbered BZ and belong to family CT, so 29 prefixes span 28 families.

Family	Rules	What it governs
SK	28	The sketch document: shapes, layers, groups and mirroring, and the ground they rasterise into.
DR	17	Dressing a built world: where a tree, boulder, building or body of water may rest, and what it may not close off.
PL	16	The plan: pieces, placements, walls and the spawns and objectives a board is laid out around.
WX	13	The room frame stamped for a spawn or wool: shell, pad, door, iron, and where it stands on its piece.
OB	12	Objectives as they are played: where a goal may stand and float, what it is built of, and how the match can end.
CT	10	The middle: what the mid is carved into, the holes rotation leaves in it, and the build zones docked there (the BZ-numbered rules).
HS	10	House style: the materials and parts a building is authored from, part by part.
WL	8	Wool siting: a wool's distance from its spawn, its siblings and the frontline, and the approach into its room.
EX	6	The export gate: what must hold before a map document and its world are written.
IM	6	World import: the host fetched from, what arrived, and whether it holds a world the studio can read.
RL	6	Relief: whether the elevation is the landform the island claims, and whether it can be walked.
RQ	6	The request contract: which status a malformed, unknown, conflicting or unreadable request answers with.
HJ	5	Wing joints: the ways two rectangles of one building may meet, and the roof that follows.
PT	5	Terrain painting: how a theme states its materials, bands and sampled patterns.
ST	5	Structure pieces: what a wool-room or spawn piece defines, and the caps on what it may raise.
FR	4	The frontline: split or wide, how it docks, and how much of a face a crossing spans.
GO	4	Destroy-goal distances, measured by walk: to its own spawn, to the enemy's, and between goals.
SP	4	Spawn siting: where a spawn stands on its lane, and what its egress and its door face.
DC	3	Destroyables and cores: casing and interior, float against leak, and the material a size is worth.
G	3	The board envelope: corridor width, the void gaps a route hops, and map size against player count.
HP	3	The footprint a placed building is made of: how many rectangles, how thin, how much ground.
ED	2	The edit document: the references it must resolve, and the state it may be applied in.
LN	2	Lanes: the corridor width a board is built at, and the run before a junction or a dead end.
CO	1	Compose: descriptors that are well-formed and name a board the composer has no shape for.
EL	1	Land seams: the step a player cannot walk up, and the palette that steps by it.
EZ	1	Editable ground: standable ground nobody can edit, outside the zones a map seals on purpose.
PC	1	Piece connectivity: what counts as a join between two pieces, and what does not.
SH	1	Shops: the references a shop makes that the document has to define.
Total	183	28 families

5.3 Three Severities

This is the part of the system that has cost authoring runs the most, and it is the first thing an agent should learn.

A refusal

stops the request. The work did not happen. It arrives in the error envelope's findings, and nothing rode along with it.

A decline

says that a piece of what was posted is *not in what was built*. The request succeeded. Something you sent was dropped.

A complaint

is a remark. The request succeeded and the studio has an observation about it.

Both surviving severities ride on a success, under one key:

A 2xx JSON object answers warnings when something was complained about or declined, and carries no such key when nothing was. ...A caller reads warnings ?? [] and is done.

The consequence is blunt. A driver that reads only the status code throws away half of every answer. A board can store at 200 with a floor missing under its walls, with a theme that painted nothing, with a field it misspelled silently unread, and the only place any of that is said is warnings. One run report names this the dominant failure class in the whole corpus and calls it, exactly, a 200 that isn't a yes.

The distinction between the two is carried on each finding's severity, and the documentation is explicit about why it could not be carried anywhere else: `warnings.some(w => w.severity === "decline")` is *something I sent was dropped*, which no status code and no other field says.

The Findings Read

GET /map/{slug}/findings is asked on every run beside the others, because it is the only read that answers SK9, the gate that knows two shapes on one layer stacked and the lower one is not in the world. It is a decline, the channel every other route publishes on keeps complaints alone, and so a board can store at 200 with a floor missing under its walls and nothing anywhere says so.

5.4 The Refusal Envelope and Status Codes

A refusal on the wire looks like this:

```
{ "error": "invalid house style",
  "message": "...",
  "findings": [ { "rule": "HS1", "message": "...",
                  "severity": "refusal", "field": "doorHead.block" } ] }
```

error names the *gate*, such as *objective placement*, *unknown gamemode* or *plan not compilable*, and never the fault itself. message is the findings' sentences joined together.

Status codes stay the gate's own, and each means one thing: **400** a document wrong as posted, **404** a subject the route names and the studio does not have, **409** well-formed but conflicting with the map's state, **422** cannot be processed, **500** the studio's own fault. Each route declares which of them it can answer, 95 of 149 operations carrying at least one, and a test fails if a documented row and its route stop agreeing.

The edge answers in the same shape as the gates. No route under `/api` answers a failure in a shape of its own, and none answers one with no body at all. A request the framework itself could not bind becomes RQ1 per field under *request will not read*, naming each field *as the wire spells it*: `region_id`, not `regionId`.

5.5 Faults of the Request Itself

Six rules are about the request rather than the map, and they exist because the uniform shape held everywhere except at the door. Three are worth knowing by name.

RQ1 is *the document could not be read*: a 400, carrying the field where the reader knew one.

RQ3 is *a field went unread*, and it is a complaint on a success rather than a refusal. It is the rule that catches a typo: `roof.pich`, `wings[1].ptch`. The check walks down the posted value, matching by the name the *document* uses rather than the name the C# type uses, and it runs over the body as posted, before any merge. This is the studio's answer to the failure mode where an agent posts a field the reader silently drops and believes it took effect.

RQ5 is *the request conflicts with what is stored*: a 409, and the one refusal where nothing is wrong with the request at all. Every document carries a revision, a read answers it as an ETag, and a write may state it back as an If-Match. A request with no If-Match states no precondition and writes. Protection is opted into by having read first, which is the only way it can mean anything.

5.6 One Question at Four Grains

The most interesting structural property of the rule system is that the same question recurs at four resolutions, with a different id at each, and each id can only see what its own level knows.

Reachability is the worked example.

Table 5.3. One question, *can this be reached?*, asked at four grains. Read down the column: a board can pass every gate above and still read a third dead.

Rule	Grain	The question it asks
WX6	one piece of the plan	can a door be cut into this room at all
PL9	the whole plan	can a capturing team's spawn reach this wool
EX1	the built world	is the ground the rasterizer produced connected
DR-PASS	one building	does this building leave eight blocks to get past it
DR-SLOPE	one footprint	is this building's footprint level enough
(none)	a measurement	is reachable ground ground any journey actually uses

The documentation draws the conclusion, and it is the best argument in the system for why there are four documents rather than one:

A finer grain exists only because a coarser one does ... Collapsing the four documents into one would not simplify the rules. It would remove the vocabulary each rule uses to say precisely what is wrong.

Ids are grouped by what they are *about*, never by which gate happens to ask. The same objective rule is asked at the compile gate against a plan's pieces and at the export gate against the ground the rasterizer produced, and a rule that changed its name between the two would be two rules.

5.7 Bands and Targets

Chapter 3 stated this as a finding; here is the mechanism. Rules divide by *kind*, and the two kinds most often confused are the target and the band.

A **target** is prescriptive and per-request: this compose wants exactly two frontline runs, connected. A **band** is descriptive and measured off the corpus: authored maps run one to seven frontline runs. The distinction is load bearing, and the documentation states the failure precisely. A band pressed into service as a target silently turns an observation about the corpus into a requirement on the generator.

Several rules therefore read authored bands whose values depend on what the board is played for. A frontline width cap of six to eight cells is the wool board's figure; on a board played for cores or destroyables there is no width cap at all, because a wide front feeds a wool lane and a destroy board has none. One run put it as the most transferable thing it learned: three boards of experience on one objective family taught it nothing about the next.

This is why GET /api/rules/terms exists. A driver asks what band is enforced on *this* board rather than writing down a number that was true of the last one. The 31 scored terms and their bands are appendix B.

5.8 Nine Representative Rules

The law is large and mostly technical, but a handful of its rules carry most of what a reader needs to know about what the studio thinks a map is. These are quoted as the law states them.

The corridor ladder. G2 fixes the whole board's scale: the map's corridor width is the size band's, stated in blocks (12 at nano, 14 at micro, 16 at milli and centi), with the wool approach one rung under it. Measured over 359 built CTW maps.

The jump, not the bridge. WL12 is the newest wool rule and the one an adapting agent breaks most often:

A bay or a hole beside a goal is at least 16 blocks across. Negative space is crossed by **jumping** long before it is crossed by building ... Stated in **blocks**, never in cells: a floor stated as a cell count moves with the grid scale, and a jump does not care what the grid was.

The floor is 16 blocks where the space touches a wool room or spawn and 12 where it touches neither, because a hole in a team's own ground is crossed on purpose.

Holes are the rotation device. CT8 is the clearest statement in the law of a rule that exists for a gameplay reason rather than a geometric one. The closure encloses internal void pockets in every seed measured, and the author's reason is that a loop around a hole gives alternative routes between lanes without retreating through a single chokepoint. It is not mandatory, since not every map has to have one though most do, so a holeless plan is flagged and never blocked.

The strait. CT12: on a two-team wool board the direct crossing between the two team islands spans 15 to 40 blocks. Closer is two halves all but merged; farther is a crossing nobody bridges.

The unwalkable seam. EL1: a land seam of two blocks or more is ground a player cannot walk up. The consequence is what the rule is for. On a board built from the authored height palette every non-flush seam is a step nobody walks, so the seam wants a ramp or a flight carved into it, and the check's output is the work list for that pass rather than a fault to redraw around.

The mid never touches a wool. BZ6, short and absolute: otherwise players bridge freely through the mid straight into the point and there is no gameplay direction.

A zone is bounded by the ground it docks. BZ9 forbids the *overhang*: a stretch carried beyond the last ground it meets, which connects nothing and reads as no man's land a player walks into and stops.

The defence triangle. WL10 bans a specific failure: a front-near wool whose spawn is far, beside a back wool whose spawn is adjacent, so that the defender always guards the wrong door.

The destroy goal's ratio. G01: a destroy goal sits three to four times as far from the enemy's spawn as from its own, measured by the walk over the surface. Under the band the goal falls to a rush before a defence can form; over it the attack is a march across the whole board into a set defence.

Distance Is Measured as a Walk

A distance between two things on a board is the walk between them over the walkable surface. It is eight-connected, a diagonal costs what a player walks rather than two steps, a climb is charged the blocks it takes to place, a fall is counted but not charged, and the route goes around voids. Never the straight line through the air. Every distance in every rule is this.

Ground: Outlines and Their Construction

6.1 Islands and the Height Convention

The ground level deals in **shapes** on **layers**, grouped into **groups**. A shape has an outline, a floor and a thickness, and where its ground touches or overlaps another shape's in the same group, the two build one **island**.

The first thing to get right is the height convention, because it is off by one from the obvious reading and an agent gets it wrong every time otherwise: `base_height: 9` puts the top block at $y = 8$, not $y = 9$. A column read at such a shape answers grass at $y8$, dirt at $y7$ and $y6$, stone from $y5$ to $y1$, and bedrock at $y0$.

6.2 Four Forms of Outline

6.2.1 Polygons and the Rectangle

A polygon is a ring of points. A rectangle is not: it states bounds rather than points, and so it is restated as a four-point polygon before any point of it can move. That restatement is invisible in the document and load-bearing in practice.

Three routes edit one point without disturbing any other:

```
PATCH /api/map/{slug}/sketch/shapes/{id}/vertices/{index} {x, z}
POST  /api/map/{slug}/sketch/shapes/{id}/vertices      {after, x, z}
DELETE /api/map/{slug}/sketch/shapes/{id}/vertices/{index}
```

The insert has index arithmetic that bites. A point added after: `0` lands at index 1, and every later index goes up by one, so a pair of points previously at 2 and 3 migrates to 3 and 4. An agent replaying a list of edits by index, computed against the original ring, will scramble the outline.

6.2.2 Height as a Per-Vertex Field

`anchor_heights` is one thickness per vertex, in the same ring order, and the surface between them is interpolated across the face. A six-point island carrying `[9, 12, 16, 17, 17, 14]` is the plane $9 + x/4 + z/4$ in the island's own frame. That is a tilt, authored with six numbers.

6.2.3 Curves and Bézier Handles

Four ways to state an outline that is not made of straight edges. A **circle** takes a centre and a radius and the studio draws it as a 64-sided polygon. A **lasso** is, in the document, simply a polygon. The difference is the canvas, which traces it freehand and simplifies the trace at a four-block tolerance. And any polygon edge can be bowed with Bézier handles.

The Bézier contract is worth stating exactly, because every part of it is a place to go wrong:

controls is keyed by a vertex's index as a string, and every handle is an **absolute point on the board**. The edge from vertex i to vertex $i + 1$ is a cubic curve from vertex i , pulled by `controls[i].out` and `controls[i+1].in`, to vertex $i + 1$. So a vertex's out bends the edge after it, and its in bends the edge before it. The studio samples every curved edge 16 times, and the curve still passes through every vertex.

Bowing one edge is two handles:

```
"controls": { "0": { "out": [74.67, -6] }, "1": { "in": [81.33, -6] } }
```

Rounding a whole ring is a closed-form recipe rather than a hand fit: at each vertex P with neighbours *previous* and *next*,

$$in = P - \frac{next - previous}{6}, \quad out = P + \frac{next - previous}{6}.$$

Limits on a Bézier Handle

Keep each handle at least as far *along* its edge as it is *away* from it, and keep the bulge under about a third of the edge's length. Past that the curve crosses itself.

6.2.4 Polylines

A polyline is a line with a width: its points are a centreline, and the band radius blocks to each side of it is ground. The order of operations is the fact to know:

The points are **smoothed first**, by a centripetal Catmull–Rom spline with eight samples between each pair, and only then offset radius blocks to either side, so four points draw a curve rather than three chords. The ends are cut square.

```
{ "id": "line-solid", "type": "polyline", "operation": "add", "floor": 0,
  "base_height": 9, "vertices": [[0,10],[13,3],[27,17],[40,10]],
  "radius": 3, "stroke_edge": "solid", "stroke_seed": 7 }
```

`stroke_edge` takes three values with numbers attached. `solid` keeps one width. `rough` lets it swing up to 45% either way with the two sides wandering apart, seeded by `stroke_seed`. `tapered` narrows it to 35% at the ends. A polyline also takes `anchor_heights`, which are heights at its points, with every block taking the height of the nearest place along the line. A lane can therefore rise.

Two facts separate a polyline from things that look like it. A polyline whose ends lie *inside* two islands joins them into one, which is how a land bridge is drawn. And, in the distinction that costs an afternoon, a polyline shape *is ground*, whereas the stroke prop, which looks alike, repaints the top course of what it crosses and adds no ground at all.

Self-Overlap in a Polyline

The band is one outline filled even–odd, so where its two sides cross they cancel and build void. The studio names it SK25. Four points across 26 blocks at radius 3 lap in every bend; the same four points spread over 40 blocks clear. A coil is drawn as one polyline per part–turn.

6.3 Overlap: Height and Paint

Where two shapes cover a cell, two questions are settled separately, and conflating them is a common error.

The height and the paint of an overlap are decided separately. Where two shapes cover a cell, the taller one gives the column its height, and the paint goes to the **smallest themed shape whose top is that tallest top**.

So on a flat island where a yard and a meadow both stand at 9, the yard being smaller means the overlap paints as the yard. A lawn smaller again paints as the lawn inside the yard. Height is decided by tallest; paint, among the shapes that tie for tallest, by smallest.

Occlusion Within a Layer

On one layer the taller shape takes the whole column, so a lower shape drawn inside a higher one simply is not there, and the studio names it SK9. A sunken place inside a raised one is drawn as shapes *around* it, or on a layer of its own. This is the same rule chapter 8 states from the other side.

6.4 Faces and Band Stacks

Where a shape's ground drops, whether over lower ground or over the void, the exposed side is a **face**, and a theme can stripe it with a band stack.

One property of that stack is not what it looks like:

A band stack does not cycle: past its last band, "ending": "repeat" hands every further course to that last band.

So striping a face sixteen courses tall with a four-course pattern means writing the four courses out four times. A stack that reads as four bands in the author's head is twelve bands in the document.

The other property is that only the courses standing *above* the neighbouring ground are face. The same yard, stepped up from a meadow on one side and dropping to the void on the other, reads four striped courses on the first and the whole stripe from top to bedrock on the second.

6.5 Five Forms of Ascent

Getting from one height to another is its own small craft, and the studio offers five forms: a ramp (one polygon with anchors at each end), a stair (one polygon with deeper treads), a turned flight (two polygons with a landing between), a run of steps (one rectangle per tread), and a floating flight (one raised slab per tread, with air under each).

One rule governs all of them and it is the most counter-intuitive fact in the whole ground model:

anchor_heights are heights at a polygon's corners, and every block reads the slope at its own centre and rounds. A flight therefore comes out even **only when the anchors sit half a riser past its first and last tread**: a 2-deep stair whose treads are 1 to 7 states 0.5 and 7.5. Whole-number anchors 8 and 1 over 16 blocks build treads 1, 2 and 3 deep. The ramp works with whole numbers because one block up per block rounds cleanly.

And a consequence that decides how a flight can be authored at all: the studio's canvas rounds anchor heights to whole blocks, so the half-block form can only be written into the document. Drawing an even stair by hand means one rectangle per tread instead.

Relief: The Shaping of Ground

7.1 Relief as a Solve

Relief is how flat ground becomes terrain. It is stated per group, and it is a *solve*: the studio takes a small number of statements about where the ground must be, and computes a height for every cell in between.

The single most important property follows from that:

A relief replaces the top of every column it solves, and the heights it solves come from its marks and pushes. A solved shape's own `base_height` therefore decides nothing about where the ground ends up.

The technique cards prove this rather than assert it. The same relief is applied to two plots whose shapes are drawn at different heights, 13 and 9 in one and both 9 in the other, and the built ground is identical: 2304 columns compared, 0 differ. An author who tunes a shape's drawn height to influence a solved surface is tuning a number nothing reads.

7.2 Constituents of a Relief

```
"relief": { "stepped": {
  "base": 11, "reach": 0, "step": 1,
  "marks": [
    { "id": "upland-bench", "kind": "area", "h": 13, "bevel": 2,
      "ring": [[-28,-26],[-20,-30],[-8,-29],[-3,-22],
        [-4,-12],[-10,-7],[-22,-8],[-28,-14]] },
    { "id": "lowland-bench", "kind": "area", "h": 9, "bevel": 2,
      "ring": [[-46,8],[-38,5],[-20,6],[-6,5],
        [-2,10],[-8,15],[-26,14],[-44,15]] } ],
  "pushes": [
    { "id": "knoll", "amount": 1, "falloff": 3, "crown": 3.5,
      "roughness": 0, "seed": 1,
      "ring": [[-32,-32],[-21,-32],[-18,-25],[-22,-18],[-32,-19]] } ] } }
```

A **mark** pins ground. An area mark holds the middle of a region at a stated height, and its bevel softens the edge. A **push** raises a region by an amount over a falloff, with a crown and an optional roughness.

Table 7.1 is the full set. A mark is placed rather than sampled, for the same reason a prop is: a decision about where the high ground is decides how the map plays, so it belongs to the person making it rather than to a noise field.

A line mark's `r` reaches either side of the centreline, so the band it writes is *twice* it: a stated 6 holds a twelve-block strip. The rim is optional, and it is what keeps a hill inside its shape. Without one, marks alone decide the whole surface, so a shape carrying a single high mark rises to that height everywhere and runs off its own edge.

The most useful sentence about marks is about the space *between* them:

Table 7.1. The five marks. Each pins part of the footprint at a chosen height; the solve fills between them.

Mark	Pins	Says
point	a disc of a given radius	a summit, a hollow, a spot height
line	a band reaching r either side of a polyline, optionally with several heights along its run and a tread held flat	a ridge, a valley floor, a shoulder falling as it runs, a road switching back down a hill
area	every cell inside a ring, at one height or one per vertex, with an optional graded bevel inside its edge	a bench, a mesa top, a sunken floor, a shelf cut into a hillside
rim	the footprint's own outer rings	where the land meets the void
scarp	a band either side of a drawn line, at two heights, with the face between them left free	a break of slope, and the mark that decides where players can go

Each area mark holds the middle of one meadow at its own height, and its bevel softens the edge. The ground between the two benches is pinned by nothing, so the solve grades it. **That free band is the connection.**

Relief is therefore authored by pinning as little as possible. Pin the two things that must be at particular heights and let the solve invent the slope between them; pin everything and there is no slope, only a set of plateaux.

Agreement of a Push's Gradients

A push publishes two gradients: its skirt, which is amount / falloff, and its crown. A knoll reading 0.33 up its skirt and 0.31 to its crown is smooth. Where the two disagree the push steps at its own outline instead of rising into the ground, and the studio raises RL6.

7.3 Relief Scope: Three Treatments

The question is what a relief does to a shape that already has a height. There are three answers and the difference between two of them is the difference between a join and a cliff.

No scope at all. The shape joins the solve.

Its drawn height is discarded and the relief decides where its ground goes. This is the default and it is right for terrain.

hold. The shape keeps its height and is still arrived at.

Its cells stay in the solve but are pinned at the height the shape states, so the ground around it is solved *knowing where it has to arrive*.

exclude. The shape is a hole in the solve.

The relief bends round its edge exactly as it bends round the void, and nothing climbs to meet it. The shape keeps its faces.

The distinction between the last two is measurable, and the technique card measures it. Along one line, a shape drawn at 12 dropping to ground drawn at 8:

Table 7.2. The same shape under hold and under exclude, read along one line. hold makes the join; exclude leaves a face.

z	$\dots - 1$	0	1	2	3	4	5	6	7
as drawn	12	8	8	8	8	8	8	8	8
hold, with the relief	12	12	11	11	10	10	9	9	8
exclude instead	12	8	8	8	8	8	8	8	8

exclude on the upland would not make the join. An excluded shape is a **hole in the solve**, the relief bends round its edge as it bends round the void, and nothing climbs to meet it.

The corresponding failure runs the other way. A built thing, such as a yard, a plinth or a platform, wants exclude, because it should keep its faces. The same layout built without it puts the yard's top at 11 instead of 17: the relief simply replaced the height it was drawn at.

7.4 Joins Between Grounds

This is the craft rule the whole chapter exists to support, and it is stated in the authoring discipline rather than in the API.

A board is two or three or four *grounds*, each stated by the one instrument that can state it, and every join between them is chosen rather than left over. A beach and the dunes behind it are two reliefs on two layers, held at different heights, because relief solves per layer and the fields are shifted into world Y separately, so two reliefs can meet. One relief across both is one field, and there is no join at all.

That is the difference between a board that reads as a place and a board that reads as one field with a gradient in it.

Table 7.3. A shore board, worked through: each ground stated by the instrument that can state it, and each join chosen.

The ground	Stated by	Why that instrument
the beach	a relief with flow: marks pinning only the waterline, a grain, a low step	it has to read as deposited, so nothing is pinned that does not have to be
the dunes behind it	a second relief, on its own layer, held a little higher	two reliefs can meet; one relief across both is one field and no join
the yard the houses stand on	a shape carrying relief_scope: "exclude"	the footprint leaves the solve, so the two tiers meet at a <i>face</i>
beach → dune	the two reliefs' own meeting, across the layer seam	the join has a shape because two fields made it
dune → yard	an authored flight: height_mode: "level", anchors, skirt: 0, a material rather than a theme, run at least twice the rise	a relief graded across the seam deletes the boundary; a flight states it

One Map, Several Grounds

Plains everywhere is not simplicity. It is one instrument.

7.5 Shapes Laid on Solved Ground

Sand on the beach, gravel on the tide line: something that paints rather than shapes. Such a shape carries a theme, a `base_height` equal to the ground it lies on, and, in the part that goes wrong, **no** `relief_scope` at all.

Two reasons. A shape owns the paint only on a cell it forms the surface of, so one stated lower runs under the ground, paints nothing, and says so on a 200. And a `relief_scope` is a statement about *height*: follow seats the shape on the solved field and then pins its whole ring there rigidly, which over a terrain patch is a floor. The strand solves level to the block, and every mark inside it is overwritten in silence.

7.6 The Relief Read

One call answers what the marks did to *each other*, which no render can show: `POST /api/map/{slug}/sketch/relief/read` given the stored layout.

It answers, per group: the cell count, the low and high, the relief, walk and scramble tiers, the faces, the cliffs, and four fields that are the reason to ask it at all.

silentMarks

every mark that landed nowhere. A mark that pinned nothing is invisible in every picture.

seams

the pairs that meet on a step. A seam naming a *shape* rather than a mark is a `relief_scope` pinning ground the marks were meant to shape.

level and largestField

the two numbers that say whether the ground has a shape at all. Over about 0.45 and 0.13 respectively, it is a table with edges.

landform

what the solve thinks it made: *rolling*, *plain*, and so on.

Relief and Symmetry Folding

Any symmetry mode folds the relief across the centre, a group's `mirrors: false` does not change that, and a layout stating no `mirror_mode` takes `rot_180`. A board meant to solve unfolded has to say `mirror_mode: "none"`.

7.7 The Limits of Relief

The repository's author states the danger directly: relief should be used, and solid structural areas matter. A board whose every square is rolling has no ground a fight can happen on, and an agent reaching for relief to make a board interesting will reliably overdo it.

The counterpart discipline is a count. Run over a finished spec: how many reliefs, how many marks and of what kinds, how many pushes, how many shapes at each height mode, how much made ground, how many polylines, how many made layers, how many copied trees. A zero in that list is not a fault. Four zeros is a board that used one instrument and called it terrain.

Layers and Literal Depth

8.1 One Span per Column

A sketch layer is a slab, and the one sentence that governs it is this:

A layer holds exactly one span per column, and where two adds contest a cell **the taller replaces the shorter outright, floor included**.

Those last five words are, as the repository’s own account of the technique puts it, the whole of what makes sculpture possible. They mean that a stack of nested shapes writes an arbitrary height field with no subtraction anywhere and no per-column authoring. Draw a wide low disc, then a narrower taller one on top of it, then a narrower taller one again, and the result is a stepped mound. Nothing computed a mound; at every column the tallest add present is simply the one that survives.

It also means the word *slab* is misleading:

A layer is not a flat slab. It is one arbitrary *height field*, a (floor, top) pair per column, and a stack of layers is a set of those fields with air between them. Any solid whatever can be written in that form.

8.1.1 The Algebra of a Single Layer

Within a layer, a group of shapes resolves as

$$((\text{adds} - \text{subtracts}) \cup \text{override-adds}) - \text{override-subtracts}.$$

Two consequences follow, and both catch authors out. An *ordinary* add loses to a subtract over the same column regardless of which was authored first, because the subtract side of the algebra runs before either shape’s override flag is looked at. And the two add sets each settle among themselves by height before the override plane overwrites the ordinary one, so an override add replaces the column it lands on whatever its height is. That is what puts a one-block floor inside a twelve-block wall, and a threshold through a doorway, with neither shape cut against the other.

The Limit of Nesting

A bowl falls inward, and the outer disc that should keep only its own ring is also the tallest thing over the middle, so nested discs come out as a flat plate. A falling field needs shapes that do not overlap at all.

8.2 A Storey as Rock with Holes

This is the part that most often has to be unlearned. The instinct, drawing a second storey under a map, is to draw the rooms. That produces nothing, and the reason is the one-span-per-column rule:

A sketch layer carries one span per column, so a room drawn on its own leaves every other column of that storey empty. The board's lower half is air and the ground above it floats. What is stated is the **rock**, over every column the board has, and the rooms are the holes in it.

Building an undercroft therefore takes three coordinated moves.

8.2.1 The Lift

The ground has to be raised to make room underneath it. That means the plan states a higher surface, "globals": { "surface": 22 }, and the finish thins the slab to its top courses, leaving the rest free.

Moving only one of the two fails in a way that is confusing to diagnose. If the surface is lifted without telling the plan, the spawns stay at their old height while the ground goes up, and the buildability check then reports every placement as standing over open void, because the marker is a dozen courses under the ground it names.

8.2.2 The Layer Below

```
{ "id": "under", "name": "Undercroft", "base_y": 0, "below": true,
  "shapes": [
    { "id": "rk1", "type": "rectangle", "operation": "add",
      "floor": 0, "base_height": 14, "theme": "rock",
      "min_x": -50, "min_z": -50, "max_x": 50, "max_z": -30 },
    { "id": "rk2", "...": "min_x -50, min_z -30, max_x -30, max_z 30" },
    { "id": "rk3", "...": "min_x 30, min_z -30, max_x 50, max_z 30" },
    { "id": "rk4", "...": "min_x -50, min_z 30, max_x 50, max_z 50" },
    { "id": "hall", "type": "rectangle", "operation": "add",
      "floor": 0, "base_height": 8, "theme": "room",
      "min_x": -30, "min_z": -30, "max_x": 30, "max_z": 30 }
  ],
  "groups": [ { "id": "under", "mirrors": false, "shapeIds": [ "..." ] } ] }
```

below: true inserts the layer under the compiled ground rather than over it. Every shape in it is an **add**: the rock is banded round its hole rather than cut with a subtract. Four rectangles ring the hall; the hall is a fifth.

The depth arithmetic is then by construction rather than by any field. The rock runs from floor 0 for fourteen courses, meeting the landmass's underside at $y = 14$. The hall is a shorter span in the same columns, floor 0 and eight courses, so its floor is stood on at $y = 7$ and it has six courses of headroom under the rock's ceiling. Stated generally: a storey's headroom is the gap between two spans and needs no field at all.

8.2.3 Stamping Without Mirroring

The group carries mirrors: false, and that is not a detail.

The storey covers the whole board, so its own rot_180 image lies over it; fanning it would put every column's ground back twice. A layer that covers only one half is fanned and one that covers both is stamped once.

The cost of stamping once is that the layer then has to be its own image, which is to say symmetric by construction rather than by machinery.

8.3 Three Failure Modes

8.3.1 Subtracts Across Layers

The natural way to make a hall is to cut it out of a slab. The studio refuses that, twice:

```
POST /map/from-documents 422
SK13 's0' fills 3600 column(s) that 'cut1' takes away -- from (-30, -30)
    -- 's0' is on layer 'ground' and the subtract on 'under', and a
    subtract reaches only the layer it is on
SK13 'descent' fills 56 column(s) that 'cut1' takes away -- from (-22, 2)
```

The reason is semantic rather than technical. A subtract states that the column is *empty*, and the landmass over it then fills what the storey below said was void. So the recipe is banding: split into bands at every hole edge along one axis, split each band along the other, and drop the parts that fall inside a hole.

The sculptor's version of the same move avoids the problem at the outline level: an outline is filled even-odd, so a ring is drawn as the outer ellipse, slit inward, the inner ellipse the other way round, and closed: one polygon, no subtract, no refusal.

8.3.2 Silent Occlusion

This is the failure to be most afraid of, because everything reports success:

Writing the rock as one rectangle over the whole board and the hall as a shorter add in the same columns compiles, builds and exports with the gate **open**, and there is no hall.

No gate fires, and the studio is right not to fire one: two adds at one floor are ordinary ground, and the taller winning is exactly what *height is the tallest add* means. That is precisely why the banding is the technique rather than a detail of it.

8.3.3 Verification by Count

A missing column in a storey is invisible in every render, because the rock above and below it hides the gap. So it is not inspected; it is counted. The undercroft showcase states its check as a number: over the board's 10,000 columns, none is bare and none carries two spans.

The honest cost of the technique shows up when the holes get complicated. A hall is four bands. A maze of a hundred corridors and links leaves **213 rock bands** between them.

Banding Versus Overlay

The rock is banded round the hall because the two differ in *height*, fourteen courses against eight, and on one layer the taller add wins the column outright. A basin and a hall that are the same eight courses in the same columns win nothing from each other: the geometry is identical either way and only the paint differs. Getting it the wrong way round is silent in both directions: a banded-out basin is the same world, and a shorter add inside a taller one is no room at all.

8.4 Override and Floor

An underpass makes the mechanism measurable. A deck and a cut occupy the same cells. Without override, a column read at the deck answers *0 solid blocks, void*: the ordinary add lost to the subtract.

With override: true but no floor, the studio refuses:

```
POST /map/from-documents 422
[refusal] SK13 'deck' fills 260 column(s) that 'gullet-cut' takes away
-- from (-28, -5)
```

The reason is that `floor` defaults to zero, the very bottom of the shape's column, so an override add with no stated floor spans from the world's floor to its own top, and once again fills what the subtract declared void.

`floor` is the field that decides, and it decides by setting where the column's own span *starts*, not by any relation to the ground it replaces. With `"floor": 13`, `"base_height": 4` the same column reads four blocks, *y13* to *y16*, and nothing recorded beneath. The rule that separates a lid from a fill is then exact: an override add resting *above* the subtract's floor moves that span up and records nothing beneath it, and the gate is silent on it; the same shape at the subtract's own floor is the refusal above. The difference between the two documents is one field.

8.4.1 The Scope of the Constraint

It is often said that the studio has no floor underneath a shape. That is not quite it, and the accurate version is more useful:

A column carries one span *per sketch layer*, not one span. ...So the honest statement is narrower and more useful: **one layer cannot stack, and stacking is what layers are for.**

8.4.2 The Stairwell

The two mechanisms combine in the move that connects a storey to the one above it:

```
{ "id": "descent", "type": "polygon", "operation": "add",
  "override": true, "keepClear": true,
  "floor": 8, "base_height": 14, "theme": "stair",
  "height_mode": "level", "skirt": 0,
  "vertices":      [[-22,2], [-18,2], [-18,16], [-22,16]],
  "anchor_heights": [ 1,      1,      14,      14 ] }
```

Two numbers matter and they are the same number. The flight's floor is 8, the hall floor's own top, so it rests on the storey below rather than driving through it. And an override add is what cuts the shaft; a subtract there would be the refusal above again.

One rasterizer fact is worth carrying away from this, because it explains a class of ugly output that looks like a bug. A ramp at one course per cell only rasterizes cleanly because the surface is sampled at the cell's centre and *floored* into the column. Rounding to nearest instead puts every sample of a 1:1 grade exactly on the boundary, and the flight comes out as a beat of noughts and twos.

8.5 Paint Order in a Stack

Layers are written bottom-up and the painter walks them in document order, which produces one failure worth stating plainly: a storey listed after one that stands over it finds no stone left to paint. An undercroft goes *before* the compiled ground in the document, and below is what puts it there.

Two further properties of a stacked board catch authors out. A plain layer's paint bands run from the bedrock course whatever its `base_y`, since only a layer marked as a *made* thing is painted over its own span, so every ground theme on such a board wants a fill that is not trivial. And a placement names its storey explicitly; naming none takes the top surface, which on a roofed goal is the roof.

8.6 Made Layers

A layer marked `kind: "made"` is out of the stacking rules: it is painted over its own span, and it is exempted from the walk that checks a board's pairs and the walk that checks its reachability. The reason is stated as a fact about the shape rather than as a convenience. A solid standing in a hill has no gap to lose, and its roof is not a stair somebody forgot.

A made layer also carries `part_of`, naming the thing a run of layers is sliced from, and may carry `seat: "ground"`, which settles the run onto the terrain together rather than each slice finding its own height. That is how a sculpted landmark, whether a dome, a spire, a tapered tower or a gatehouse, is authored: as ordinary sketch shapes, circles and polygons with a floor and a height, rather than as stamped block soup.

Terrain: Themes, Materials and the Slope Axis

9.1 A Second Pass over a Finished World

Painting terrain is not part of building it. It is a second pass over the realized world, run after every structure has been stamped, and it adds no geometry at all. It reads the terrain the world builder placed and rewrites its surface.

Where the stampers write rooms, cubes and objectives onto the terrain, this pass dresses the terrain *itself*, bucket by bucket, through whichever theme applies; unthemed, every bucket stays the raw stone it already is.

One invariant makes the whole model safe to state simply: **the painter touches only stone**. Terrain arrives as a bedrock floor at $y = 0$ with a stone column above it, and every structure the stampers add is bedrock, wool, obsidian, anything but stone. So the pass runs last and rewrites stone blocks only. Bedrock is excluded by construction rather than by a special case.

Two refinements of that invariant were each learned as a defect. The test is over the whole block and not over its id, because stone's id is shared by granite, diorite, andesite and their polished forms; a plinth in polished diorite under a red prop came back red until the check compared the pair rather than the id. And the ground a room stands on is not a stamp, so it is painted like any other ground: a room levels its own footprint by filling upward in stone, and the result comes out in whatever the board's theme paints there.

9.2 Four Buckets

Every column splits into at most three painted parts and one kept part. The four names are the shared vocabulary that every later stage speaks.

Rim

The edge lip: a single block, the top of an edge column. Not a volume. A map's rim is therefore a one-block-wide line tracing every lip, quartz against the grass.

Wall

The exposed riser below the rim, painted only on the face that is actually exposed. Stone below the shallowest drop stays stone, because it is buried behind the neighbour that made the drop and painting it would tint blocks nobody can see.

Surface

The top of the interior, as a layered stack to a configured depth rather than a fixed single block. One block of stone by default; grass over two dirt, three deep, in a meadow finish.

Fill

The interior body below the surface. This is the *required* bucket: it claims whatever no other enabled bucket took, so a theme is never partial.

A freestanding nine-tall column against the void therefore reads, bottom to top, one bedrock, seven wall, one rim.

9.2.1 Edges over Eight Neighbours

A column is an edge when any of its *eight* neighbours is void or carries a strictly lower surface. The four-neighbour test is wrong here, and the reason is geometric: at a reentrant corner of an L or a comb, the corner block's only lower neighbour is diagonal, so a four-neighbour test leaves a one-block hole and the rim reads as broken. The extra four neighbours are exactly the reentrant corner cells, so the rule thickens nothing and only completes the turn.

How wide a net the lip casts is itself authorable, through three nested tests. *drop*, the default, lips only drops. *boundary* traces the whole plateau outline, adding the edges that face something *taller*, such as a room or an approach wall. *void* narrows instead of widening: a column is rim only where a neighbour is off the footprint entirely.

The difference matters on stepped ground. A staircase of five shapes each a block above the last is a drop at every tread, so *drop* draws a lip on each one and the body reads as five plateaus that happen to touch. *void* caps the outside and leaves every tread bare, so it reads as one body with a rim around it.

9.3 Materials and the Band Stack

A material is either simple or composed. The composed one is the interesting case:

`LayeredMaterial` is a `BandStack`, an ordered run of (`material`, `thickness`), plus the **axis** it is read along. The stack states its bands and what happens where they run out, and deliberately not the distance, because the distance belongs to whoever is measuring it.

That last clause is the design decision the whole section rests on. There is one type with the axis named, not four types, because the bands, the thicknesses and the run-out rule are identical whichever way the stack is read and only the distance differs.

Table 9.1. The four band axes. The stack is the same object in each row; only what is measured changes.

Axis	The distance	What it draws
depth	courses down from the top of the bucket (the default)	grass over two dirt; a wall's banded riser
inward	steps in from the landmass's void-facing edge	a cobble rim, two rings of stone brick, then a field
height	courses up from a stated world <i>Y</i>	a hull's waterline; a tower banded by course, whatever the ground beneath it does
slope	how steeply the ground is inclined here, 0 to 89 degrees from level	meadow on the flat, coarse dirt on the shoulder, bare rock on the face of the same hill

9.3.1 Running Out

ending decides what happens past the last band, and beyond says what shows there. Under repeat the last band claims everything further along, which is right where the stack owns its whole space.

A ring stack usually does not own its whole space: it is meant to run a few rings in and then stop, so it ends handOver, and beyond answers for everything past it.

This is the mechanism behind a trap worth stating in its own right, because it produces documents that look wrong and are not. A stack does not cycle. Striping a face sixteen courses tall with a four-course pattern under repeat means writing the four courses out four times, because past the last band the last band claims everything.

9.3.2 Why the Height Axis Exists

depth and inward both measure from the shape the block is in, so a colour change along either splits nothing. A colour change stated as *geometry* is a different matter: a layer holds one theme, so banding a form by splitting it into a layer per colour costs a layer per band, and that, rather than the shape, is what a sculpture costs.

Measured over nine models, the layer count a shape alone needs against the count it actually took: a starship 2 against 4, a robot 5 against 16, and a Rubik's cube, which is a solid box with one run per column, 1 against 7. Read along height the same banding is one material on one layer.

9.4 The Slope Axis

This is the load-bearing one, and it is the axis a board should be finished along.

The slope axis is an angle mask, and a thickness on it is a span of degrees.

A stack of grass at 30, coarse dirt at 15 and cobblestone at 45 is therefore meadow up to 29 degrees, coarse dirt from 30 to 44, and bare rock above.

```
{ "kind": "layered", "axis": "slope", "stack": { "ending": "repeat", "bands": [
  { "material": { "kind": "solid", "id": 2, "data": 0 }, "thickness": 30 },
  { "material": { "kind": "solid", "id": 3, "data": 1 }, "thickness": 15 },
  { "material": { "kind": "solid", "id": 4, "data": 0 }, "thickness": 45 } ] } }
```

The angle comes from Horn's 3×3 gradient over the neighbouring surface tops, so a ramp climbing one block a cell answers 45 exactly and level ground answers 0. It is a fact about the *surface*, carried by the whole column, which is why a face's fill and the block on top of it resolve to the same band and a hillside comes out finished top to bottom.

Nothing Else Distinguishes a Hillside from a Field

The wall bucket is the exposed riser, and a surface climbing one block a cell has no riser at all, because its drop is one course and the top course already claims it. Every other axis therefore paints a 45-degree hillside and a flat meadow alike. A board finished on plan pieces or on height bands is painted flat from above however much relief is under it.

The void is deliberately not a slope. A neighbour off the footprint, or one carrying a structure, reads as level with the cell itself, so a coastline is flat ground that stops, and the ground beside a building does not take the building's roof for a gradient.

9.4.1 The Window Is Two Cells

The gradient is read two cells either side rather than one, and the reason is that a gentle slope in a voxel world is a staircase.

Ground falling one block every four cells is four flat treads and a one-block riser, and a window one cell wide reads each riser on its own, 27 degrees on the riser and nothing on the treads, so a uniform gentle grade comes out speckled with whatever the theme paints at 27.

The tell is in the distribution rather than in the picture: read one cell either side, a board authored at a one-block step puts a quarter of its ground in a spike at 20 to 29 degrees with almost nothing at 30 to 39, which is a board reporting its own quantum rather than its shape.

The arithmetic for why two is the right number is worth recording. A *sustained* slope reads its true angle at every window, so one block per four cells is 14 degrees at windows of 1, 2, 3 and 4, and widening costs a real slope nothing. What widening costs is a face, whose drop is spread over more cells: the lip of a six-block drop reads 72 degrees at a window of one, 56 at two and 45 at three, where it can no longer be told from a walkable ramp. The isolated one-block contour that should be suppressed falls from 27 degrees to 14 to 9.5 over the same range. Two is where the contour is already under any threshold an author would set and the face is still unmistakably a face.

9.4.2 Where the Cuts Go Is Read, Not Guessed

GET /map/{slug}/incline answers how much ground stands in each ten degrees. It is the only thing that says whether a band cut lands where the author thinks it does, and ?window= answers the same board read wider, so both decisions a mask needs are made without an export.

Two corpus episodes show what happens without it. A first cut at 24 degrees put an entire hill onto one band, and the board came out grey with green only at its edges; recut at 32 after reading the histogram, it was right. On another board the reading answered 77% of cells under ten degrees, 22.6% in the teens, 0.5% above twenty and nothing past forty, which is to say that a board about to be finished by angle had almost no angle in it.

9.5 Composition, and the Commonest Mistake

The axes compose, and this removes the need for several knobs that would otherwise be necessary. A surface material that is a depth stack whose first band is one course thick, and is *itself* an inward stack, rings the top course and leaves the courses beneath it alone.

That same composition is what keeps a grass ring one block thick rather than three. A palette that repeats grass down every course is, by the repository's own account, the most-repeated authoring mistake in it.

Structures: Houses, Rooms and the Course Stack

10.1 A Stamp Is One Course

The phrase recurs throughout the studio's documentation and it is a claim about architecture rather than about blocks. A shell is not a hard-coded building. It is a stack of courses per part, and any shipped shell is one edit of that stack away from any other.

A cage and a spawn differ only in their band material, their door and their contents, and everything below is one course stack away from being something else.

Table 10.1 is the whole of the two shipped shells, numbered from the floor. Reading the two columns against each other is the argument: three fixed layers are ordinary courses, the coloured band is a team-tinted course, and the light slit is a course of air.

Table 10.1. The two shipped room shells. A wool cage and a spawn cube are two bound styles rather than two code paths.

Layer	Wool cage	Spawn cube
8 (roof)	bedrock, centred hole, proportional and capped at 4×4	bedrock, centred hole
7	bedrock	bedrock
6	missing: a light slit around all four walls	missing: a light slit
5	bedrock	bedrock
4 (colour strip)	wool, in the room colour	stained clay, in the team colour
3	bedrock	bedrock
2	bedrock	bedrock
1 (doors begin)	bedrock with a door opening	bedrock with a door opening
0 (floor)	bedrock with a wool pad, the wool spawn point	bedrock with a wool pad, the player spawn

There is a second, unrelated sense of the same phrase, and it is worth separating because both appear in the painting model. When the painter excludes a stamp, the exclusion is of the *course*, not of the column: the painter keeps a stamp's own blocks and paints the ground under it.

10.2 Parts, and What a Style May Not Decide

A shell is parts and overrides. A wall and a foundation's plate are each a part: a band stack plus an extent saying how far the part goes. Over them sit the two things a style may not decide, and each is excluded for a stated reason. The **pad** is excluded because the exported wool or spawn location is derived from it. The **doorway** is excluded because it is the entry contract.

A Statement About One Piece Lives on That Piece

What a building stands on is a foundation, what stands above the eave is a roof style, and the way in is a doorway. A field that only *looks* like part of one is the trap: which wall the doorway is cut through is the building's *front*, not the doorway's, because it is the same wall a shed roof falls toward and a porch stands against.

10.2.1 Direction Is Load-Bearing

A stack is read from its part's own base outward: a floor *downward* from the course players stand on, a wall and a roof *upward*. That direction is not a convention. A floor that grew upward would lift the pad, and lifting the pad would move the exported point.

The last band repeats past the end of the stack, so an extent can move without the stack being re-authored. Repeating is the stack's own statement rather than the axis's, and the two are kept separate for a reason already met in chapter 9: repeating is right where a stack owns its whole space, and handing over is right where it is a band inside a larger one. A repeating band does not reseed a pattern.

Air Is a Gap, Not a Block

A part's air course is skipped rather than written, so no style can erase what another stamp already placed. A doorway's air *is* written, because it cuts the opening out of the wall the same pass just built. Windows follow the doorway's rule for the same reason.

10.3 The Roof Is a Height Field

Six roof forms exist, and they are one object with one formula.

Every roof is one height field: for each cell of the roof's plan, how far the roof's own surface stands above its base there, and how many courses that column writes to close the step down to its neighbours. The stamper walks that plan once. The forms differ in a single formula over the same two distances, how far the cell stands from each wall line, and in nothing else.

Table 10.2. The six roof forms, as one formula over two distances.

Form	Rise over a cell, per block travelled
Flat	the base course, everywhere
Gable	the smaller distance across the building's <i>shorter</i> side, times the pitch
Hip	the smallest of all four wall distances, times the pitch
Shed	the distance from the front wall, held to the shorter side's run, times the pitch
Gambrel	as Gable, at pitch plus one for the first quarter-span in from each eave, then at the pitch
Saltbox	as Gable, but the two distances across the shorter side climb at different rates: the front's side at pitch plus one, the other at the pitch

A hip over a square footprint is a pyramid. A shed and a saltbox need to know which wall is the front, and the front is the wall the doors are cut through.

10.3.1 No Roof Climbs with the Long Side

The constraint that keeps this usable is that every form measures across the *shorter* side. Without it, a 10×60 building would carry a lean-to sixty courses over its wall: a tower with a door in it, produced from a footprint an author drew flat. A shed therefore rises at its pitch until it has risen the shorter side's worth, then runs flat the rest of the way, which is a lean-to that levels off.

10.3.2 Eaves, and Where the Fall Stops

Distances are measured from the wall line and are allowed to go negative, which is what makes the eave part of the slope rather than a separate thing bolted on. The fall then stops two courses down, and the bound is derived rather than chosen. One course is what a pitch of 1 needs to read at all. Past two, the eave hangs in front of the wall it shelters rather than over it: at an overhang of 2 and a pitch of 4, an unfloored eave lands two courses *below* the floor the building stands on.

Walls climb to meet the roof wherever the roof stands above them, which is the gable's two ends, the shed's back wall and both its flanks, and nothing at all under a hip, whose slopes come down to the wall line on every side. What they climb in is the **gable**, a part of its own; unbound, it is the wall's top course carried up, so a wall that bands as it rises goes flat the moment it turns into a gable. The **verge** is a different piece again, being the roof's own outermost ring: on a flush roof it is the raking edge directly over the gable, and under an eave it moves out to the overhang.

10.3.3 Halves, and What a Slab Roof Is

A roof laid in cubes steps two halves per block travelled. The field is measured in halves, and naming a roof slab on the style puts the roof on one: it then climbs half a block per block, laying that slab on every odd step with the roof's own material filling the cubes between. The halving has to floor rather than truncate, because the eave's rise goes negative below the base plane.

10.3.4 One Material, and One Refusal

A roof is one material, and the slab is that material in halves. The body and the verge are each a single block, and a pattern is refused in either, because a roof is read as one plane from below and a scattered material across it is several blocks in one surface.

Logs are a special case with a tidy explanation. A log's data value records which way it lies, so a log named as a plain solid has none: every one of them stands upright and shows a sawn face out at the slope. A *laid* log takes the axis of the surface it is on, and for a roof that axis is the ridge. A laid log carries its own whole-course rise and no slab is cut from one, so a roof slab over a log roof is refused.

A Roof Has No Thickness Knob

The height field is the reason: a column writes as many courses as the step down to its neighbours needs, so thickness is not a free parameter. Two fields did once carry such a number, and nothing ever read either. Both were dropped rather than implemented.

10.4 Wings: A Building of More Than One Rectangle

Every roof is a height field over one rectangle, and a building is not always one rectangle. A building plan is one or more touching **wings**, and everything below the eave reads the plan's own cells, so an L, a T or a U is built as one house on one outline.

A wall stands wherever the plan is exposed, diagonals included. Where two wings meet, the two walls running into the turn touch along one vertical edge and nothing else, and the cell behind that edge, which has building on all four sides of it, is the corner. It takes a post like any corner, so an L stands on six.

A Storey Is a Plan of Its Own

That is the whole of how a building of unequal wings works. A wing carries a storey count, the plan at a storey is the set of wings still standing there, and every pass reads the plan for the storey it is on. A two-storey hall with a single-storey range attached is therefore not a special case; it is one plan at the ground floor and a smaller plan above it.

Dressing: Placing Rather Than Sprinkling

11.1 The Central Decision, and the Rewrite That Produced It

Dressing is the third pass over a realized world, after the structures are stamped and the terrain is painted. Where the painter adds no geometry and only rewrites materials, this pass exists to add geometry: trees, boulders, flora, paths, buildings.

The obvious design for such a pass is a set of density fields assigned to shapes, the way a terrain theme is assigned. That was the first implementation, and it was replaced. The reasoning is the clearest statement in the system of why map-making resists automation:

A tree is cover and a boulder is a wall, so where each one stands decides how the map plays, and a decision about how a map plays belongs to the person making it rather than to a noise field. So every part of this stage is a thing put somewhere: a route is dragged, an area of cover is traced, a tree and a rock are clicked into place, and each carries its own knobs. The fields that remain are the ones *inside* a drawn area, where placing them one at a time would be data entry rather than authoring.

That line, between a decision that affects play and a decision that is merely tedious, is drawn once and then holds. Which tree stands at the mouth of a lane is authored. Which blade of grass fills a square metre of meadow is not.

11.2 Running Last Is What Makes It Tractable

The pass reads the finished, painted world. That ordering is load-bearing rather than incidental: the painter has already decided, per cell, what the surface *is*, so soil accepts flora and quartz does not, grass can be replaced by a stroke and a monument's wool cannot. A pass running earlier would have to predict all of that.

The one elevation model it needs is the first air height above each column, and the one it is given is the tops of everything on the board *that is not a made thing*.

A Made Thing Is Not Ground

Nothing that rests on the board treats a made thing as ground. The failure that established this is instructive: a cloud drawn at $y = 78$ over a car park was the top of every column beneath it, so the goal there read the cloud and was stamped at $y = 83$, over a build ceiling of 68. The same elevation feeds the structure-site check, which would otherwise report a shed under a balloon as standing on a fifty-block plinth.

Two consequences follow for a stacked board. A prop carries an optional layer and rests on that layer's own surfaces; naming a layer the board has no ground on is declined. And the claim book that stops two props sharing a cell is keyed on the layer as well as the cell, so two props share a cell only where they share a layer.

11.3 Keeping Ground Clear

A shape can declare that it is not ground to dress. A shape marked `keepClear` puts its own columns into the mask exactly and with no margin, so a road still runs to a gate while the wall either side of it keeps its top course.

The failure without it is worth stating because it is not obvious. A stroke repaints whatever it crosses. A channel, whose water line is the *lowest* surface its band crosses, cuts every other column in the band down to that line. On a wall standing seventeen courses over a river, that is a hole through the wall rather than a bank.

The keep-out mask itself gathers spawns, wool rooms, stated structures, built columns, the shapes that marked themselves clear, and every door's approach. It is not the map contract's protection regions, and destroyables and cores are deliberately not in it.

11.4 Six Primitives

Noise

The same deterministic hash-from-cell the terrain patterns use, never a random number generator. A density field decides where flora goes, a low-frequency field gathers flowers into meadows and trees into groves, and a per-cell value is the dice a worn stroke rolls.

Mask

Eligibility, read from the painted surface and from the keep-out set. The whole of it applies to anything that stands on ground; only the structure part applies to a stroke, which stands on nothing.

Placement

A *point* for props that stand somewhere, a *drawn outline* for ones that cover a stretch, and a *dragged rectangle* for the one whose stamp takes a footprint. A marker seats on the ground and can therefore only be dropped on rasterized terrain; an area has no such limit.

Reshaping

A selected prop wears one grip per point. Reshaping never changes how many points a prop has.

Stamp

A three-dimensional volume seated on the surface and written cell by cell. A prop seats on the *lowest* column of its own footprint, so it sits into a slope rather than floating over the low side.

The fan

Every prop is placed once and stamped at every image of its orbit, in the prop's own local frame, with each offset turned by that image's transform.

Two details in that list repay attention.

Only the Feet Are Asked About

What a prop *rests* on needs real ground; what it merely reaches over does not. A cell of the prop standing where something already is is skipped, and nothing else happens.

The fan carries a mistake worth naming, because it is invisible in a top-down render and obvious in play. Mirroring only the anchor leaves both teams with the same unmirrored prop shape, so a

boulder with a lobe to its east has a lobe to its east on both sides of the map. The stamp is also all-or-nothing per image: if one image's ground is missing or kept clear, that image is skipped rather than clipped.

11.5 Clipping, and the Piece of Tree Floating Next to a Building

A prop stamped against an obstruction can end in one of three states, and the studio distinguishes them because only one is a fault.

Truncated is a prop with a face taken off it. Measured against a wall two blocks off its centre, an erratic of size seven loses 36% of its 1,089 blocks and severs none of them. That is acceptable: it reads as a boulder against a wall.

Cut in two is a prop whose obstruction took out the blocks joining a piece of it to the rest. Nothing declines, the tree is in the world, and there is a piece of tree floating next to a building.

The measurement is taken against the prop *as it would have stood on open ground*, and connectivity is judged by face adjacency, because a block joined to its neighbour at a corner alone has air on all six of its own faces and is seen straight past.

11.6 Placement Is a Query

Chapter 3 and chapter 15 both arrive at this point from different directions, and it belongs here as mechanism. The studio can answer where a prop or a building may legally stand, and an agent that asks converges immediately where an agent that guesses does not.

Across three separate runs, declined props on successive dressing passes ran seven, four, two, zero; then six, five, three, none; then five, five, five, six. In every case the pass that reached zero was the one where every position came off the studio's own seats mask rather than off the agent's judgement. One run states the final tally plainly: 78 placed, 0 declined.

The same read answers a question about the board rather than about the prop. Asked whether a composed board had room for buildings, the seats read answered that a 12×8 house seated in *zero* cells, a 9×7 in *zero*, and a 7×5 in *seventy*. That is a fact about the board's grain, and no picture of it would have said so.

11.7 Trees That Are Copied Rather Than Grown

A tree may be a recipe the studio grows, or it may be *copied*: a body recorded block for block out of a real world and filed in the library under a name. A placement then names that style. The distinction matters for the look of a board more than almost anything else in this chapter, because a grown tree is recognisably generated and a copied one is recognisably built.

One caution from the corpus, which no gate reports: nothing asks what block a prop is seated *on*. The site check asks whether there is ground under a prop, not what that ground is. One run found thirty-five of ninety-eight trees not seated on soil, including trees on ice, on coal ore, six on a ruin's masonry, and one planted in another tree's leaves. The only read that answers it is one column per prop.

The Intent, the Contract and the Export

12.1 What the Intent States

The intent is the third of the four documents: teams, spawns, protections, build regions, objectives, and how each objective is captured. It is the only one of the four that is about the *match* rather than about the ground.

Storing it and projecting it into the PGM document is one call and one idempotent pass. PUT /api/map/{slug}/intent writes the intent and derives from it the teams, kits, regions, filters, apply-rules and spawns that the document needs. Chapter 3 gave the scale of that derivation: a plan stating four wool rooms and their markers produced four wools, four monuments, 22 regions, 34 filters and 11 apply-rules, with nothing authored and nothing corrected.

12.2 Naming, and a Trap in It

PGM has no <monument> element. The destroy objective's element is <destroyable>, and the word *monument* is already taken throughout the studio for the CTW wool-monument, which is the block a capturing team places wool on.

The Objective Is a Destroyable, Never a Monument

Colloquially a destroyable *is* the DTM monument. In types, columns, identifiers and JSON keys it is not, because the word already names a different thing in the same system.

This paper uses *monument* in the colloquial sense when describing how a map plays, because that is what players call it, and *destroyable* whenever the subject is the document.

12.3 The Two Destroy Objectives

Both are goals owned by one team and destroyed by everyone else, and both are structurally simpler than a wool, which drags a room region, a spawner, a dye item, team filters, apply rules and a coupling to the spawns.

12.3.1 The Destroyable

<destroyable owner name region materials [completion] [id] [modes]>. PGM builds the set of blocks inside the region that match materials, and the goal completes when completion of them are broken.

completion defaults to 1.0. Four attributes are required and everything else defaults.

12.3.2 The Core

<core team region [material] [leak] [name] [id] [modes]>. material defaults to obsidian and leak to 5.

The core builds two block sets from the world inside its region: the casing, being the blocks matching material, and the lava. It leaks when a lava block reaches a height of `region.min.y` minus leak minus one, within fifteen blocks horizontally of the core bounds.

One inconsistency is worth noting because it shows in the document: the core's owning attribute is `team` rather than `owner`, which is a known inconsistency in PGM itself. The studio mirrors the XML and calls the field `owner` in its own code.

12.3.3 Why This Caps the Team Count

Section 1.2 stated the consequence and this is the mechanism. Neither objective has the wool's per-capturing-team fan-out. The shape is **one object, one owner, and $N - 1$ attackers**.

A wool at four teams gives each team three distinct things to capture, so each team has an objective of its own and a winner is well defined. A destroyable at four teams gives each team one thing to lose and three things to break, and nothing in that arrangement says who won. It is a property of the goal's structure rather than a limit anyone imposed.

12.4 The Float, and What the Corpus Actually Does

Both objectives float above the terrain, which is why no carving or negative-space primitive is needed to seat one. The studio's generator floats a destroyable three to five blocks, with a default of four, and a lone floating pillar is a normal output rather than an error. An author's cap bounds the other end at twelve, and the reasoning behind that cap is that a goal high enough to need a tower under it is reached by a build project rather than by a raid.

The corpus does not do this, and the difference is worth stating because a detector expecting a gap will miss most real structures. Measured on the declared blocks themselves across 571 single-material structures, the median air gap beneath is **zero**, and **424 of 571 rest directly on something**. Only 68 float three to five blocks. The floating pillar is the authored ideal; the corpus overwhelmingly seats its destroyables on terrain, a pedestal or a build.

Float Is an Offset, Not a Height

The studio's `float` counts from the ground as built rather than from a plan-nominal surface. A destroyable or a core is never given a Y at all: the structure floats its stated offset over whatever the relief actually left under its column, resolved once the layout is rasterized, which is the only place that ground is known. A goal placed on a mesa stays on the mesa however the mesa's own height is later solved.

Because the compiler resolves nothing about height for these two markers, they are also the one marker kind that may name no plan piece at all. The position reads as an absolute board position, and the ground it rides can be an authored sketch shape with no piece behind it. A landform is then authored once, as the shape it is, rather than twice: once as a plan rectangle purely to give a marker something to ride, and again as the polygon it actually is.

12.5 Flooring Is Per Field

One detail of the contract has cost hours more than once, and it looks like an inconsistency until the reason is known.

Coordinate Flooring Is Per Field, Not Global

The wool's <location> is floored and the destroyable's <block> is not, because PGM floors one itself and never the other. Applying one rule globally in either direction produces a document that is wrong in half its objectives.

12.6 The Export Gate

The export is where several checks first become fatal. GET /api/map/{slug}/xml answers the document, gated on the pre-flight checks, and GET /api/map/{slug}/export answers the world.

Four refusals are worth knowing by name. EX2 is no spawn of any kind. EX3 is a kind that went in and did not come out, which is the harder failure of the two because it means something was accepted and then lost. EX4 is an objective with no team. 0B20, an unknown gamemode, is checked first on every export regardless of how the map originated.

Which gate runs does not depend on which door the caller came through. A sketch-originated map's whole chain runs inside the same composer as a plan-originated one's.

The corpus records this as an expensive property rather than a neutral one. Two of those refusals fire only at export, so the first time a board hears about them is a 409 answered *after the whole world has been built*. One run hit them three times between them, at a cost of three full build cycles.

12.7 What Is Delivered

A sketch-originated map exports a ZIP carrying map.xml, level.dat and the region files at the archive top. A map that arrived through the import path exports the XML alone.

The world format is Anvil in its numeric block form, which is the 1.8 to 1.12 encoding with nibble-packed sections. That matches the studio's supported range, and it is the reason the studio refuses maps outside it rather than accommodating them: a malformed or out-of-range map is rejected rather than handled by weakening the schema.

Reading a World Back

13.1 Why the Reads Are the Interface

An agent cannot look at the map. Everything it learns about what it has built comes through a read, and the quality of those reads therefore bounds the quality of the authoring. This is the chapter that makes the rest of the system operable.

The studio offers about thirty of them. They divide cleanly into pictures and numbers, and most have both forms: a PNG for a person, and a `?format=text` twin that answers the same question as characters or a table. The rule the corpus settled on is that the text comes first. A picture answers *whether* something came out. A number answers *whether it is right*.

13.2 Every Read States What It Cannot Answer

The property that distinguishes this surface from an ordinary set of debugging endpoints is that each one documents its own blind spot. The studio's published schema gives every read a summary and, beside it, a clause headed MISLEADS naming the wrong conclusion a reader is likely to draw from it.

This is unusual enough to deserve examples, because the clauses are not boilerplate. They are the specific mistakes that have been made.

Column. A column through the middle of a house reads floor, air, roof. The walls are at the perimeter; that is a correct building, not a broken one.

Top-down render. It keeps no Y at all, so a riser, a ramp's step heights, a stamped room's floor and a goal's clearance are none of them in it.

Transect. A transect answers exactly the line asked for and nothing either side of it. A barrier two cells off the stated points is invisible to it.

Reach. It names nothing as wrong. Scenery, a side observer island and a shelf over the ceiling all read exactly like a shape stranded by accident, and only the author can tell them apart.

Chapter 15 recounted several runs that redesigned correct boards on the strength of a reading, and every one of them was a case of a reader taking an answer as a verdict when the instrument had been asked a different question.

13.3 The Workhorse

`GET /map/{slug}/column` answers one or more columns from bedrock to sky with every block named. The studio's own description of it is unambiguous about its status:

THE WORKHORSE: every picture beside it is a projection, and this is what is actually at a coordinate, which is why it is the read to reach for when a picture and a document disagree.

One run recorded that nine of its ten faults were found by reading a column and counting courses, with the renders confirming them afterwards. When two reads disagree, the column is the one that is not a projection.

13.4 A Shape Is a Profile, Never a Point

A column answers about a coordinate. Many claims an author wants to check are claims about a *shape*: a bank, a wall, a stair, a basin. Those need a **transect**, which walks a polyline block by block and answers numbers.

Each station carries the ground, the storey stood on, the water and the highest block in its column, and the step from the station before it, classed as walked, scrambled, a barrier or a drop. Two parameters matter: every thins the stations, and beside lists every distinct claim within a stated number of cells of the line, which is the answer to the transect's own blind spot.

This read settles the class of question that arithmetic gets wrong. The corpus carries several ramps whose anchors were computed in absolute terms and landed ten or more blocks below a relief-solved hill, showing as a barrier. None was visible in any render. Each was one transect.

13.5 Four Answers in Four Units

The **walk** is the read that prices a journey, and its design deserves note because it refuses to summarise:

Whether the place can be reached, how far it is in blocks, how many blocks a player must place to get there, and how many falls over the free height it takes. Four answers in four units, none weighed against the others, so a caller reads the field its own rule is stated in.

The climb at a rise of δ costs $\delta - 1$ blocks placed, and bridging void costs one per cell. *aim* picks which route is measured: the shortest (*travel*), the one placing fewest blocks (*reach*), or the least edge-hugging of the routes within ten blocks of the shortest (*comfort*).

Refusing to combine the four into a score is the right decision for a system whose rules are stated in different units. A rule about how far a goal stands from a spawn is stated in blocks; a rule about whether a fresh kit can reach a wool is stated in blocks placed. A single number would serve neither.

13.6 The Instruments, and What Each Is For

13.6.1 The Incline Read, and Why It Cannot Disagree with the Paint

incline deserves a note because of a property it has by construction. Each cell's glyph is the *tens* of its angle, so a mask cut at 20 and 45 has both boundaries visible on the page as the step from 1 to 2 and from 4 to 5. Beneath the grid it prints how much ground stands in each ten degrees, which is the reading a cut is actually chosen on.

It runs the same formula the painter resolves against, so the two cannot disagree. The one caveat is stated in its own misleads clause: a wider window answers a different question from the one the paint was chosen by.

13.6.2 Two Reads Keep Y

Only section and column keep elevation. Every other picture is a projection onto the plan, which is why a top-down render can show a board that looks complete and conceal a storey, a riser, a

Table 13.1. The principal reads. Every one has a ?format=text twin unless noted.

Read	The question it answers
column	what is actually at this coordinate, block by block
transect	what a line across a shape does, station by station
slopes	the worst step to a neighbour per cell, classed walked, scrambled or barrier. A cliff reads as a line, a ramp as a band crossing it, and an overdone relief as a page
walk	what a journey costs, in four units
incline	how steeply each cell is inclined, and how much ground stands in each ten degrees
coverage	where the ground is lived on and where it is dead, with the dead patches named by coordinate
reach	which standing ground no player can get to, and why
themes/census	every ground cell counted by the theme that paints it
render/mirror	whether a board somebody believes is symmetric actually is
render/section	a vertical cut with a Y scale
render/topdown	what was built and where, one subject per image
plan/flow	what the board asks of the two sides, and what that leaves unreachable
findings	everything wrong with this map right now, from every gate its stored documents can reach

ramp's tread heights or a goal's clearance. An author checking anything vertical has exactly two instruments.

13.7 Findings, and the Gate That Runs Nowhere Else

GET /map/{slug}/findings answers what is wrong with a map right now from every gate its stored documents can reach, and names the gates it did *not* ask along with the route that pays for them. It is the read that answers a question no other channel carries, because a decline published on one route's warnings is not published on another's.

Its pair is GET /map/{slug}/state, which answers where the map has got to, which artifacts it holds, and which moves those artifacts allow, each with its route. An agent that has lost track of a map asks those two and needs nothing else.

13.8 What the Reads Do Not Do

Three limits are worth stating plainly, because each has produced a wrong report.

A read names nothing as *wrong*. Several of them partition a board and hand back the partition; deciding which part is a defect and which is scenery is the author's. Dead-ground calibration is meaningless on a board whose objective the coverage model does not yet understand, and on control-point boards it has misreported badly for exactly that reason.

A read answers about the world as it stands, not about the world as played. Nothing in this surface can say whether a map is good.

And a plan-tier read is not a world-tier read. Plan-level distances are computed over pieces and build zones; the same journey over built ground can be shorter or longer, and on a board with a maze in it one run measured plan-tier ratios of 2.19 to 5.31 where the built world answered 1.70 to

2.77. The plan-tier reads are a floor and nothing more on a board authored that way.

Teaching Material: The Technique Cards

14.1 A Third Party in the Division of Labour

The paper has so far described two parties. There is a third, and it is easy to miss because it does not run: the body of worked examples an agent reads before it authors anything.

This matters more here than it would in most systems, because an agent authoring blind cannot learn a primitive by trying it and looking. It learns by reading an example, and what it reads shapes what it builds. A teaching corpus that demonstrates one instrument per map teaches an agent to reach for one instrument per map. The repository has learned this the expensive way and has changed its teaching format accordingly, which makes the change itself worth documenting.

14.2 Two Formats, and What Each Holds Constant

The repository carries two bodies of worked examples.

The older is a **showcase** of twenty-three boards, one technique each. Its method is to hold everything else fixed: every board forks one control board, a plain hundred-by-hundred destroy board that scores zero with no violation and no complaint, and changes *only* what its technique needs. The diff is therefore the lesson, with nothing else in it.

The newer is a set of six **technique cards**. Its method is more radical: it deletes the board. A technique card has no teams, no spawns, no objectives, no gamemode and no plan at all. Its whole document is a setup, its themes, a map theme and its layers. There is nothing to diff against because there is nothing else present.

The size difference follows from that and is not incidental. The six cards come to 392 KB. The twenty-three showcase boards come to 12 MB, because each carries a built world. The showcase's own README promises examples short enough to read in full and small enough to paste; the cards are what actually delivers it, with four complete outlines stated in 4,395 bytes.

14.3 Five Differences That Change What Is Learned

14.3.1 Removing the Symmetry Confounder

Every technique card states `mirror_mode: "none"` and every group in it states that it does not mirror. The showcase cannot: it is made of real boards, so it must spend a section of its README on what the symmetry fan decides, and it records three boards whose pieces landed in each other because of it, at the cost of a rebuild.

Mirroring is a real subject and an agent must learn it. The point is that it becomes a subject one can learn *separately*, rather than a tax on learning anything else. The relief card still states the fact, that any symmetry mode folds the relief across the centre and a group's own setting does not change it. It simply does not make the reader pay for it while learning about relief.

14.3.2 Comparison Inside One Render

This is the largest pedagogical change. The showcase is one technique per map, so a reader comparing two treatments reads two folders and holds the difference in their head. A card puts the variants side by side in one world: five flights up the same eight blocks; the same island flat and at four heights; three relief plots sixty-four blocks apart; four outlines in a row.

The controlled comparison is then *in the picture*, and the relief card takes it further by making it quantitative. Its two plots differ only in the heights their shapes are drawn at, and the committed check reads: every one of one plot's 2,304 columns has the same top as the column sixty-four blocks east of it, and none differ. That is a proof that a solved shape's drawn height decides nothing, rather than an assertion of it.

14.3.3 The One Mistake as a Standing Section

Five of the six cards carry a heading titled, literally, *The one mistake*. It names the failure, its rule id, and the threshold at which it fires. The set is a good summary of where a terrain author actually goes wrong:

Table 14.1. The mistake each card guards against, with the threshold where each one begins to bite.

Card	Rule	The mistake
combined shapes	SK9	a lower shape drawn inside a higher one on the same layer is not in the world
polylines	SK25	a band that turns tighter than it is wide laps itself. Four points across 26 blocks at radius 3 lap in every bend; the same four spread over 40 clear
curved outlines	—	a handle pulling further from its edge than along it draws a loop rather than a bulge
ramp and stair	—	whole-number anchors 8 and 1 over 16 blocks build treads 1, 2 and 3 deep. A flight comes out even only with anchors half a riser past its first and last tread
relief on shapes	—	a made shape with no <code>relief_scope</code> joins the solve. The same yard reads <code>y11</code> instead of <code>y17</code>

The showcase scatters the equivalent knowledge through prose and collects it in a thirty-row index at the end. Making it structural and local is what lets an agent find it at the moment it is about to make the mistake.

14.3.4 Verification Committed as Text

The repository's own rule is to read the text before the pictures, and the cards implement it literally. Every card ends with a section naming the committed read that proves each of its claims: a census, a column, a section, a heightmap, a set of transects, a walk, a slope grid, a comparison, a relief read.

Those are the API's actual answers, verbatim, re-checkable, and diffable across a studio upgrade. The showcase's verification is a folder of pictures plus a table of numbers typed into its README, which cannot be re-run.

The strongest instance is the relief card's comparison file, which prints the same rows three times: as drawn, with `hold`, and with `exclude`. The counterfactual is committed alongside the fact, which is a standard of evidence most documentation does not attempt.

14.3.5 Teaching the Revision Surface

The showcase is document-in, world-out: one call drives a whole spec. That teaches an agent to author, and teaches it nothing about *revising*.

One technique card is the only place in either body that demonstrates the incremental editing surface, with every request and its response committed: moving a vertex, inserting one, deleting one, and setting a shape's anchor heights. It also captures the arithmetic that bites, which is that an insert lands one past the index it names and every later index moves up by one. An agent replaying a list of edits computed against the original ring will scramble the outline, and this is the only example anywhere that would have warned it.

14.4 What the Older Format Still Holds

The honest framing is narrower than *the cards replace the showcase*. The cards cover geometry and paint, on ground with nothing on it. The showcase covers the entire non-terrain half of the system: objectives and their gate rules, plan-tier authoring, walls, dressing, room styles, coverage, traversability, symmetry and the export gate.

So the cards supersede the showcase for the terrain-authoring primitives, and the showcase remains the record for everything board-shaped. That is already how they are used in practice: one orchestrated run, briefing twelve authoring agents, records giving them the cards in preference to the showcase, alongside a ground family chosen to stop twelve boards coming out as twelve greys.

14.5 Why the Format Is Part of the System

It would be possible to treat all of this as documentation hygiene. It is not, and the reason returns to the paper's argument.

The studio's knowledge reaches an agent through exactly three channels. The **API** tells it what may be asked. The **refusals** tell it what may not be built, and chapter 5 showed how much design is carried in them. The **worked examples** tell it what is worth building, and that third channel is the only one with any opinion about quality.

Nothing in the first two channels would stop an agent building a board that used one instrument and called it terrain. No rule fires. Every gate passes. The board exports. The only thing that prevents it is having read an example that put five treatments side by side and made the difference visible, and the only thing that prevents the subtler version of the same failure is a card that says, in a section of its own, which mistake this instrument invites.

PART III

Two Maps, End to End

The two boards at every stage, and what an authoring session looks like from the outside, reconstructed from a recording of every call its author made.

The Anatomy of an Authoring Session

15.1 Authoring Without Sight

An agent building a map cannot see it. It has no viewport, it cannot walk the ground, and it cannot glance at a hillside and notice that the slope is wrong. Every belief it holds about what it has built is a number that the studio handed back, or an inference from one.

That single constraint produces most of the craft in this chapter. Almost every recurring failure across the corpus of run reports reduces to one of three mistakes: believing a success that was not one, believing a picture instead of a measurement, or believing a number the agent computed itself. The disciplines that have accumulated against those three are the substance of an authoring method.

15.2 Success Without Effect

The most repeated finding in the corpus, and the one that has cost the most builds, is that a 200 is not a yes.

Every 2xx carries warnings, and a decline means a piece of what you posted is not in the world. SK9, SK10, SK15, WX11, HS7 and the DR family all arrive on a 200. The status code is half the answer.

Table 15.1 collects instances. Each one stored at 200, passed the export gate, and produced a world missing something the author believed was in it.

One run's summary of the studio-side defects it had provoked states the systemic shape: six of the seven were one shape, which is that something was dropped and nothing said so. And the sharpest sentence in the corpus on why that matters:

A gate that drops something without saying so does not just cost a build cycle. It teaches the wrong contract, and the wrong contract then gets written down.

That is not hypothetical. Five documents carried a wrong claim about author identifiers before the repository's author caught it, each one written in good faith by an agent that had watched a request succeed.

15.3 Projection Versus Measurement

The studio renders a great many pictures, and every one of them is a projection. A projection answers whether something came out. It does not answer whether it is right, and reading one as though it did is the second failure class.

The corpus rule is blunt: when two reads disagree, column is the one that is not a projection. One run recorded that nine of its ten faults were found by reading a column and counting courses, with the renders confirming them afterwards.

Four corrections are worth stating because each looks like a defect in the map and is in fact a defect in the reading.

Table 15.1. Things that silently did not happen. Every row stored successfully.

What was not in the world	What said so, if anything
Eleven piers and kerbs, all on one layer, where a layer holds one span per column	SK9 on a 200, export gate open
The enemy team's bridge to its own island, because zones is not fanned by the plan's symmetry	nothing refused; the evaluator scored 0; a grid render showed ten cells of enclosed void
Both wool rooms, built as bedrock boxes because the finish key is wool and the document said cage	nothing. A key the reader does not know is dropped in silence
A gill stated as a lower piece enclosed by higher ones, where the taller add wins every column	nothing until a transect was read across it
A flight, deleted by a raise 0 paint patch drawn after it	nothing. Paint goes down before flights
A wall, buried by terrain, because a wall's height comes from the plan and not from the relief	nothing
One column of deck, standing as a six-block wall across the only way into a lower storey, because a polygon's far bound is exclusive	nothing, and invisible to every read but a transect
A path, drawn at width zero because the polyline was written with width and points instead of radius and vertices	SK4 on the dressing preview, and nothing else
Twenty-six courses of city, painted as viaduct rail, because a plain layer's bands run from the bedrock course whatever its base_y	nothing. The census counts the surface and was right

A top-down material render draws the top *solid* block, so water reads as its own bed. One run concluded a lake was missing on that evidence. A traversability render marks an approach wall's cobweb course as impassable although the gate does not, so a board with an approach wall reads isolated; measured across twelve boards, every board with such a wall read isolated except one, and five that read badly all passed the export gate. A transect taken through a height filter reads short, which produced a report that a causeway was six blocks low. And a symmetry error of zero is a true statement about *one solved field*, not about the four copies of it, which is how one run trusted a symmetry reading for four builds while two of four quarters had no relief at all.

Two Reads That Are Both Right

On one board, the cross-board reachability reading reported 0 of 108 rows crossable while the walk tier reported a single connected place holding 99.8% of the ground. Both were correct: they ask different questions. A reader who does not know which question an instrument asks will take one of its answers as a verdict on the map.

15.4 Computed Values and the Duty to Read Back

The third failure class has a single tell, and the working skill states it directly: a number you derived yourself, used as if it had been read back. If you computed it, measure the thing you computed before building on it.

The instances are instructive because the arithmetic was not wrong so much as answering a different question. An agent assumed a river's eight-block drop was the same on both banks; it was not. An agent computed a push's crown gradient at 0.62 and measured 0.30, twice, because the half-width a crown divides by is the ring's own medial distance and no arithmetic in a specification

predicts it.

The best example is about distance itself. An agent proved, by a perpendicular-bisector argument, that a particular board's lopsided wool could not be closed, and wrote the proof into the specification. Then it asked the studio for the walk, which read 169 and 165. Moving the near wool twenty blocks west, onto an outwork across a strait, moved the *walk* by thirty-five blocks, because the walk has to come back round the water.

Straight-Line Arithmetic Is Not a Walk

Every distance in every rule is measured over the walkable surface, and the studio will compute one on request at no cost. An agent reasoning in Euclidean distance about a board with a hole in it is reasoning about a different board.

15.5 Misreading the Evaluator

Two failures are about the scoring surface rather than the map, and both are failures of interpretation.

The first is reading a score upside down. The evaluator's score is a *distance*: zero is a board with no term outside its band. One run narrowed a board, watched the score fall from 0.8 to 0, and read that as worse.

The second is more expensive and is the most instructive single episode in the corpus. An agent read "kind": "hard" on two scoring terms as meaning *this will not build*, and rebuilt a board at twice its size to satisfy them. Nothing refuses on those terms. What followed was not one mistake but a cascade: doubling the cell doubled the spawn protection region past its cap, which split the corner into three, which took the wool room to 20×20 with an 18×18 cage, which needed four shelf pieces to hide the resulting plinth, which produced 1,236 blocks of dead-level pad per quarter and 4,944 on the board, or 21% of its ground. Driven afterwards at the original scale, the board pre-flighted clean and exported at 0.0% dead against 1.1% doubled.

The agent's own verdict on the episode is the part worth quoting, because it names a failure mode specific to systems that write their own documentation:

And I did not merely make the mistake, I canonised it. *A cell is a scale and an arrangement is a design* went into the review and into the README as though it were a principle. It is a sentence I invented to justify a change.

15.6 Missing, Unreachable, Mistaken

The discipline that has done most to keep the corpus honest is a three-way verdict that every recent report applies to anything it could not do.

Missing

There is no mechanism. The system genuinely cannot do this.

Unreachable

The mechanism exists and the surface hid it.

Mistaken

The mechanism exists, is documented, and the agent did not find it.

Mistaken is the usual verdict, and the reason is a property of how surfaces are indexed:

Search the surface by what a thing would *do*, not by what you would call it. mistaken is the usual verdict because the surface is indexed by name and a capability is wanted by function, so one grep of the obvious word answers nothing and reads as proof.

Two gaps filed in a single run illustrate it exactly. Ground cover was written up as absent; the mechanism exists under the name *flora*, and the word was never searched because the question had been phrased as grass coverage. An intra-team build zone was written up as unmodelled; it is on the wire three times over, under a transient-link clause, a defender-egress bridge and a scored term with a stated band. None of the three contains the word *zone*, and the search was for *zone*.

The judgement that follows is the strongest argument in the corpus for the verdict discipline:

A gap filed against a surface that has the feature is worse than no gap. It is a capability the next run also will not use, and nothing will ever contradict it.

The ratio is worth recording as a number. Of nine things one run could not do, five were the schema being right and the agent reading it wrong, three were a deliberate rule saying no and saying why, and one was the surface genuinely unable to answer.

15.7 Placement as Query, Restated

Chapter 3 introduced the inversion; this is its cost curve. Across three separate runs, the count of declined props on successive dressing passes runs seven, four, two, zero; six, five, three, none; and five, five, five, six. In every case the pass that moved the number to zero was the one where every position came off the studio's own seats mask rather than off the agent's judgement.

An agent that places by eye and corrects converges slowly or not at all. An agent that asks which seats are legal and chooses among them is finished in one pass. This is the clearest instance in the system of the general principle that the studio knows things the agent cannot see, and the productive move is to ask rather than to guess and be corrected.

15.8 The Corrections That Reads Bought

It is worth ending on the positive form, because the reads are not merely a check. Several of the largest design improvements in the corpus came from a measurement contradicting a belief, and the change that followed was small.

Moving one wool room from a ring's near corner to its far one took the board's dead ground from 25% to 5%, with nothing else on the board moving. Widening the middle third of a bar, having found that widening the whole corridor made things worse, took unreached ground from 9% to zero. A first cut of the slope bands at 24 degrees put an entire hill into one band and produced a board that came out grey with green only at its edges; the incline reading said where the bands actually fell, and a recut at 32 fixed it. On another board the same reading answered that 77% of cells were under ten degrees and 0.5% above twenty, which is to say that the board the agent had planned to finish by angle had almost no angle in it.

None of those were caught by a gate. All of them were caught by asking the studio a question whose answer the agent could not have computed.

Glossary

Terms are defined where they are first used in the text. This collects them, so that a reader who begins at Part III can follow it without the middle.

Abutment

The interval where two footprints genuinely touch, obtained by intersecting the two rectangles. Never a point and never a node.

Agent

In this paper, the operator driving the studio's HTTP API, which for the maps described here is a language model in a terminal.

Band (paint)

One entry in a band stack: a material and a thickness. What the thickness counts depends on the stack's axis, so on the slope axis it is a span of degrees rather than a count of blocks.

Band (rule)

A descriptive envelope measured off the corpus, against which a scored term reports a distance. Contrast *target*.

Band stack

An ordered run of bands plus the axis it is read along, and a rule for what happens past the last band.

Bucket

One of the four parts a painted column splits into: rim, wall, surface and fill.

Box

A typed bounding envelope the composer's allocator places, carrying a footprint and a land target. Five kinds: spawn, hub, wools, frontline, mid. Boxes exist only during composition.

CTW

Capture the Wool. Two or four teams; each team's objective is capturing the others' wools.

Cell

The plan's coarse grid scale, four blocks by default.

Complaint

A finding on a successful request that remarks on something without saying anything was dropped. Rides on warnings.

Composer

The studio's board arranger. It emits a plan, not a layout, and builds two-team boards only.

Decline

A finding on a successful request saying that a piece of what was posted is not in the world. Rides on warnings and is distinguished by severity.

Descriptor

A composed board's identity: players, teams, symmetry, cell, seed, composer version and schema. Reproduces the board rather than storing it.

Destroyable

The DTM objective's element and the studio's name for it. Never called a monument in code, because that word names the CTW wool-monument.

DTM / DTC

Destroy the Monument; Destroy the Core. Two-team in practice, because the objective has one owner and $N - 1$ attackers and nothing determines a winner beyond two.

Face

In the plan model, the front-most row of a unit's terrain and the only part the mid band docks. In the paint model, the exposed side where a shape's ground drops.

Fill

The bucket claiming whatever no other bucket took. Required, so a theme is never partial.

Finding

A rule id, a sentence carrying the measured numbers, and what it is about. Optionally an edit stating a mechanical fix.

Footprint

Total box area, meaning terrain plus build zone plus gap. A build zone costs footprint but not land.

Grant

The corridor width a consumer was given across an abutment. A selection, where an *offer* is a capacity.

Hub

The designation with no terminal, placed first, taking what the budget leaves and never under a third of it. Publishes the width each free run of its edge can carry.

Intent

The third document: teams, spawns, protections, build regions and objectives. The one about the match rather than the ground.

Island

The union, over land interfaces, of the terrain pieces that touch.

Land

Walkable terrain area, set by the player count and spent by every emitted piece.

Layer

A slab holding exactly one span per column, where a taller add replaces a shorter one outright. A stack of layers is a set of height fields with air between them.

Made layer

A layer out of the stacking rules: painted over its own span and exempt from the pair and reachability walks.

Mark

A patch of a footprint held at a chosen height, which a relief solves between. Five kinds: point, line, area, rim, scarp.

Mid band

The neutral crossing between the two fanned halves. Laterally the hull of the opposing front faces; in depth, the gap.

Offer

A run published with the width it can carry. Bounds a search rather than filtering its output.

Orbit

The set of images a symmetry mode fans an authored unit into.

Plan

The first document: rectangles on a cell grid, what each is for, and where the objectives sit.

Push

A relief instrument raising a region by an amount over a falloff, with a crown. Its two gradients must agree or it steps at its own outline.

Refusal

A finding that stopped the request. Arrives in the error envelope; nothing rode along with it.

Relief

The interior half of the height model: a field authored out of marks and solved into a surface between them.

relief_scope

What a relief does to a shape that has a height. Absent, the shape joins the solve; hold pins it and the ground is solved to meet it; exclude makes it a hole the relief bends around.

Rim

The edge lip: a single block at the top of an edge column.

Run

One contiguous interval of real material on one edge, measured from that edge's origin. Not an edge and not a side.

Seat

The single integer offset at which a neighbour's extent begins along the hub's edge. Because the search runs in one integer, nothing is built and then moved.

Size band

One of four named player ranges a board is built for: nano 6–13, micro 14–21, milli 22–31, centi 32 and up. Two counts in one band compose to the same budget.

Sketch layout

The second document: real geometry at block resolution, being outlines, heights, relief, paint and props.

Slug

The identifier a map is stored under. Loading the same documents twice under one slug is a reload, not a second map.

Stamp

A structure written onto terrain, as a stack of courses per part rather than as a fixed building.

Strait

The direct crossing between two team islands on a two-team wool board, bounded at 15 to 40 blocks.

Target

A prescriptive, per-request constraint a compose holds and verifies. Contrast *band*.

Term

A scored measurable. Hard terms are an acceptance gate; soft terms score distance from a band.

Theme

A named set of materials per bucket, applied to ground.

Unit

One team's composed half. The board is that unit repeated through its symmetry.

Wool lane

The corridor a wool room owns, found by stacking a fixed-width band outward from where the room meets terrain until it reaches void, build or a crossbar. Only wools stack a lane.

The rule catalogue

Every rule the studio can raise as a finding, generated from `GET /api/rules` on the running studio rather than transcribed. One table per family, families in descending size. A rule's *means* is abridged to its opening claim where the full text runs long; the whole of it is on the wire, and `GET /api/rules?rule=<id>` answers one.

Family SK — 28 rules

Rule	What it means	Concerns
SK1	A relief the recompile does not keep. Either a group the author had drawn relief onto no longer exists to carry it, because the board fused differently — or the merge carried the stored relief over one the posted body states for the same group, so the board builds the terrain it already had.	plan, terrain
SK2	The board spans more ground than the studio will realize: the extent every shape and its symmetry images cover, measured in columns, is past what a build can walk. Refused before anything is built, because the cost is paid per column of the extent whether or not ground is drawn there — a board that large does not fail, it takes the machine with it.	plan, studio
SK3	The document names something that is not there — a shape kind nobody has, a mirror mode nobody has, a group listing a shape id the layout does not carry, a relief keyed to a group that does not exist. A complaint rather than a refusal: the board still builds, and it builds without whatever the name was for, which is the thing worth saying out loud.	plan, terrain, theme
SK4	A shape draws no ground: a polygon or lasso with fewer than three vertices, a circle or path of no width, a rectangle with no area. The shape is in the document and contributes nothing to the board, which reads exactly like a shape that was never drawn.	plan, terrain
SK5	A shape states a column the world cannot hold: a floor below bedrock, a top above the world roof, or a thickness that is negative. The build clamps it into the world rather than refusing, so what stands is not what the document asked for.	plan, terrain, world
SK6	Nothing has been stored to finish. The map is at the sketch stage and no layout has been written for it, so there is no document to rasterize into world geometry. 422.	request, plan
SK7	The stored layout rasterizes to no ground at all. Every shape in it draws nothing — an empty document, or one whose shapes are all of the kinds SK3 and SK4 report — so finishing it would write a world with no land in it. 422, and the one place the sketch's complaints become fatal: finishing is what declares the drawing done.	plan, terrain
SK8	A board is finished carrying no finish: no theme registry, no relief and no props. Ground alone is a legitimate board — a test piece, a shape being tried — so this is a complaint and never a refusal. It exists because it is the one silence this stage kept: SK3 names a shape citing a theme the layout does not carry, SK4 a shape drawing nothing and SK7 a layout rasterizing to no ground, and all three need something stated to disagree with. ...	terrain, world
SK9	Two shapes on one layer stack over the same ground, and the lower one is not in the world. A layer is a slab: it holds one span per column, and a taller add replaces a shorter one outright, floor included. So a floor with a roof drawn over it on the same layer builds as the roof alone, over open air, and reads as a board the author never drew.	terrain
SK10	Two layers are driven into each other by more than the one course a stack shares at its seam. A layer is a slab and the stack is what puts air between two of them, so where their spans meet they build as one solid mass: the gap the layers were drawn to have is not in the world there, and nothing under the upper slab can be stood in.	terrain

continued

Family SK, continued

Rule	What it means	Concerns
SK11	A mass of standable ground under open sky that no route reaches from the rest of the board. Ground under a roof is a room and says nothing; ground with sky over it and no way onto it is either a second landmass the author meant or an upper level whose stair was never drawn, and only the author knows which.	terrain, world
SK12	Two groups answering to the same id. A group id is the key a relief is stored under and the handle a placement names, so a board carrying it twice has no single answer to either — and which way it goes wrong depends on where the second one is. On one layer the last group solved takes the terrain over its own cells and the ones before it build flat....	terrain
SK13	An add covers ground a subtract takes away. A subtract is how a board states its negative space — the void a plan’s buffer pieces compile to, the hole a composed footprint leaves — and a hole is never scenery: what a body encircles is ground players go round, and a board’s walls are drawn to guard it....	terrain
SK14	An override add states a top its group’s relief will solve straight through. An override add says the column is its own, floor and all — it is what a wall, a flight of stairs, a crop bed or a stepped mound is drawn as — and a relief replaces the top of every column of its group. Only a shape naming a height_mode stands out of that field, and only a relief_scope keeps its ground out of the solve, so a made thing carrying neither is built to whatever the relief says and the author’s number is nowhere in the world....	terrain
SK15	A shape states a theme over ground another shape stands taller on, so the theme is on none of it. Two override adds over one column is not a fault — the taller wins it, which is what “the tallest add is the height” means — and paint follows the shape that forms the surface, so the taller shape’s own theme is what shows and the shorter one’s is nowhere on the columns the two share....	terrain, theme
SK16	A made thing asks to be seated on the ground and has none under it. A layer that seats takes its floors from the lowest solid column of its own footprint, so a thing whose footprint covers no ground at all has nothing to measure against and stays at the height it was drawn. A complaint: a sculpture hanging in open sky is a legitimate board — a balloon, a ship in the air, a thing on a spire — and the word for that is simply to state no seat....	terrain
SK17	A shape no group on its layer lists. A group is the unit the symmetry orbit is fanned by — the build reads each mirroring group’s shapelds and copies those shapes onto their images — so a shape no list names is drawn once, on the side it was drawn on, and has no image anywhere. Nothing else catches it: the shape is in the document, it rasterizes where the author put it, the drawn mirror outline still covers it because a group’s outline is the union of the ground it fused and not the shapes it lists, and half a landmass is missing only in the world....	terrain
SK18	A made thing and a built thing holding the same courses. A made thing — a layer stating kind: “made” — is laid by the rasterizer, before anything is stamped, and every stamper writes where it is told: a wool cage, a spawn cube, an objective and a dressing-placed building all seat on the terrain’s surface, which is every column’s top with the made things taken out....	structure, feature
SK19	A placement naming a recipe the document does not state. A tree, a boulder and a building each carry a key into the layout’s own dressing.styles registry — what is placed is a position, what stands there is a recipe named once — and a key the registry has no entry for names nothing at all. Every read of the dressing refuses it already, naming the placement and the key, so the world is never built from one....	feature
SK20	The layers of a stack are not in the order their ground stands in — a later layer starts below an earlier one. A layer’s position in the list is its draw order and base_y is its height, so the two say different things and a document where they disagree reads as a stack that is not the one it builds: a reader walking the list top to bottom meets the storeys out of order, and a strip drawn from it puts the cellar above the roof....	terrain
SK21	A bend that could not pull every point it inserted. A coast is drawn by resampling an outline’s long edges and pulling each inserted point to the side the bend asks for, and a point whose two offsets both land on the wrong side — a neck or a notch narrower than twice the wander — stays on the edge it was cut from....	terrain

continued

Family SK, continued

Rule	What it means	Concerns
SK22	A shape stating a height per vertex that its kind has no reader for, so the world builds one uniform thickness and the author's numbers are nowhere in it. <code>anchor_heights</code> is interpolated across a footprint as a TIN over the shape's own ring, which only a polygon or a lasso has: a rectangle and a circle state bounds rather than vertices, and a polyline's vertices are its centreline rather than its footprint — the band around them carries many more points, and a thickness graded along one would have to interpolate along the arc instead. ...	terrain, world
SK23	A theme scoped to a shape that has no interior column, so the only buckets that ever paint it are the rim and the wall and the theme's own surface is nowhere on the board. A theme is a recipe for ground: which of its five buckets a block takes is decided per column by whether that column is an edge, and every column of a shape whose whole footprint touches the void is an edge under every <code>rimEdges</code> setting there is. ...	terrain, world
SK24	A shape stating both a theme and a material. The two answer one question — what paints this shape — at two grains, and a document holding both says nothing about which was meant: the build takes the material, because it is the narrower statement, and the theme the author also wrote is read by nothing.	terrain
SK25	A polyline whose band crosses itself. The two offset edges are stored as one outline and an outline is filled even-odd, so where a centreline winds back beside itself the two windings cancel and the lap comes back as void: a spiral drawn as one stroke builds with a gap between every turn, and a hairpin tighter than the band is wide builds with a hole in its elbow.	terrain
SK26	A tilted shape whose climb ends at a drop. A flight is one shape at any gradient, so nothing in the document says where it arrives — and a flight that tops out level with the ground beside it for one cell and then falls is walkable by every measure taken of its treads and is not a way up anything. Read at both ends, in the shape's own direction of slope: the high end wants somewhere to arrive and the low end wants somewhere to step on from. ...	terrain
SK27	One landform painted a theme per step. A plan component spanning several surfaces compiles to one shape per surface — a stepped island becomes stacked plateaus, each addressable by its own id — and a theme scoped per plateau paints one hillside as two or three grounds with a hard line at every riser. ...	terrain, theme
SK28	A group that declines the fan while standing wholly inside one orbit image. The orbit is fanned per group, so a group stating mirrors: false is built once, where it was drawn — which is right for a landmark on the symmetry centre, because such a thing is already its own image, and wrong for anything a team owns. ...	terrain

Family DR — 17 rules

Rule	What it means	Concerns
DR-BANK	A body of water that dug a shaft rather than filled a hollow. Its line is one plane across the whole run — by default the lowest surface it crosses — and every bed column standing above that line is emptied down to it. <code>depth</code> bounds how far below the line the bed goes and nothing bounds how far above it the carve reaches, so a pond drawn across a slope comes out as a straight-sided pit as deep as the ground falls, whatever depth was asked for. ...	feature, terrain, world
DR-CLAIM	A prop rests on ground something already standing has claimed — a channel, a road, a building or an earlier prop. The pass places in priority order and the first claimant keeps the cell, so this is the collision itself rather than a near miss; the finding names the cell and what holds it. A building holds the ground it stamps <code>andStructureClearance</code> blocks of ring beyond it, so a prop seating under an eave is this fault rather than a silent build. ...	feature, terrain
DR-CROSS	A building stands across a route: the road it covers carries on out the other side, so what was one way through the board is now two dead ends at a wall. A road that simply ends at the building is a porch and stands — that is what a road running to a door is — and the two are told apart by what is left of the stroke once the footprint is out of it: one run of paving is an end, two or more is a crossing. ...	structure, feature, terrain

continued

Family DR, continued

Rule	What it means	Concerns
DR-CUT	A prop the world cut down. A prop seats on its feet — the cells of its lowest course are what the keep-outs, the claims and the ground are asked about — and everything it reaches over is written only where it meets air. So a prop that stands clear of a building may still have most of its body inside one, and the blocks that land are whatever the wall left: a rock flattened along a face, or a crown cut off from its own trunk and standing in the air on the far side. . . .	feature, terrain
DR-DOC	The rule this refusal carries into the export gate, so a caller can act on the id rather than parse the sentence.	request, feature
DR-DRY	Water standing against a hole in its own basin. A pool fills the bed it carves, and the hollow it sits in is very often dug by something else — a relief mark, a shape's own floor — so where that hollow reaches further than the bed does, the extra is excavated and never filled: a trench as deep as the water is, running alongside it, with the water standing against open air and nothing said. . . .	feature, terrain, world
DR-FACE	A block of a copied body naming a face it has nothing to cling to. A copied body is written block for block with the data it was cut with — that is what makes it a copy — so a block whose data is a direction has to point at something the body actually holds. A vine states every side it clings to at once; a side naming air is a curtain hanging on nothing, and a pair of opposite sides is that same fault seen from the front, as a vine with two faces in one block. . . .	feature, material
DR-KEEP	A prop rests on a column the map keeps clear of everything: a spawn point or its room, a wool room or its monument, a structure the plan stated, a column whose surface is built rather than terrain, or the lane in front of a spawn or wool-room door. The finding names which of those it was and the cell it happened at.	feature, spawn, objective
DR-LAYER	A prop names a layer this board does not have ground on. A stacked board carries a surface per storey and a prop may say which one it rests on; naming one that is not there is not the same as naming none, so it is declined rather than quietly seated on the top surface — which is exactly the storey the author was saying they did not mean.	feature, terrain
DR-PASS	A building leaves no way past itself: a side of it carries fewer than 8 blocks of passable ground alongside its whole run (the author's number). Every side is asked, because the lane a building stands in is the ground players arrive on rather than a way round it — a house in the middle of a route corks it however far the route runs on ahead. . . .	structure, feature, terrain
DR-ROAD	A prop rests nearer to the road than its kind's standoff allows: a tree 3 blocks, a boulder 2, measured from its resting cells to the nearest paved cell (Chebyshev; exactly-at-distance stands). The numbers are each kind's own RouteStandoff — the author's ruling, stated on the type so there is exactly one place they live.	feature, terrain
DR-SITE	A prop has no ground to rest on: one of the cells it rests at is off the map's terrain altogether, so there is no column to seat it in. A building is held to every cell of its footprint — it seats on its lowest column, so one cell on land and the rest over void builds a house hanging off a corner — and the finding names the first bare column it stopped at.	feature, terrain
DR-SIZE	A building's box is smaller than 5×5 blocks: a footprint four blocks deep is a wall with a roof, not a building anyone enters, and a corpus run produced eight of fourteen that way. Measured on the plan's bounding box, so a multi-wing building is judged as the one building it is.	structure
DR-SLOPE	A building's own footprint is not level enough to stand on: the ground across it rises by at least the height of the building itself — its wall courses plus the rise of its roof. A building seats on the lowest column of its plan and the terrain standing over that floor is carved out of it, so a footprint spanning more relief than the building is tall comes out with its uphill wall entirely below the ground beside it — a house hidden in a hill rather than one dug into a slope. . . .	structure, feature, terrain
DR-STEEP	A boulder standing on a face rather than on ground. An erratic is a mass left where the ice dropped it, which is ground flat enough to hold one; a rock pinned to a steep hillside reads as neither — the slope is already the feature there, and the rock sitting on it only interrupts a line that was doing the work. . . .	feature, terrain, world
DR-TONE	A boulder built out of nothing but the tones of the ground it stands on. A rock is an erratic — a mass carried here and left — so it reads as a rock by not being made of the field it sits in, and one whose every tone family the ground already states has no silhouette at any size: it is a patch of the same ground standing up. . . .	feature, material, terrain

continued

Family DR, continued

Rule	What it means	Concerns
DR-WAY	A prop closes the way between two of the places the map is played between — its spawns, its monuments, its goals — or sends that way further round than a player will go. Measured by walking the terrain between every pair of waypoints and walking it again with the prop's footprint taken out of the ground: a pair that had a route and now has none is a way closed, and a route surviving more than ten blocks longer is the same fault at a lesser degree, ten being how far out of their way a player goes. ...	structure, feature, terrain

Family PL — 16 rules

Rule	What it means	Concerns
PL1	No generating piece: there is no land, so there is nothing to build.	plan
PL2	No spawn: PGM has nowhere to put a player and the map cannot be entered.	plan, spawn
PL3	The plan states no objective of any kind — a complaint, since which goal a map carries is the author's. All four families count: a wool, a destroyable, a core, and the capture points a board states a count of. It is a statement about the plan and says nothing about the match. A board can state its goals downstream instead, on the intent the configure tool and the API write, and a plan-tier rule cannot see that — so what this reports is a plan with nothing in it to win on, not a map that cannot be won. ...	plan, objective
PL4	Two pieces claim the same ground at incompatible heights, so there is no coherent surface.	plan
PL5	A placement names a piece the plan does not have.	plan
PL6	A placement stands on a buffer, which is reserved empty space and produces no terrain.	plan
PL7	A placement falls outside the piece it names.	plan
PL8	A spawn room cannot seat every monument its team will capture.	plan, spawn, objective, structure
PL9	A wool cannot be reached from a capturing team's spawn at all.	plan, objective, spawn
PL10	A destroyable style names something that is not a style.	plan, objective, style
PL11	A wall is drawn on a pair of pieces that share no land interface.	plan, structure
PL12	A connected landmass mixes fanned and non-fanned pieces, so it has no coherent orbit image.	plan
PL13	A bedrock wall is drawn on the wool room's own interface, so the wall and the room stand through each other and the room can barely be entered. The wool's own edge is never a wall seat.	plan, structure, objective
PL14	A wool colour names something that is not a dye.	plan, objective
PL15	The document states a shape version this build does not read. A marker's at is an offset in blocks from its piece's minimum corner, and an earlier version stated the same field in cells — the same numbers, a different distance — so a document that names another version is refused rather than read under the wrong unit.	plan
PL16	A capture-point count the board's own symmetry cannot lay out. A point belongs to nobody, so it has to be the same walk for every team, and the only positions that are lie on the board's axes: the centre of symmetry, which is one point, and a ring the orbit fans from a single side point, which is as many as the orbit has images. ...	plan, objective

Family wx — 13 rules

Rule	What it means	Concerns
WX1	The shell's footprint is the piece rect inset one block on every side. The ring of clean floor around a room is part of what a piece promises, so a 10×10 piece carries an 8×8 shell and a 10×20 piece an 8×18 one; the shell takes the rect's own orientation, and the fanned rect orients the orbit images.	plan, structure
WX2	The footprint cannot hold a room of the least legal span. A wall is what the interior is inset by and what the pad keeps its clearance to, so the span a shell needs is four blocks more on each axis than the span a pad and its chest corners need on open ground.	plan, structure
WX3	The marker's block-lattice parity differs between axes, and the pad is always square.	plan, structure, spawn
WX4	The pad keeps at least one block of clear floor to every wall. A marker sitting too close has its pad shifted inward by the minimum that restores the clearance — the exported point moves with it — and a 3×3 is chosen only where it still fits after that shift. An interior with no room for the pad even shifted is refused.	plan, structure, spawn
WX5	The exported spawn or wool location is the pad's centre, after any WX4 shift. The world is the ground truth the map document has to agree with, so the point follows the pad rather than the marker that asked for it, and it snaps to the half-block lattice rather than to an integer.	structure, spawn, objective, world
WX6	A wool room with no seam and no abutting build zone has nothing to enter it by.	plan, structure, objective
WX7	A door's width follows the wall it is cut into. An odd interior wall centres a 3-wide door; an even wall takes 4 once the interior is at least 6 across and narrows to 2 at the 4-across minimum. The invariant under the numbers is that a door is at least one block narrower than the interior on each side, so the corner cells a spawn cube seats monuments in are never opened to the outside.	structure
WX8	An iron cube is IronSpan blocks square and stands outside the room shell, inside the piece, holding IronGap blocks of clear air to the wall — the standing room a player has to get round it. It fits where its marker puts it or it does not: the room keeps the footprint WX1 gave it and never yields an edge, and the cube is one size whatever the marker's parity. ...	plan, structure, spawn
WX9	Every structure marker resolves to placeable or not, and an unplaceable one is not an error the export throws on: it stamps nothing, the room takes its full WX1 footprint, and the marker stays on the board where the author put it. Validation flags it and the structure preview draws only what will be placed, so the iso view never shows a cube the export declines.	structure, world
WX10	A bound room style builds a shell taller than the build ceiling — BuildCeiling's clearance over the ground it stands on. A room's shell is authored geometry and is subject to no cap of its own, so a tall storey stack swallows the goal marker that hangs MarkerOver blocks over that ceiling, and the map's own sky sign ends up inside the building it points at. ...	style, structure, objective
WX11	A stamped structure stands well above the ground beside it. Its foundation fills the column under its whole footprint and levels it at the footprint's highest, so where a neighbouring cell sits well below that floor what the building meets the world with is a sheer face of bedrock — a wall nobody drew, at a height nobody chose, which a player cannot climb and no other read reports. ...	structure, world, terrain
WX12	The footprint reaches outside the piece it stands on. A piece is one rectangle at one surface, so a footprint inside it is on ground by construction and crosses no interface; one that reaches past it is over whatever the neighbour happens to be, or over the void.	plan, structure
WX13	The room a region raises is larger than a room. A shell's footprint is the region inset one block on every side wherever the document states no footprint of its own, so an oversized region does not overflow anything — it is the building, and the hall comes out as long as the region is. Read off the resolved frame rather than off a plan piece, so a hand-authored intent meets the same cap a compiled one does. ...	structure, spawn

Family 0B — 12 rules

Rule	What it means	Concerns
OB14	A destroyable and a core are authored at orbit order 2 only — one team's to defend and every other team's to break, which only means anything with two teams.	objective, intent
OB17	The three places a goal may not stand: over the void, inside a spawn room, or inside a wool room — everywhere the map's own rules would leave its blocks unbreakable.	objective, structure, terrain
OB19	A tree, boulder or building stands inside a goal's clearance: the ground its structure covers grown by four blocks, and never nearer than ten blocks to the marker itself (the author's numbers). The prop is declined — a goal is what the map is for, and a prop is removable, so the map is built without it rather than refused for it.	objective, feature
OB20	A declared <gamemode> is outside PGM's own closed enum, so the map fails to load however clean everything else is.	intent, world
OB22	A goal is authored floating further over the ground than a goal may float. The stated float is a distance from whatever the relief leaves under the column, and without a maximum it puts a goal anywhere — including where reaching it is a build project rather than a raid.	objective
OB23	A goal's own structure reaches above the height players may build to — the build ceiling over the terrain and the buildings the map actually builds, which a goal is deliberately not counted among. The blocks above it can still be broken, so the map is not unwinnable; a goal standing over the line players may reach by building is a goal contested from nowhere.	objective, world
OB24	Two goals are built into the same blocks. A destroyable stamped inside a core's casing is one structure serving two objectives: breaking either is breaking the other, and the blocks a match is played for belong to whichever stamper ran last. The usual cause is the symmetry rather than the author's hand — a goal drawn at one position and a second at the position the orbit maps the first onto, so each lands on the other's image and nothing in the plan looks wrong. ...	objective, world
OB25	A wool's monument was authored somewhere and the world carries it somewhere else. The capturing team's spawn structure stamps the monument block it is won on, so the exported location is the air cell that stamp produced and a stated one is not in the world — three write paths take a location, every read answers it, and the export replaces it.	objective, world
OB26	A destroy map with no way to end. A monument or a core is obsidian because obsidian reads as a goal — opaque, slow, unmistakable — and that same slowness is what lets the defending team hold one for the whole match: the attacker's pick is what drops the obsidian the defender rebuilds with, and PGM lets the owner repair unless the map says otherwise. ...	objective, intent
OB27	A capture point that ends the match the moment somebody takes it. required is a Goal's attribute and PGM defaults it to true at proto 1.4.0 and above, which is every map the studio reads, so a point that leaves it off is a goal a competitor completes by standing on it — and GoalsVictoryCondition ends the match the instant one competitor holds all of its required goals. ...	objective
OB28	A capture point that pays into no score. ControlPoint.tickScore looks up ScoreMatchModule every tick, and PGM builds one only for a document carrying a <score> element — so a point naming a points rate or an owner-points bonus on a map that declares no score pays nothing at all, for the whole match, with no error anywhere. ...	objective
OB29	A capture point built out of blocks PGM will not recolour. A point's display regions are filtered through ColorUtils's colour-affected set — wool, carpet, stained clay, stained glass and its pane, a banner — and a block outside it is left exactly as it was found. Colour is the map's only signal that a point was captured, so a pad of hardened clay, stone or planks parses, exports, loads and shows nothing at all. ...	objective, material

Family CT — 10 rules

Rule	What it means	Concerns
BZ5	[expert, retired as a prohibition] Build zones may touch spawn pieces — a zone at the spawn is a real motif: the defender-egress bridge (four-team-wool-two-sided : the spawn's second exit is a bridge mainly for defenders rotating to their wool; attackers push the other crossings). No proximity rule; the old lint is dropped. ...	—
BZ6	[author] The mid build region never interfaces a wool piece — and never contains a wool position. Otherwise players bridge freely through the mid straight into the point and there is no gameplay direction. A zone touching a wool room exists only as the <i>deliberate</i> WL8 two-approach variant (BZ5's wool note), never as mid-band spillover.	—
BZ9	[author] Fit — a zone is bounded by the ground it docks, not by a width. What it may not do is overhang : carry a stretch beyond the last ground it meets, which connects nothing and reads as no man's land a player walks into and stops. Oversized mid regions spanning the whole board width "AND more" are that fault at its limit. ...	—
BZ11	[author] One zone for a compact middle. Several zones merging into one region whose union is itself a plain rectangle are a stitch one zone would have drawn — the author reads one crossing, the player a patchwork (the four-zones-over-a-30-block-middle fault). An L/T-shaped union is not stitching : rectangles are the only shape a zone comes in, so a region that turns a corner needs several. ...	—
CT1	the mid is an interface, not a cut. There is no team-separation cut — the symmetry axis already plays that part. What the author shapes as "the middle" is the interface between the two team territories , and that interface always connects through bridge zones. Its physical forms, author-labeled across the corpus: • Clean — one connected build region holding 0..n mid islands ...	—
CT4	the island-size gradient . Measured over the 90 fanned islands of the ten-seed corpus (<code>docs/generator/seed-stats.md</code> , "Island gradient sweep"): islands grow with distance from the centre — pooled Spearman(area, centroid-distance) +0.61 , holds per-seed in 8/10 . Stepping-stone candidates are islands fully submerged in a build zone, or small (≤ 100 -block) islands with exactly two ...	—
CT5	carve the mid, cut the team side. "Cut" is the wrong picture for the middle — it can hold many islands; the mid operator is carving : shaping the interface's islands and zones directly into one of CT1's forms. Cutting belongs to the team side : severing a piece from its parent isolates it behind a bridge — the isolated wool (WL4), the isolated spawn (SP6) — and, deliberate variants aside, each team side stays internally land-traversable after cutting. ...	—
CT8	internal holes are the rotation device [corpus, author; amendment 2026-07-04]. The closure encloses internal void pockets — holes — in 12/12 seeds (2–13 per fanned board, 4–72 cells... sizes in proxy cells; always in symmetric orbit multiples). "These holes are what enables player rotation" (author): a loop around a hole gives alternative routes between lanes without retreating through a single chokepoint. ...	—
CT9	[author] The frontline rotation hole. A split frontline + hub + band encloses a rotation hole in the tip-gap: the hub caps it above, the band (or a bridge) below, and a buffer gap between band/bridge and the hub is the void (mirror-mid ex -6 hole -6 between the tips under hub -6 ; ex -8 bridge -8b splits the gap into hole -8a/8b ; ex -11 hole -11). ...	—
CT12	[author] The CTW strait is 15–40 blocks. On a two-team wool board, the direct crossing between the two team islands — a build region carrying the pair with no third landmass in it — spans 15–40 blocks (the walk over the empty cells). Closer is two halves all but merged (the flush-fan fault produced literally one landmass); farther is a crossing nobody bridges. ...	—

Family HS — 10 rules

Rule	What it means	Concerns
HS1	A block named for a geometric role — which way a stair climbs, which half a slab fills — is not that kind of block: <code>doorHead.block</code> , its <code>fillBlock</code> under <code>upperSlab</code> , a window's block under <code>stairLattice</code> , arched or <code>slabBanded</code> , or <code>roofSlab</code> itself.	style, material

continued

Family HS, continued

Rule	What it means	Concerns
HS2	A doorway does not clear the least height a door may, once its head is written into the top course.	style, structure
HS3	A roof is not one material. Its body and its verge are each a single block — never a pattern, so nothing spreads a voronoi across a roof — and the half-course slab continues the body in the body's own material. A bare log or a ground material is not a roof material at all, and a slab named as the whole-block body with no half-course companion builds a roof with a gap in every course....	style, material
HS4	A part built of two blocks is built of two materials — a door head whose stair and whose slab fill are cut from different stone, a window whose block and whose host disagree. The two blocks are one line of the building and read as one thing or as a mistake.	style, material
HS5	An ore is used as a building material. An ore is stone with something in it — it belongs in the ground a map is dug out of, and in a wall, a post or a beam it reads as a mistake rather than as a material.	style, material
HS6	A door is cut through a wall that is not there. A storey whose wall is air over the doorway's own courses — a house on stilts, an open undercroft — has nothing to cut, so a doorway and its head are a lintel standing in mid-air.	style, world
HS7	A footing round a plate one course deep. The footing is what a foundation stands on, so over a plate that is a single course it is a one-block rim round a building with no foundation under it — noise at every wall, on every house, and the commonest thing wrong with how a village reads.	style, world
HS8	A porch whose canopy climbs past the wall it is attached to. A canopy is seated by its own lowest course clearing the doorway it fronts — which is what keeps the way out of that door walkable, and is the reason it is not seated under the eave, where on a tower it would ride the wall up and leave a colonnade open to the sky — and its ridge then follows by however far the form happens to rise....	style, structure
HS9	Beams over a wall that is carrying nothing. A beam end is the end of a floor timber, left long the way a log building leaves them — so the course it comes out of has to be that timber, which is a laid log running along the wall. Over a course of brick or clay the ends are eight logs sticking out of masonry with nothing behind them, which is not a detail but a mistake about how the building is put together....	style, structure, material
HS10	A house on stilts standing on a floor. The point of a stilt storey is that the ground runs on underneath it — a bank, a mire, a shore — and a plate laid across the footprint puts a plank rectangle on that ground and stops it. Nothing is wrong with the building; what is wrong is that the ground it was raised to leave alone has a lid on it.	style, world, terrain

Family WL — 8 rules

Rule	What it means	Concerns
WL1	At the far/back end of a dead-end lane, inset ~5. Two-sided wools are authored twice (four-team-wool-two-sided : two stepped land seams into the room; mirror-big-board).	—
WL2	On a different lane than the spawn; $\text{wool} \leftrightarrow \text{spawn} \geq 20$ — all 17 corpus pairs pass, tightest 22.4 (the base seeds), typical 36–58, up to 101 on the big board.	—
WL7	Separation between a team's wools, measured over 8 multi-wool pairs: 46–143 blocks (46.1 / 58.3 / 64 / 70 / 75 / 85.6 / 95.5 / 143). Working minimum ≈ 45 . That sweep was taken with a cardinal walk; the same routes cost up to a quarter less under the octile walk the preamble states, so the band reads high by that much and the sweep behind it is not in the repository to re-run (amendment 21)....	—
WL8	[expert; amendment 2026-08-31 — the term is retired] Wool approach routes: the default is a single chokepoint route ; real maps sometimes add alternative routes to the wool (and then a build zone may touch the wool room — see BZ5). This is a description and no longer a scored term. The wool-ringed-hole term that carried it decided its verdict on piece names : a hole ringed around a wool passed only where the ring's ids shared the room's prefix, so south-rim beside room wool-south failed hard and wool-south-rim — the same rectangles — passed....	—

continued

Family WL, continued

Rule	What it means	Concerns
WL9	[author] Spawn↔wool balance . A team's wools sit comparably far from the spawn : the spread (max — min) of the per-wool spawn→wool traversal distances stays modest. A large spread means one wool is trivially defended (the spawn on its doorstep) while another is left to fend for itself. Bands learned from the teaching seeds (spawn-wool-spread)....	—
WL10	[author] The spawn-wool-frontline triangle . Two reads, both by surface traversal: (a) each wool keeps a real distance to the frontline edge — the seam where the mid build band meets the land, the line an attacker crosses (wool-front-distance , measured at the most exposed wool); (b) across a team's wools the triangle stays balanced : per wool, the <i>defence deficit</i> is its spawn distance minus its frontline distance, and the deficits' spread stays modest (wool-front-balance)....	—
WL11	[author] A wool room's approach steps by 1 level or takes a ramp : an entry seam at $\Delta \geq 2$ is un-walkable bare, the same reading SP8 takes of a spawn's egress and for the same reason (EL1's palette steps by 2). The player who crosses that seam is the attacker — a team is kept out of its own wool, so nobody defends through the door — which is what makes the step worse here than at a spawn: it is met at the end of the run that decides the map, either as a wall to build up or as a drop that cannot be climbed back out of....	—
WL12	[author] A bay or a hole beside a goal is at least 16 blocks across . Negative space is crossed by jumping long before it is crossed by building: a short gap between a frontline and a wool room, or between a spawn and a wool room, lets a player tower at the near edge and jump in, and the approach the board was drawn around stops being walked at all....	—

Family EX — 6 rules

Rule	What it means	Concerns
EX1	Some part of the map cannot be walked to from the rest of it.	world, terrain
EX2	Nobody can enter the map: it declares no spawn of any kind.	intent, spawn
EX3	What the intent declared is not in the document about to be written.	intent, world, studio
EX4	A map with something to win has nobody to win it: it declares an objective and no team.	intent, objective
EX5	The observer platform stands higher than the intent asked, because the board builds something over the point it named.	intent, world
EX6	The map names nobody who made it, so the observer platform's authors board has a heading and nothing under it.	intent, world

Family IM — 6 rules

Rule	What it means	Concerns
IM1	The url names a host the import does not fetch from. The allowlist is the studio's SSRF guard — an import fetches server-side, so an arbitrary host would make the studio a proxy into whatever its network can reach. 403.	request, studio
IM2	The host answered, and not with the archive: a 404, a 403, a 5xx. Nothing about the request is wrong, so it is reported as what it is — 502, the upstream's answer rather than the studio's.	request, studio
IM3	The archive states more bytes than the import will fetch. 413, before anything is downloaded.	request, studio
IM4	What arrived is not a zip: it does not begin with the header. 415, before extraction, so a login page served as HTML fails here rather than inside the unpacker.	request

continued

Family IM, continued

Rule	What it means	Concerns
IM5	There is no world in it. The import reads region/*.mca and nothing else, so an archive or folder without them carries nothing the studio can scan. 422.	request, world
IM6	The folder is a map already: it carries a map.xml. Importing originates a new map from a world, and one that already has a map document is a map to open rather than a world to originate from. 422.	request, world

Family RL — 6 rules

Rule	What it means	Concerns
RL1	The ground is not the landform the island says it is. The island states what it is meant to be — a plain, rolling ground, hills, a mountain — and the solved surface measures as something else, which is a board that got away from its author between the marks and the field.	terrain
RL2	The elevation is there and was never graded. Ground that rolls keeps more two-block scrambles than barriers; ground where every height change is a wall keeps the reverse, whatever its range — a quarry and a mountainside carry the same elevation and are not the same ground.	terrain
RL3	Two marks' ground meets on a step taller than a player can scramble. A mark pins every cell in its band exactly, so two placed to describe one slope describe a wall instead — and the wall reads back as terrain, a step and a face and a barrier cell, with no mark's name on any of it. Neither mark looks wrong on its own, which is why this has to be said rather than seen.	terrain
RL4	A mark pinned no cell at all. It was placed off its group's footprint, or over ground a shape took out of the solve, so the surface is the surface it would have been without it — and nothing else says so, since a mark that did nothing leaves nothing behind to notice.	terrain
RL5	The elevation was graded everywhere and left nowhere to stand. RL2's twin: where that one names ground whose every height change is a wall, this names ground that is all transition — under a third of it level, so a player crosses the whole board and can stop nowhere on it. It is an angle rather than a step, and that is why nothing else reports it: a surface graded end to end is walkable end to end, one connected place with no ledge, and the step histogram calls it perfect. ...	terrain
RL6	A push climbs at two different rates and the landform has a step at its own outline. A push rises at amount / falloff over its skirt, from the ground outside in to the drawn ring, and again at crown / deepest from that ring in to its medial axis — and where the two disagree what stands there is a cliff with a hill on top of it, whatever its height. ...	terrain

Family RQ — 6 rules

Rule	What it means	Concerns
RQ1	A posted document could not be read — absent, empty, malformed, or naming a kind that does not exist. The caller's to fix, so it answers 400.	request
RQ2	Not the caller's fault. Something escaped an endpoint that no gate refused, which is a defect in the studio; it answers 500 and stays a 500, because dressing a bug as a bad request sends the author looking for a mistake they did not make. What it buys is that the caller gets the one envelope every other refusal arrives in rather than a .NET stack trace, and the trace goes to the log where it belongs. ...	studio

continued

Family RQ, continued

Rule	What it means	Concerns
RQ3	A complaint, never a refusal. The document carried a property the reader had nowhere to keep — a typo, or a field from a document this is not. It rides on the success response under warnings because the work did succeed and the rest of the document was read; refusing it would refuse every snapshot written before the last shape change, since a stored document legitimately holds retired names an upgrade carries forward. ...	request
RQ4	The route names a subject the studio does not have — a slug no map is stored under, an id no library row carries, an artifact a map has not produced yet. It answers 404 with a body, because an empty one cannot say whether the identifier was wrong or the route was, and only one of those is something the caller can correct.	request
RQ5	The request is well-formed and conflicts with what is stored: a name already taken, or a row something still binds. It answers 409, and the finding names what is in the way — the maps or styles still referencing it, so the caller can act rather than guess.	request
RQ6	A document the studio stored will not read back — a plan row, an artifact, a snapshot written under a shape no upgrade claims. It answers 422 rather than 500 because it is data rather than a defect and the author clears it by writing the document again; and deliberately not RQ1, which would blame the request that merely asked to read it and send the author looking at what they just posted.	request, studio

Family HJ — 5 rules

Rule	What it means	Concerns
HJ1	Two rectangles share blocks. A plan states its ground once.	structure
HJ2	They touch over part of an edge only, so neither joint can happen along it.	structure
HJ3	Both ridges run along the shared edge — two ranges side by side, meeting in a gutter.	structure, style
HJ4	Both ridges run into it — one longer range, which wants drawing as one rectangle.	structure
HJ5	The wing reaches further along the shared edge than the hall reaches across it, so its roof stands taller than the one it is meant to run into.	structure

Family PT — 5 rules

Rule	What it means	Concerns
PT1	A block that surfaces ground is painted below the course it surfaces. Grass, podzol, mycelium and farmland are each exactly one course thick — what is under them is soil — so a bucket deeper than one course filled with one writes it into every course of its depth, and the ground comes out made of its own skin.	theme, terrain
PT2	A pattern states a band, a stop or a side and carries no material in it. The member reads as present and holds nothing, so the painter meets it with no block to write — and it meets it while the world is being built, long after the document was stored.	theme, terrain
PT3	A sampled pattern's brush is finer than the blocks it paints. A cell size or a field scale is the period a pattern varies over, in blocks, so below two it changes faster than the ground it is laid on can show: every block is its own feature, the pattern resolves to noise at any distance, and no palette rescues it. ...	theme, terrain
PT4	A sampled field paints a bucket a player reads from the side and states no vertical period, so every block in a column resolves the same and the face comes out in vertical stripes. A field of the plane is a fabric for ground seen from above; a wall or a fill is seen edge-on, and the one thing that gives a face its grain there is the field varying with height.	theme, terrain

continued

Family PT, continued

Rule	What it means	Concerns
PT5	A theme tints its ground by team over land more than one team enters. The tint is one colour per canonical island, which is what makes it readable — a player standing anywhere on a landmass knows whose it is — and an island two teams’ spawns stand on has one colour for both, so the whole of it wears whichever team the ownership resolved. ...	theme, terrain

Family ST — 5 rules

Rule	What it means	Concerns
ST1	<i>Wool room piece</i> (optional): defines the full room region and sizes the stamped cage — the shell footprint is the piece inset one block, per the WX rules (docs/world-export/structures.md). Nothing under it is sealed — the room region’s own enter filter is what keeps an enemy out, at any depth; a redstone line with a torch at either end lies on the last block row at each of the room’s entry interfaces — every terrain↔room land seam and every abutting build-zone edge (WX6) — the conventional marker for where entrance protection begins. ...	—
ST2	<i>Spawn piece</i> (optional): defines the spawn region and the ground the spawn room is framed on (the WX footprint rule). Iron placed inside it is auto-renewed in the generated XML (load-bearing for gameplay); lint: every iron marker’s whole cube belongs inside a spawn piece, since one half in renews half of itself and one outside is a block a team mines once. ...	—
ST8	[author] <i>Approach wall geometry</i> : the interface a wall bars is a 10–20 block lane mouth (a wall across a 30-block face bars a room, not a lane), and the wall stands about 15 blocks in front of the wool room’s entrance — judged against the nearest parallel entry seam, since a wall on a side interface defends a flank and its entry distance means nothing. ...	—
ST9	[author] <i>Building cap</i> : the building a role piece raises is at most 20×20 blocks — the footprint the shell stands on, which is what a player walks into, so past it the room is a field with a roof. The cap is on the rectangle the export actually stamps: the one the placement states, or the one WX1 defaults from the region where it states none. ...	—
ST10	[author] <i>Region cap</i> : a wool-room or spawn piece is at most 20×30 blocks , in either orientation. The piece is the protection region <i>and</i> the ground the building stands on, so its long axis is the approach a room is entered along; past 30 it stops being a room’s ground and becomes a field of immunity a team cannot be fought out of. ...	—

Family FR — 4 rules

Rule	What it means	Concerns
FR4	[expert, split] Two distinct “angles of attack”: • Team approaches — ways to reach the enemy’s side: 1–3 (corpus: 1 on six seeds, 3 on three); 1 is acceptable only if it is wide. • Wool approaches — ways to reach a wool room: WL8.	—
FR6	[author] Split vs wide frontline; the band docks flush . A frontline is either split — two tips with a gap, hung off a hub (the common form: mirror-mid ex-6 frontline-6a/6b off hub-6) — or wide , one broad face (ex-2/3/4 , 6–8 cells). The mid band docks flush against the front edge: across both tips of a split frontline (band-6 spans frontline-6a..6b), or aligned to the corners of a wide frontline (band-3 kept aligned to frontline-3 ’s ends). ...	—
FR8	[author] A crossing spans the face it docks against . Measured per piece side as the frontline share of the exposed run: authored crossing faces read 1.00 across the corpus (the BZ8/BZ9 fit), the worst incidental partial 0.40 , and the funnel fault 0.25 — a 10-block zone on an 80-block face, invisible to players as a front. ...	—

continued

Family FR, continued

Rule	What it means	Concerns
FR9	[author; amendment 2026-08-31] A crossing is at least 15 blocks of frontline. The share FR8 measures cannot see a front that is narrow in blocks and wide in proportion: a 10-block frontline on a 10-block exposed face reads 1.00 and is still a funnel. Lint fires on any face carrying a frontline under 15 blocks. ...	—

Family G0 — 4 rules

Rule	What it means	Concerns
G01	[author; amendment 2026-08-16] A destroy goal sits three to four times as far from the enemy's spawn as from its own, by walk. Per goal: the traversal from the owning team's nearest spawn and from the enemy's, both over the fanned closure (pieces $\cup$ build zones — the cross-team leg crosses the axis, so the un-fanned surface would cut it), ratio enemy $\div$ own held to [3.0, 4.0]	—
G02	A team's own destroy goals stand 35–65 blocks apart, by walk. Two goals a team defends from one position are one goal with two names, and two so far apart that no position covers both make the defence a shuttle rather than a stand. The measure is the walk between them over the fanned closure, in the octile unit the preamble states — the destroy-side counterpart of WL7, which separates a team's wools. ...	—
G03	Opposing destroy goals stand 85–150 blocks apart, by walk. This is what the contest spans: under the band the two objectives are close enough that a rush reaches one before the other team has formed, and over it the attacker crosses a board's width into a set defence and the match settles into a stalemate. ...	—
G04	[author] A destroy goal sits 40 to 90 blocks from its own spawn, by walk. The measure is GoalDistances' own-spawn leg — the same walk G01's ratio is built from, judged here on its own rather than against the enemy's. Under the band the spawn is already standing on the goal, leaving nothing to defend; over it the spawn cannot reinforce before the goal falls. ...	—

Family SP — 4 rules

Rule	What it means	Concerns
SP1	The frontline→wool path never passes through the spawn (protection regions enforce this anyway); it may pass around it on a wide or split lane.	—
SP2	Near the back of its lane — otherwise the space behind spawn is dead space with no purpose. (The current lint approximates “back” per-piece and misreads spawns placed mid-chain; honest measurement needs lane chains — G24.)	—
SP8	[author] A spawn's egress steps by 1 level or takes a ramp : a seam at $\Delta \geq 2$ ahead of the door is un-walkable bare (EL1's palette steps by 2, so bare seams are un-walkable by construction; bridgid-ii was hand-carved). Lint over the spawn piece's forward land seams; a cliff at the spawn's <i>back</i> is a legitimate wall and is not the egress. ...	—
SP9	[author] A spawn door stands ≥ 15 blocks from bare void , measured along the door's own line. A build zone counts as ground — the gap-only spawn (SP6) whose door opens onto its egress bridge is an authored motif; a <i>buffer</i> is exactly the declared emptiness this keeps off the doorstep (the entrance-void fault: a 25-deep drop at the door face).	—

Family DC — 3 rules

Rule	What it means	Concerns
DC1	A core casing has no interior left to hold lava — a solid casing is a goal that can never leak, so the match it is the objective of cannot be won.	objective, style
DC2	A core's float and leak are one knob: together they say how far players must dig under it, and either alone says nothing.	objective
DC3	A destroyable is built in a material its size is wrong for: obsidian is worth at most three blocks, so a cube or a plus-section column carrying it is a grind rather than a raid — or the material names nothing the studio builds at all, which writes into the map.xml verbatim while the blocks come out obsidian, and a declared material matching nothing in its own region is a goal at zero health (OB3).	objective, material

Family G — 3 rules

Rule	What it means	Concerns
G2	[corpus, revised] The map's corridor width is the size band's , stated in blocks: 12 at nano, 14 at micro, 16 at milli and centi — and the wool approach one rung under it at 10 · 12 · 14 · 16 . Measured over 359 built CTW maps (docs/world-scan/map-size-ladder.md): the modal local thickness of a map's ground runs 8 · 10 · 14 · 16 from the smallest maps up to the top of the ladder, the quartile a working lane sits at runs 8 · 12 · 14 · 16, and ground narrower than 8 blocks is 1.5–4% of a map at every size, which is the floor authors build to. ...	—
G5	[expert, refined] Void gaps between <i>individual</i> landmasses: 10–20 for the crossing a route <i>depends on</i> — but a longer hop is fine when the same buildable region offers a shorter alternative (four-team-towers-big 's 25 sits beside 15- and 20-hops in one region). Lint therefore judges a region's minimal crossing, not every pair. ...	—
G8	[corpus, revised] Map size is driven by the intended player count — maxPlayers is an <i>input</i> to the board envelope, not an afterthought — and the count's job is to name a size band . A map is not built for one count: it works across a range, and the ranges are the author's — nano 6–13 players a team, micro 14–21, milli 22–31, centi 32 and up. ...	—

Family HP — 3 rules

Rule	What it means	Concerns
HP1	No rectangles at all — there is no building to place.	structure
HP2	A wing is not two opposite corners, or is too thin to hold two walls and an inside.	structure
HP3	The wings cover more ground than a placed building may take.	structure

Family ED — 2 rules

Rule	What it means	Concerns
ED1	A reference the document cannot resolve. An apply-rule or a filter naming a region or a filter that is not in the registry and is not a builtin, or a filter naming itself — which would be a cycle of one. The wiring is what makes a PGM map mean anything, so a rule pointing at nothing is a rule that silently never fires.	request, world
ED2	An edit the document is not in a state to take. The payload is well-formed and everything it names exists; what it would produce cannot mean anything — a group with fewer children than its type takes, a compound with no children to remove one from, an apply-rule carrying no region, filter or action, a build-void enforcement with no build region to enforce.	request, world

Family LN — 2 rules

Rule	What it means	Concerns
LN1	Width 10 base — piece min-dims across 150 corpus pieces: 5-wide $\times$ 54 (the step/ledge idiom, not corridors), 10×81 , 15×15 , $20+ \times 0$ (the “>20 near mid/spawn” case is authored via assembled footprints, not single pieces). The stretch in front of a wool stays $\leq \sim 16$.	—
LN2	Length 20–50 before a junction or dead end; a lane may include a turn/twist (the L-shape case).	—

Family C0 — 1 rules

Rule	What it means	Concerns
C01	The descriptor is well-formed and names a board the composer cannot emit: a spawn profile outside the two forms it builds, a hub arm count it has no shape for, a frontline form off its menu, a unit whose hub publishes no offer to consume. The sentence carries which, because the emitter that stopped is the only thing that knows. 422 — the request was read and acted on, and there is no board at the end of it. ...	plan, studio

Family EL — 1 rules

Rule	What it means	Concerns
EL1	[corpus; amendment 2026-08-31] A land seam of $\Delta \geq 2$ is ground a player cannot walk up , and the rule is the list of those seams. The plateau step unit is 2 — the authored surface palette was base 9 + even steps up (9/11/13/15/17/19, all odd), extended below base by the traced big-board-...-parallel-mid (frontline at 7 and 5) and down to 3 by mirror-tiny-map-cliff , so the palette is 3–19 and the 137 measured land-interface deltas are even by construction: histogram $\Delta 0 \times 47$, $\Delta 2 \times 73$, $\Delta 4 \times 10$, $\Delta 6 \times 4$, $\Delta 8+ \times 3$	—

Family EZ — 1 rules

Rule	What it means	Concerns
EZ1	Ground a player can stand on and nobody can edit, outside the zones a map protects on purpose. A spawn and a wool room are sealed by design and are excluded by the rule that seals them — they restrict entry, and a place you may not walk into is not a place you were expecting to build in. What is left is ground the author can reach, cannot touch, and did not say so: a canopy hanging over the void outside every build zone, a crag the void rule closed along with the air around it, a platform whose supporting column never reached $y=0$	terrain, intent

Family PC — 1 rules

Rule	What it means	Concerns
PC-C	Two pieces meet at a single point and along no edge, and nothing else joins them. A corner is never a connection — a point has no walkable corridor mouth — so the board reads as one area where players find two, and the diagonal is the sneaky crossing that is not there. Suppressed where the pair already reaches the same land component through real interfaces, since the corner is then redundant rather than misleading....	plan

Family SH — 1 rules

Rule	What it means	Concerns
SH1	A shop reference names something the document does not define, so PGM refuses the whole map at load rather than opening a menu without it. Three references are read: a keeper's shop, which throws "No shop with id '...' could be found"; a keeper's region, and an icon's action, which are feature references and throw when nothing resolves them.	intent, studio

APPENDIX B

The scored terms

The 31 terms the evaluator scores, from `GET /api/rules/terms`. A hard term is an acceptance gate; a soft term scores distance from a band. *Band from* says where the envelope came from: a measurement over the corpus, or an author's ruling.

Term	Rule	Kind	Band	Band from	Learns
band-wool-clearance	BZ6	hard	—	none	—
gap-hop-band	G5	hard	—	none	—
lint-g2	G2	hard	—	none	—
lint-pc-c	PC-C	hard	—	none	—
spawn-wool-floor	WL2	hard	—	none	—
structural-integrity	STRUCT	hard	—	none	—
band-count	CT1	soft	0–3	envelope	yes
dead-share	G8	soft	0–0.1204	envelope	yes
enclosed-void-count	CT8	soft	0–15	envelope	yes
fill-ratio	G8	soft	0.2007–0.5417	envelope	yes
frontline-count	FR4	soft	1–8	envelope	yes
frontline-width	FR6	soft	1–16	envelope	yes
goal-spawn-distance	G04	soft	40–90	authored	no
goal-spawn-ratio	G01	soft	3–4	authored	no
isolation-cut-count	CT5	soft	0–6	envelope	yes
lane-width	LN1	soft	10–30	envelope	yes
max-chain-length	LN2	soft	25–110	envelope	no
neutral-stepping-count	CT4	soft	0–4.5	envelope	yes
opposing-goal-distance	G03	soft	85–150	authored	no
own-goal-distance	G02	soft	35–65	authored	no
route-interference	CT8	soft	0–0.875	envelope	yes
spawn-wool-distance	WL2	soft	29–176	envelope	yes
spawn-wool-ratio	WL9	soft	1.0312–1.2203	envelope	no
spawn-wool-spread	WL9	soft	0–73	envelope	yes
team-stepping-count	CT4	soft	0–2	envelope	yes
uncrossed-middle-void	CT9	soft	0–0	envelope	yes
wool-front-balance	WL10	soft	0–116	envelope	yes
wool-front-distance	WL10	soft	22–147	envelope	yes
wool-front-ratio	WL10	soft	1–1.439	envelope	no
wool-front-remoteness	WL10	soft	25–130	envelope	no
wool-wool-distance	WL7	soft	50–227	envelope	yes

The endpoint surface

231 operations over 174 paths in 22 areas, generated from the studio's own OpenAPI document.

Method	Path	Summary
/map (126)		
POST	/api/map/from-documents	load a whole map from the three documents it is made of: the plan it was drawn from, the layout that was built, and the intent it is played for. The way back in for a map authored elsewhere....
POST	/api/map/import-folder	import a local xml-less world (B8 "open a folder"): resolve <root>/<slug>/region via MapsRoots (only configured roots — no client path), create the map row, and scan into MariaDB....
POST	/api/map/import-url	B8 import-from-url (docs/tools/configure.md, the Import phase). Server-side: fetch a zipped Minecraft world from an allowlisted host, safely extract only region/.mca, create the map row, and scan it into MariaDB (reusing WorldFeatureWriter)....
GET	/api/map/{slug}	the full map document (the xml_data.json shape), reconstructed from the relational rows.
GET	/api/map/{slug}/block-seat	?x=&y=&z= — whether one block can hold a thing placed into it, read off the same vertical segments column-floor takes. A monument is the block a player puts a wool into, so the authoring step asks both halves at once: the block itself must...
GET	/api/map/{slug}/column	One or more columns bedrock-to-sky, every block named, as text. THE WORKHORSE: every picture beside it is a projection, and this is what is actually at a coordinate — which is why it is the read to reach for when a picture and a document disagree....
GET	/api/map/{slug}/column-floor	?x=&z=&y= — the terrain floor Y at a single column from the vertical segments: the top of the highest solid segment at or below the reference y (the floor a thing at y rests on), falling back to the segment top nearest y....
GET	/api/map/{slug}/core-suggestions	[?box=x0,y0,z0,x1,y1,z1] — the cores the ingest scan proposed for this map, read straight out of core_candidate. No world access and no scoring pass: a core's signature is unambiguous enough that the gather already knew the casing, so the ...
GET	/api/map/{slug}/coverage	where the ground is lived on and where it is dead: the traffic corridors between every pair of waypoints, each waypoint's own ring, the prop-decorated fringe, and the dead patches named with coordinates. ?format=png answers the same grid as the coverage picture....
GET	/api/map/{slug}/destroyable-suggestions	[?box=x0,y0,z0,x1,y1,z1] — the destroyables the ingest scan proposed for this map, read straight out of destroyable_candidate. box is optional and narrows the list to masses intersecting it, which is how the configure step scopes a detect to the area the author drew....
GET	/api/map/{slug}/editability	which columns a player may edit and what makes each one editable, plus what the pass has to say about the result (EZ1).
GET	/api/map/{slug}/export	the Configure export action. For a sketch-originated map (one with a stored sketch layout) it returns a ZIP holding map.xml, level.dat and region/.mca at its top — a world directory's own contents, which is what a server is handed — a ...
GET	/api/map/{slug}/findings	everything wrong with this map right now, as far as its stored documents can say, plus the gates a read cannot reach and where to ask them. The operation is MapFindings; this is the door to it.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/map/{slug}/incline	How steeply the ground is inclined, cell by cell, as characters — the angle a slope band is picked by (TP24). A cell's glyph is the TENS of its angle, so 0 is ground under ten degrees from level, 4 is forty to fifty and 8 is a face...
GET	/api/map/{slug}/intent	the map's declarative authoring intent (docs/pgm/new-map-authoring.md), empty if none yet.
PUT	/api/map/{slug}/intent	store the intent the author edited and regenerate the map from it. Replaces the stored intent wholesale, which is what makes a deletion in Configure stick.
PUT	/api/map/{slug}/intent/from-plan	store a freshly compiled intent, carrying the map's authored slices onto it first (IntentCarry), then project it exactly as a normal PUT does. The plan owns the map's structure, so a rebuild is meant to replace its teams, spawns, wools and build zones....
PATCH	/api/map/{slug}/intent/rooms/{reference}	put one room piece where the sketch just dragged it. The sketch draws a spawn, a wool room and the building inside one as locked annotations projected out of the intent, so a drag on that canvas has to arrive back here or the picture and...
GET	/api/map/{slug}/islands	the detected island polygons (from the islands_json artifact).
GET	/api/map/{slug}/kit-reach	can a fresh spawn bridge to each wool with only the placeable blocks its spawn kit grants? (budget-aware traversability).
PATCH	/api/map/{slug}/metadata	what a map is called, what it states and who wrote it. The operation is MapMetadata; this is the door to it.
GET	/api/map/{slug}/monument-seat	whether each wool monument's block can hold the wool won on it: clear of anything standing in it, and with a block on one of its six faces to be placed against. Both faults stop the wool going in, so both are errors.
GET	/api/map/{slug}/monument-suggestions	?box=x0,y0,z0,x1,y1,z1[&style=pedestal,label,cap] — score the pre-gathered monument_candidate rows inside the author's box for the declared style (F9). No world access: loads the candidates and runs MonumentSuggester.Score. box is required (the author marks the monument area); style defaults to Any,Any,Any....
PATCH	/api/map/{slug}/observer-spawn	set/replace the observer spawn.
DELETE	/api/map/{slug}/observer-spawn	remove the observer spawn.
GET	/api/map/{slug}/origin	{ sketch: bool }; whether the map originated in the sketch tool (has a stored sketch layout). The Configure wizard reads this to auto-wire the monument step away for sketch-origin maps (their monuments are derived at export, not authored).
GET	/api/map/{slug}/plan	the stored plan blob for a plan-stage map, or {} if none.
PUT	/api/map/{slug}/plan	replace the stored plan blob (the plan editor's saved state).
GET	/api/map/{slug}/plan/ascii	the stored plan as a grid of characters, one per proxy cell. The map-scoped twin of GET /plans/{id}/ascii: the same render, reached by the slug an authoring driver already holds rather than by a candidate id. ?every=N downsamples. 404 when the map or its plan is missing, 422 when the stored document cannot be read....
GET	/api/map/{slug}/plan/flow	what the board asks of the two sides, and what that leaves unused, in prose. The companion to the coverage picture and meant to be read before it: a render says where ground is dead and only the flow says why....
GET	/api/map/{slug}/preflight	the Review phase's pre-flight gate (new-map-authoring.md §6). Runs the four generated-map checks and reports the export verdict: Round-trip + Mirror — pure codec/categorizer checks (Preflight)....
GET	/api/map/{slug}/reach	Which standing ground no player can get to, and why — the patches, their sizes, their lowest course and the box to stand in. It is the traversability picture's own partition read as numbers: the navigable components, minus the one the board...
GET	/api/map/{slug}/regions	derived region facets + category counts.

continued

Endpoint surface, continued

Method	Path	Summary
POST	/api/map/{slug}/regions	create a primitive region (optionally tagged with the editor draft_step so it shows in that activity until it's wired — E10).
POST	/api/map/{slug}/regions/group	wrap regions in a compound.
GET	/api/map/{slug}/regions/tree	category-grouped nested region tree (canvas render input).
POST	/api/map/{slug}/regions/ungroup	dissolve a compound.
PATCH	/api/map/{slug}/regions/{regionId}	rename / bounds / coords.
DELETE	/api/map/{slug}/regions/{regionId}	delete a region (returns an undo snapshot).
POST	/api/map/{slug}/regions/{regionId}/counterpart	create the symmetry counterpart(s) of a region (F3). Body: {mode, center:{cx,cz}?}; centre falls back to the confirmed symmetry artifact.
POST	/api/map/{slug}/regions/{regionId}/orbit	fill the symmetry orbit of a region (F3), i.e. create every counterpart implied by the map's confirmed symmetry (rot_90 → 3 turns, mirror/rot_180 → 1). Used right after a draw so an authored region appears in all symmetric positions. ...
GET	/api/map/{slug}/render/heightmap	Elevation as tone, with contour lines every contour blocks. The read for whether a relief solved into the shape it was drawn as, and the one that shows a flat pad butted against a hill as the ruled edge it is. ...
GET	/api/map/{slug}/render/mirror	The board against its own symmetry: the columns that agree with their image, and the ones that do not. The read for whether a board somebody believes is symmetric actually is.
GET	/api/map/{slug}/render/section	A vertical cut with a Y scale: axis which way it runs, from/to its extent, at the other coordinate, depth how far behind it to project. One of only two reads that keep Y, and the one that shows a layer stack, a roof idiom and a riser's courses. ...
GET	/api/map/{slug}/render/structures	The building census by block material, for a world the studio did NOT build — a scanned map, or a corpus one. MISLEADS: It finds roofs by material and cannot see a town this studio built (B149): a cottage roofed in stained clay reads as ground. ...
GET	/api/map/{slug}/render/surface	The paint, read as the tone families TerrainPalette. Families names — so a board can be checked against the palette it was authored from, and a whole family taken where two members were meant reads as the noise it is. ...
GET	/api/map/{slug}/render/topdown	The board from above, one question per image: subject draws the terrain alone (ground), what the build recorded itself placing (structure), the planting (foliage), the goals and spawns (objectives), or all of it (combined) ...
GET	/api/map/{slug}/render/traversability	The navigable components, with the spawns and goals drawn on them. The read for whether a board hangs together as one place a player can walk. MISLEADS: Headroom is what a player's body passes through, not air: a flower, a torch, a carpet ...
GET	/api/map/{slug}/render/walk	What crossing this board charges, drawn: every passable cell shaded by what reaching it from from costs, with the route to to over the top. field picks which of the walk's answers is shaded — blocks a player must place, distance ...
POST	/api/map/{slug}/resources	iron/gold/diamond blocks (optionally in a drawn rect, body { bounds?: { min_x, min_z, max_x, max_z } }) + how many a <renewable> already covers.
GET	/api/map/{slug}/scan-summary	per-feature breakdowns for the import brief: wool blocks grouped by colour (with a swatch hex) and resource blocks grouped by type, each ordered by count.
POST	/api/map/{slug}/scan-world	scan the map's Minecraft world (<root>/<slug>/region) and (re)write its relational feature rows (wools/resources/chests/spawners/segments). The world-import half of the pipeline; the xml half is the importer / map editors.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/map/{slug}/segments	?axis=x/z — side-view depth profile (B5). Projects the map's vertical solid segments onto a 2D (primary × y) grid via SideView; feeds the Build-Regions side-view canvas (C7). Mirrors the reference get_segments.
GET	/api/map/{slug}/sketch	the stored sketch layout (the JS-origin blob), or {} if none.
PUT	/api/map/{slug}/sketch	replace the stored layout blob (the bridge's getState()). A drawing in progress is stored whatever it says. The board's own geometry is checked here and every finding rides back as a complaint, refusals included: a sketch is the author's...
GET	/api/map/{slug}/sketch/biome	the field every column answers to. 404 where the board states none, which is plains everywhere and what every board that never opened the question exports as.
PUT	/api/map/{slug}/sketch/biome	which biome each column of the exported world carries. Map-wide and answered per chunk, because a biome's tint is blended across a radius and a region drawn to a finer edge never reaches its own colour there.
DELETE	/api/map/{slug}/sketch/biome	take the field off the board, which is plains everywhere.
POST	/api/map/{slug}/sketch/columns	the whole built world as per-column runs, which is what the 3-D preview draws (docs/tools/sketch.md). The body is the live layout, the same as the paint and contour previews take, and the stored intent rides along so the structures a map has stated are standing in the picture....
DELETE	/api/map/{slug}/sketch/discard-if-empty	drop a still-pristine sketch draft (the row "New sketch" creates up front, then abandoned). The client calls this best-effort when it leaves the Sketch tool. Discards only a draft that is genuinely untouched: sketch stage, still carrying...
POST	/api/map/{slug}/sketch/dressing	what the dressing pass would place, run against the posted layout and stopped before anything is written. The body is the live layout, the same as the paint preview and the columns read take....
POST	/api/map/{slug}/sketch/finish	rasterize the stored layout into the world geometry artifacts (layer.parquet / groups.json / segments) so the draft flows into the Configure wizard. 422 only if the layout rasterizes to no ground at all....
PUT	/api/map/{slug}/sketch/from-plan	replace the stored layout with one a plan compiled, carrying the map's existing finish onto it (CarryFinish). The plan owns the board; the sketch owns its themes, room shells and dressing, and a plan cannot express any of those — so the compile path merges where SketchPutEndpoint replaces....
GET	/api/map/{slug}/sketch/groups	every group the board carries, across all its layers, with whether a relief is keyed over it. The list a caller reads before naming a group to draw into or to state terrain for.
GET	/api/map/{slug}/sketch/layers	the stack in draw order, each layer with the shapes and groups drawn on it. A layer that named itself keeps its id; one that did not is answered under its position, which is the id every other route addresses it by.
GET	/api/map/{slug}/sketch/layers/{layerId}	one layer of the stack, with its shapes and groups. 404 where the id names none.
PUT	/api/map/{slug}/sketch/layers/{layerId}	state one layer of the stack: the height its ground starts at, whether it is terrain or a made thing, and how its floors meet the ground. Creates the layer at the end of the stack where the id names none....
DELETE	/api/map/{slug}/sketch/layers/{layerId}	take one layer, everything drawn on it and the relief of every group that lived only there off the board.
PUT	/api/map/{slug}/sketch/layers/{layerId}/groups/{groupId}	state one group whole: what it is called, whether it is fanned onto the symmetry orbit, and which shapes share its ground. Creates it where the layer carries none under that id....
DELETE	/api/map/{slug}/sketch/layers/{layerId}/groups/{groupId}	ungroup. The shapes stay on the layer and are drawn where they were drawn; what goes with the group is the orbit fan and the relief keyed under its id.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/map/{slug}/sketch/ layers/{layerId}/shapes	the shapes drawn on one layer, in draw order.
POST	/api/map/{slug}/sketch/ layers/{layerId}/shapes	draw one shape on a layer, answering the id it was given. ?group= names the ground it joins, and a name the layer does not carry yet opens a group. A layer that already groups its shapes takes no shape that names none: the orbit fan and the ...
PUT	/api/map/{slug}/sketch/ map-theme	which registered theme covers every cell no shape's own scope claims. The body is {"theme": "meadow"}; a null or absent theme clears it, which paints unthemed stone. ...
POST	/api/map/{slug}/sketch/ paint	the sketch's terrain paint as a palette-indexed block-pixel payload (xs/zs/palette/color_idx + bounds), which the client expands into the colors array the block-overlay bitmap path already decodes (docs/world-export/terrain-painting.md TP10). ...
POST	/api/map/{slug}/sketch/ probe-footprint	whether a ring stands on ground, asked of the rasterised footprint the build actually produces rather than of a model of the coast rebuilt outside the studio. The two disagree by a cell or two, and a cell or two is the whole failure: a lift ...
GET	/api/map/{slug}/sketch/ props	every placement the map carries, with the recipes they name. The recipes ride with the placements because a placement referencing a key nobody can resolve is not readable on its own: a tree states styleKey and the registry states what that key is made of. ...
POST	/api/map/{slug}/sketch/ props	place one prop, answering the id it was given. A body stating a free id keeps it; one stating none, or one already taken, is minted {kind}-{n}. The placement goes on the end, because the pass runs in the order props were placed and an addition has not been placed before anything. ...
PATCH	/api/map/{slug}/sketch/ props/{propId}	replace one placement, keeping its position in the pass's order. 404 where the id names no placement.
DELETE	/api/map/{slug}/sketch/ props/{propId}	take one placement off the board. The recipe it named stays in the registry, since a key is shared by every placement wearing it.
GET	/api/map/{slug}/sketch/ relief	every group's relief, by the group id it is solved over. The key is a group rather than a shape because a relief solved per shape leaves a seam wherever two of them meet and disagree about the height they share.
POST	/api/map/{slug}/sketch/ relief	the contour overlay for whatever relief the posted layout carries, one entry per relief-bearing group: its traced lines, its height range, and its bounds. The body is the live layout, the same as the paint preview takes, so the overlay tracks unsaved edits. ...
POST	/api/map/{slug}/sketch/ relief/read	what the relief a posted layout carries charges, per group. Not a walkability score: a relief that is walkable everywhere is a field rather than a map, and a single number ranks every deliberate barrier as a defect. ...
GET	/api/map/{slug}/sketch/ relief/{groupId}	one group's relief. 404 where the layout states none for that group, which is every group whose ground is as flat as its shapes drew it.
PUT	/api/map/{slug}/sketch/ relief/{groupId}	state one group's interior elevation, replacing whatever that group carried. It does not check that the group exists. A relief is authored against a fusion the compiler produces, and whether the id still names one is the question SK1 ...
DELETE	/api/map/{slug}/sketch/ relief/{groupId}	take one group's relief off the board, which leaves its ground as flat as the shapes drew it.
GET	/api/map/{slug}/sketch/ room-styles	both shells as the stampers will read them. Resolved rather than raw, because the three states a binding has do not survive being handed back as a snapshot: absent falls back to the built-in shell, null asked for open ground, and a stated style is itself. ...
PUT	/api/map/{slug}/sketch/ room-styles/{part}	bind the shell one kind of room is stamped in. part is wool or spawn. A body of literal null is a statement, not an omission: it asks for open ground — a pad rather than a building over it, which is what a spawn on a plateau the plan already shaped often wants to be. ...

continued

Endpoint surface, continued

Method	Path	Summary
DELETE	/api/map/{slug}/sketch/ room-styles/{part}	unbind a shell, which puts that kind of room back to its built-in one. Not the same as binding null, which asks for open ground.
POST	/api/map/{slug}/sketch/ seats	where a prop of a stated kind and footprint may stand, which the five declines only ever answer backwards. The dressing pass refuses a placement for the ground it rests on — a goal's clearance, the map's keep-out, another prop's claim, a ...
GET	/api/map/{slug}/sketch/ shapes/{shapeId}	one shape, wherever on the stack it is drawn. 404 where the id names none.
PATCH	/api/map/{slug}/sketch/ shapes/{shapeId}	change one shape without restating the board. A stated field replaces what the shape carried and a stated null takes the field off, so the one call both writes a height and clears a relief scope. id is the address and is kept whatever the ...
DELETE	/api/map/{slug}/sketch/ shapes/{shapeId}	rub one shape out, and take it out of every group that listed it.
POST	/api/map/{slug}/sketch/ shapes/{shapeId}/bend	redraw one outline as a coast. The compiler emits a staircase of the plan's rectangles, which is the board's shape and not its coast. Redrawing the ring by hand states the coast twice — once in the plan and once in the sketch, free to disagree — so the bend is taken over whatever the compile produced instead. ...
POST	/api/map/{slug}/sketch/ shapes/{shapeId}/vertices	add one point to one outline, after the vertex after names. Stating no x/z puts it at the midpoint of that edge, which is a new corner half way along a wall with nothing else moved; the answer carries the index it landed at, which is the address the move route then takes. ...
PATCH	/api/map/{slug}/sketch/ shapes/{shapeId}/vertices/ {index}	move one point of one outline. Every other vertex stays exactly where it was drawn. That is the whole of the call: a board's shapes abut, and an edit that drags a ring's other points opens ground between two that were flush, which is what a whole-ring transform does and what a hand pulling one corner does not. ...
DELETE	/api/map/{slug}/sketch/ shapes/{shapeId}/vertices/ {index}	take one point out of one outline, leaving every other where it was drawn. Refused where the outline is down to its last three, since two points draw no ground, and where dropping the point folds the ring.
GET	/api/map/{slug}/sketch/ themes	the registry, and which of it is the map default. A theme that will not parse as one is left out, the same way the painter drops it: the registry is a snapshot store rather than a validated model, and a reader asking what the board paints with wants what the board will actually paint with. ...
GET	/api/map/{slug}/sketch/ themes/{themeId}	one theme, as the painter reads it. 404 where the registry carries no such id, or carries one that will not parse as a theme.
PUT	/api/map/{slug}/sketch/ themes/{themeId}	register one theme under an id, replacing whatever that id carried. A registry entry is addressed by the name an author gave it, so this is the one write in the sketch that creates and replaces through the same verb.
DELETE	/api/map/{slug}/sketch/ themes/{themeId}	take a theme out of the registry. It does not refuse over what still names the id. A shape painting with a theme the registry has stopped carrying takes the map default, and the map default naming one takes unthemed stone — both are already ...
GET	/api/map/{slug}/slopes	The worst step to a neighbour per sampled cell, classed . walked, : scrambled with a block, # a barrier — the same tiers a walk is priced in. A cliff reads as a line, a ramp as a band crossing it, and an overdone relief as a page. ...
POST	/api/map/{slug}/spawns	link a spawn to a region.
PATCH	/api/map/{slug}/spawns/ {regionId}	update a spawn link.
DELETE	/api/map/{slug}/spawns/ {regionId}	remove a spawn link.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/map/{slug}/state	where this map has got to, which authoring artifacts it holds, and what may be done to it from here. A tool asks about the map it has open so it can tell an origination from a rebuild before offering the action rather than after performing it; the moves are the same question answered outright. ...
GET	/api/map/{slug}/stroke	A drawn stroke walked end to end, down its own centreline rather than between its points: per block the ground, whether the paving reaches it, what it is made of and the step from the block before, then the worst step, the material runs and every stretch the paving does not reach. ...
GET	/api/map/{slug}/symmetry	global symmetry of the map's islands (B7). Returns the cached symmetry_json artifact, or computes it on demand from the islands_json artifact (excluding the Configure-excluded islands) and caches it with status "unconfirmed".
PATCH	/api/map/{slug}/symmetry	confirm/reject the detected symmetry (B7). Updates status ("confirmed"/"none"), an optional user-override confirmed_type, and an optional centre override. Mirrors the reference patch_symmetry.
POST	/api/map/{slug}/teams	add a team.
PATCH	/api/map/{slug}/teams/{teamId}	update a team.
DELETE	/api/map/{slug}/teams/{teamId}	remove a team and its spawns.
GET	/api/map/{slug}/themes/census	Every ground cell counted by the theme that paints it: cells per theme and their share, the distinct surface materials each carries, which theme borders which and over how many cells, and the board's whole palette count. ...
GET	/api/map/{slug}/top-surface	per-column surface colour overlay (B4). Reads the cached layer.parquet artifact, maps each column's (block_id, block_data) to a hex colour, and returns parallel xs/zs/colors arrays + the bounds. ...
GET	/api/map/{slug}/transect	A polyline walked block by block, answered as numbers: points is the line as <code>X,Z;X,Z[;X,Z...]</code> , every thins the stations, beside lists every distinct claim within that many cells of the line. ...
GET	/api/map/{slug}/traversability	spawn↔wool connectivity.
GET	/api/map/{slug}/walk	The same journey as numbers rather than as a picture: whether the place can be reached, how far it is in blocks, how many blocks a player must place to get there — the climb at a rise of delta costing delta minus one, and one a cell for void bridged — and how many falls over the free height it takes. ...
GET	/api/map/{slug}/wool-availability	per declared wool, is it obtainable?
POST	/api/map/{slug}/wool-sources	wool colours found inside a drawn rectangle (body: { bounds: { min_x, min_z, max_x, max_z } }).
GET	/api/map/{slug}/wool-suggestions	wool colours in the world not yet declared as objectives.
POST	/api/map/{slug}/wools	add a wool objective.
PATCH	/api/map/{slug}/wools/{woolId}	update a wool.
DELETE	/api/map/{slug}/wools/{woolId}	remove a wool.
POST	/api/map/{slug}/wools/{woolId}/monuments	add a monument to a wool.
PATCH	/api/map/{slug}/wools/{woolId}/monuments/{monId}	update a monument.
DELETE	/api/map/{slug}/wools/{woolId}/monuments/{monId}	remove a monument.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/map/{slug}/xml	render the map as a PGM map.xml (proto 1.5.0). The export leg of the round-trip: doc → Deserializer.FromDict → MapXml → XmlWriter.ToXml — the same path the round-trip harness (check #2) exercises across the corpus. ...
/terrain (12)		
GET	/api/terrain/biomes	the biomes a field may name, each with the grass colour it tints ground with (docs/world-export/terrain-painting.md §5b). Served rather than restated in the client for the reason the block palette is: the ids live in Biome and the colours ...
GET	/api/terrain/blocks	the blocks a terrain-paint material may be built from (TerrainPalette), each with its id/data pair, display name, picker group and swatch colour. The block picker reads it, so the colour a picker shows is the colour the export places (docs/world-export/terrain-painting.md §3). ...
GET	/api/terrain/boulder-forms	the four rock shapes, each an actual rock.
GET	/api/terrain/looks	the construction words a block's construction is drawn from, each with what it means. Served rather than restated by every caller, for the same reason the block palette is: a second copy on the far side of the wire is a vocabulary free to drift from the one the table was measured under. ...
POST	/api/terrain/material-preview	body is one serialized TerrainMaterial; returns both views of it (top-down and cut open). What a style editor re-renders on every edit — see StylePreview for why one material needs two views. ?format=png&view=plan section answers that one view as PNG bytes instead. ...
GET	/api/terrain/patterns	every material kind a theme may be built from, with what it does, the facts about a cell it varies with, and its fields with their types and defaults (MaterialVocabulary). ...
POST	/api/terrain/prop-preview	one placed prop and the terrain finish it stands on; returns a sample patch the pass actually dressed, from above and cut open. The theme is part of the request because what the paint leaves on top is what decides whether flora grows at all and what a path may repaint. ...
GET	/api/terrain/species	every vanilla species, each built.
GET	/api/terrain/stroke-styles	the five ways a stroke paves the ground it crosses, each drawn.
POST	/api/terrain/theme-map-preview	body is a plan JSON; compiles it, paints the terrain through the scoped theme resolver, and returns a top-down SVG of the map's top blocks so the Theme rail's apply step can show the themes on the actual map (TP10). 400 when the plan can't be rendered. ...
POST	/api/terrain/theme-preview	body is a serialized terrain-paint theme (a TerrainTheme JSON); returns the sample plateau painted with it and cut open, plus a top-down swatch per themeable bucket, so a theme editor previews the whole finish and each brush as it is edited (TP10). ...
GET	/api/terrain/water-forms	the three channel forms, each an actual dug channel seen from above.
/room-styles (10)		
GET	/api/room-styles	the room-style library, newest first, each with the shell it stamps.
POST	/api/room-styles	compose a room style from existing styles. 400 {error, findings} when the composed shell fails Check — a block named for a geometric role that is not that kind of block, a doorway that does not clear the least height a door may, or a roof whose own materials are wrong for its pitch or its family. ...
GET	/api/room-styles/block-kinds	what kind of block each house-style field takes, and the ids of each kind. Served for the reason the door list is: the authoritative statement is HouseBlockKinds in PgmStudio.Minecraft, which neither the client nor an agent can reach, and ...

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/room-styles/doors	the doors a room may be stamped with. Served rather than restated in the client: the authoritative list is the same DoorMaterials table the wool-room block filter is built from, so a door offered here is always one an attacker can break.
POST	/api/room-styles/preview	the shell a set of courses composes to, previewed without saving any of it. What the editor re-renders as courses are bound and knobs are turned.
POST	/api/room-styles/preview-snapshot	body is a serialized HouseStyle; returns the shell it stamps. What a map's bound style is previewed through: the binding is a snapshot rather than a library id (structures.md §9), so the picture has to come from the snapshot too — reading ...
GET	/api/room-styles/{id}	one room style with its per-part courses.
PUT	/api/room-styles/{id}	replace a room style's knobs and its whole set of courses. Refuses the same way RoomStyleCreateEndpoint does.
DELETE	/api/room-styles/{id}	forget a room style (its courses cascade; the styles stay).
GET	/api/room-styles/{id}/json	the room style assembled into the stamper's own JSON: the form the export consumes and the form a map snapshots when it binds one.
/plan (9)		
POST	/api/plan	originate a blank authored plan: create a map row at stage=plan with an empty plan_json artifact (no candidate provenance). Returns the slug; the client navigates to /maps/{slug}/plan, where the editor keeps its default blank document until first save. ...
POST	/api/plan/ascii	the board as a grid of characters, one per proxy cell, off a posted plan. The read that shows a relation between two rectangles. The other projections answer numbers and geometry, and a number cannot say that a sixteen-cell bar is reached ...
POST	/api/plan/columns	the world a plan builds, as per-column runs, which is what the plan tool's 3-D preview draws (docs/tools/plan.md). The body is the plan document itself. A compiled plan carries a full intent, so unlike the sketch side this shows the wool ...
POST	/api/plan/compile	compile a plan wire document one-way into the pair the draft pipeline consumes: { layout: <SketchLayout>, intent: <MapIntent> }. Structural validator errors block the compile (422, carrying the error findings); lint alone never blocks. ...
POST	/api/plan/evaluate	the plan editor's live rule-evaluator score + lint feed (the critic that scores a .plan.json). The request body is a plan wire document; the response is an EvaluationDto: the summed score (lower is better, 0 perfect), a valid flag (no hard ...
POST	/api/plan/feasibility	a diagnostic about the composer, not a read about the board. Could the composer have produced this plan? The question PlanEvaluateEndpoint does not ask, and the two genuinely diverge — a plan can score 0 with no rule fired and still be outside everything the emitters can build. ...
POST	/api/plan/inspect	the live derived-geometry feed for the plan editor's canvas overlays. The request body is a plan wire document (.plan.json); the response carries everything already resolved to block coordinates so the canvas draws it directly: interfaces ...
POST	/api/plan/room	?piece=<id> — the contents a spawn or wool-room piece carries: the marker the room is built around, the footprint the building stands on, and the spawn's iron cube. ...
POST	/api/plan/{planId}/author	commit a generator plan candidate to authoring: create a map row at stage=plan seeded with the candidate's plan_json (a plan_json artifact) and a plan_source_id back to the candidate. ...
/themes (8)		
GET	/api/themes	the theme library, newest first, each with the sample plateau it finishes.

continued

Endpoint surface, continued

Method	Path	Summary
POST	/api/themes	compose a theme from existing styles (the knobs + bucket→style bindings).
POST	/api/themes/import	lift a whole theme JSON into the library: one style per bucket + a theme binding them. 400, never 500, on invalid theme JSON.
POST	/api/themes/preview	the theme a set of bindings composes to, previewed without saving any of it. What the library's theme editor re-renders as buckets are bound and knobs are turned.
GET	/api/themes/{id}	a theme with its per-bucket style bindings.
PUT	/api/themes/{id}	replace a theme's knobs and its whole set of bucket bindings.
DELETE	/api/themes/{id}	forget a theme (its bucket bindings cascade; the styles stay).
GET	/api/themes/{id}/json	the theme assembled into the painter's theme JSON (the form the export consumes and a map snapshots when it applies the theme).
/boulder-styles (7)		
GET	/api/boulder-styles	—
POST	/api/boulder-styles	—
POST	/api/boulder-styles/preview	—
GET	/api/boulder-styles/{id}	—
PUT	/api/boulder-styles/{id}	—
DELETE	/api/boulder-styles/{id}	—
GET	/api/boulder-styles/{id}/json	—
/plans (7)		
GET	/api/plans	[?origin=generated authored imported] — the open-from-DB browser list, newest touched first. Summaries only (no plan JSON); the detail endpoint carries the document.
POST	/api/plans	save the plan open in the editor. Applies the fork-or-mutate doctrine in SaveFromEditorAsync and returns the resulting row. A malformed plan body is answered 400, never 500.
GET	/api/plans/{id}	one plan with its *.plan.json document, to load into the editor.
DELETE	/api/plans/{id}	forget a plan (204). Forks of it survive: the self-FK sets their parent_id null rather than cascading.
GET	/api/plans/{id}/ascii	the same fanned board as a grid of characters, one per proxy cell. A plan is a list of rectangles measured in cells, and most of what goes wrong with one is a relation between two of them — a landform wider than the band that reaches it, a wall on the only throat. ...
GET	/api/plans/{id}/png	the same fanned board as PlanSvgEndpoint, rasterized: a picture an image reader can actually open, where the vector card cannot be. Draws off PlanBoardScene, the geometry the SVG endpoint shares, so the two can never disagree about what the plan looks like. 404 when the plan is missing. ...
GET	/api/plans/{id}/svg	render a stored plan to the same board SVG the browse feed uses, so the hold tray can show a thumbnail of a persisted plan. 404 when the plan is missing.
/tree-styles (7)		
GET	/api/tree-styles	the tree library, newest first, each drawn through the grower the export runs. A tree is picked by what it looks like: six woods differ in colour and six species in shape, and neither reads off a number.
POST	/api/tree-styles	—
POST	/api/tree-styles/preview	—
GET	/api/tree-styles/{id}	—
PUT	/api/tree-styles/{id}	—
DELETE	/api/tree-styles/{id}	No question is asked, because nothing binds a recipe: a placement names a key in its own document's registry, which the pull copied, so a map keeps its trees when the row they were pulled from goes.

continued

Endpoint surface, continued

Method	Path	Summary
GET	/api/tree-styles/{id}/json	the recipe as the dressing document states it, which is what a pull copies into a map's registry under a key.
/biome-patterns (6)		
GET	/api/biome-patterns	[?kind=solid cell noise] — the biome library, newest first, each with the patch of ground it tints.
POST	/api/biome-patterns	save a new named field.
POST	/api/biome-patterns/preview	what a draft field draws, saving nothing. Body is a bare BiomeField, unwrapped, the way the material preview takes a bare material.
GET	/api/biome-patterns/{id}	one pattern.
PUT	/api/biome-patterns/{id}	update a pattern in place. A map holds a snapshot rather than a key into this library, so an edit here retints nothing already built.
DELETE	/api/biome-patterns/{id}	forget a pattern. Nothing is asked first: a map took a copy, so no board changes when the row goes.
/porch-styles (6)		
GET	/api/porch-styles	the porch library, newest first, each fronting the sample building.
POST	/api/porch-styles	—
POST	/api/porch-styles/preview	—
GET	/api/porch-styles/{id}	—
PUT	/api/porch-styles/{id}	—
DELETE	/api/porch-styles/{id}	—
/roof-styles (6)		
GET	/api/roof-styles	the roof library, newest first, each on the sample building it tops.
POST	/api/roof-styles	400 {error, findings} on anything CheckRoof names — a roof or a verge that is a log or a ground material, a slab block that is not a slab, and a slab named as a whole-block roof. ...
POST	/api/roof-styles/preview	the building a draft roof tops, previewed without saving it.
GET	/api/roof-styles/{id}	one roof with its per-part courses.
PUT	/api/roof-styles/{id}	Refuses the same way RoofStyleCreateEndpoint does.
DELETE	/api/roof-styles/{id}	forget a roof, unless a house still wears it. A part bound by a building is one the library must refuse to forget, the answer a style already gives a theme.
/storey-styles (6)		
GET	/api/storey-styles	the storey library, newest first, each as the one-storey building it makes.
POST	/api/storey-styles	400 {error, findings} when the storey's window names a block that is not the kind its form needs — a stair lattice or an arched window whose block is not a stair, or a slab band whose block is not a single slab.
POST	/api/storey-styles/preview	—
GET	/api/storey-styles/{id}	—
PUT	/api/storey-styles/{id}	Refuses the same way StoreyStyleCreateEndpoint does.
DELETE	/api/storey-styles/{id}	—
/styles (5)		
GET	/api/styles	[?kind=voronoi noise ...] — the style library, newest first, optionally one kind (the "show every voronoi" browse).
POST	/api/styles	save a new reusable style.
GET	/api/styles/{id}	one style.
PUT	/api/styles/{id}	update a style in place (edits every theme that binds it — a library edit, not a map's applied snapshot).

continued

Endpoint surface, continued

Method	Path	Summary
DELETE	/api/styles/{id}	forget a style. Refused with 409 and the names of the themes and room styles still binding it, since a style is shared by both libraries and its bindings are what the foreign key would otherwise complain about.
/maps (3)		
GET	/api/maps	[?stage=plan sketch configure edit] — the dashboard map list. stage selects a collection, and the four collections are not all the same kind of thing, because a map's layers and its progress are not the same thing. plan and sketch name...
GET	/api/maps/import-candidates	world folders under the maps roots with region/*.mca but no map.xml and not already a map: the new-map import candidates (B8 "open a local folder" source).
GET	/api/maps/stage-counts	the landing cards' tallies, each counting exactly what its list shows: sketches by the layer a map holds, the other two by the stage a map stands at.
/shapes (3)		
GET	/api/shapes/catalog	the vocabulary as cards. Unlike the browse feed this composes nothing and has no cursor: ShapeCatalog is a bounded set derived from the emitters and the tuning constants, so the whole thing is returned and the client filters in the page...
GET	/api/shapes/probe	emit one knob combination and report what the emitter said. This is the half of the catalog that a gallery cannot be: it goes through BoxFiller, so the profile check and the DockingGate run exactly as they do in composition, and a refusal is a first-class answer rather than an error...
GET	/api/shapes/probe/schema	the knob surface the panel drives: every family the emitter builds, its minimum box, which optional knobs it accepts, and whether the production menu admits it...
/compose (2)		
GET	/api/compose	the browse feed. For each seed from the cursor it composes a board, reads its StructureSummary, applies the structural sieve (wool families must-include; hub / frontline forms any-of), then — for survivors only — evaluates, applies the score/wool sieve, and renders the SVG...
POST	/api/compose/pin	keep a browse card. Re-composes the plan from its reproducible descriptor (ComposeStages, so the structural bucket key comes for free) and saves it as a generated row (SaveGeneratedAsync, idempotent by content hash) with its structure...
/configure (2)		
PATCH	/api/configure/{slug}/exclude-island	toggle one island's exclusion.
GET	/api/configure/{slug}/state	the current scan configuration.
/rules (2)		
GET	/api/rules	every rule the studio can cite, with what it means and what to do about it. A refusal carries an id and one sentence about the document it was refused over. The id is stable forever and outlives the task that added it, which is what makes...
GET	/api/rules/terms	every evaluator term with the band it is scoring against right now. GET /api/rules answers what a rule means; this answers the numbers. A soft term's band lives in one of two places — stated on the term itself where it is an authored...
/health (1)		
GET	/api/health	Liveness probe — confirms the API host and FastEndpoints routing are up.

continued

Endpoint surface, continued

Method	Path	Summary
/minecraft (1)		
GET	/api/minecraft/player	?name= uuid= — resolve a Minecraft username or UUID to {uuid, name}. The editor uses it to turn a typed username into the canonical uuid a map.xml stores (and to resolve a stored uuid back to a name). 404 means no account is called that, which is not the same as an error ...
/objectives (1)		
GET	/api/objectives/vocabulary	the destroyable designs (DestroyableStyles), the wool dyes (WoolColors) and what each optional field is worth unauthored (ObjectiveDefaults), served rather than copied onto the client because the client cannot reach PgmStudio.Domain. ...
/sketch (1)		
POST	/api/sketch	originate a sketch: create a draft (geometry-less) map + its layout artifact. Returns the slug; the client navigates to /maps/{slug}/sketch. Body: optional {name} and an optional working frame {width, depth, mode, centerX, centerZ}. ...